

A Lab Manual

on

DEVOPS LAB

R22

(III- B. Tech. – II– Semester)

Submitted to

**DEPARTMENT OF COMPUTER SCIENCE& ENGINEERING
(DATA SCIENCE)**

By

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1. Vision and Mission

CMR INSTITUTE OF TECHNOLOGY

Vision: To create world class technocrats for societal needs.

Mission: Achieve global quality technical education by assessing learning environment through

- Innovative Research & Development
- Eco-system for better Industry institute interaction
- Capacity building among stakeholders

Quality Policy: Strive for global professional excellence in pursuit of key-stakeholders.

DEPARTMENT OF CSE (DATA SCIENCE)

Vision: Develop competent software professionals, researchers and entrepreneurs to serve global society.

Mission: The department of **Computer Science and Engineering (Data Science)** is committed to

- create technocrats with proficiency in design and code for software development
- adapt contemporary technologies by lifelong learning and face challenges in IT and ITES sectors
- quench the thirst of knowledge in higher education, employment, R&D and entrepreneurship

I. Programme Educational Objectives (PEOs): Engineering Graduates will

1. Pursue successful professional career in IT and IT-enabled sectors.
2. Pursue lifelong learning skills to solve complex problems through multidisciplinary-research.
3. Exhibits professionalism, ethics and inter-personal skills to develop leadership qualities.

II. Programme Outcomes (POs): Engineering Graduates will be able to

1. Apply mathematics, science, engineering fundamentals to solve complex engineering problems.
2. Identify, formulate and analyze complex engineering problems to reach substantiated conclusions.
3. Design and develop a component/system/process to solve complex societal engineering problems.
4. Design and conduct experiments to analyze, interpret and synthesize data for valid conclusions.
5. Create, select and apply modern tools, skills, resources to solve complex engineering problems.
6. Apply contextual engineering knowledge to solve societal issues.
7. Adapt modern engineering practices with environmental safety and sustainable development.
8. Apply professional code of ethics, responsibilities and norms in engineering practices.
9. Compete as an individual and/or as a leader in collaborative cross cultural teams.

10. Communicate effectively through technical reports, designs, documentations and presentations.
11. Endorse cognitive management skills to prepare project report using modern tools and finance.
12. Engage in independent and life-long learning in the broad context of technological changes.

III. Programme Specific Outcomes (PSOs): Engineering Graduates will be able to

1. Design and develop Computer-Based-Systems using Algorithms, Networks, Security, Gaming, Full Stack, DevOps, IoT, Cloud, Data Science and AI&ML.
2. Apply data analytics to solve real world problems.

2.Syllabus**Devops LAB**

Course	B.Tech.-VI-Sem.	L	T	P	C
Course Code	22CDPC66	-	-	2	1

Course Outcomes (COs) & CO-PO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	Upon completion of course the students will be able to	PO4	PO5	PO9	PSO2
CO1	identify DevOps workflow	3	3	3	3
CO2	use eclipse and Jenkins for DevOps	3	3	3	3
CO3	develop docker image	3	3	3	3
CO4	take part in grid deployment	3	3	3	3
CO5	make use of monitoring, operations tools in DevOps	3	3	3	3

List of Experiments

Week	Title/Experiment
1	Start DevOps with a workflow that includes four phases: to do, in progress, code review, and done.
2	Setup Eclipse for DevOps.
3	Jenkins Setup on AWS.
4	Build WAR file in DevOps.
5	Ansible Setup and SSH keys.
6	Deploy the artifact on the Test Server.
7	Perform automation using Jenkins.
8	Build and deploy a grid for Chrome and Firefox based testing.
9	Create deployment resource using Kubernetes.
10	Create a docker image for any application using Docker file and push it to Docker hub.
11	Setup Grafana for Devops.
12	Setup Prometheus for Devops.

References

1. DevOps Lab Manual, Department of CSE, CMRIT, Hyd.
2. <https://www.udemy.com/course/practical-devops-for-beginners/>

Micro-Projects: Student should submit a report on one of the following/any other micro-project(s) approved by the lab faculty before commencement of lab internal examination.

1. Deploy a Containerized Web Application.
2. Develop a Version Control System/Tool: GIT.
3. Create a Monitoring Dashboard for any Application.
4. Implement a Continuous Integration/Continuous Delivery (CI/CD) Pipeline for an application.
5. Implement DevOps Lifecycle with Amazon Web Services (AWS).
6. Build a Scalable Application with Docker.
7. Create a Jenkins project that connects to a remote Jenkins server and controls it.
8. Deploy an application (with high availability) with a database
9. Create a Continuous Delivery of a Java Web Application.
10. Build and execute a selenium project.

3.Student Entry Behavior or Pre-requisites

- Students should have basic knowledge on Linux commands
- Students should have basic knowledge on basic programming.
- Student should have knowledge on software engineering concepts
- These prerequisites are taken by the students during the first two years. However during the initial sessions the topics are reviewed.

4.Course Outcomes

Course Outcome	Course Outcome Statements
CO - 1	identify DevOps workflow
CO – 2	use eclipse for DevOps
CO – 3	develop docker image
CO – 4	take part in grid deployment
CO – 5	make use of Jenkins framework in DevOps

5.Mapping of Course with PEOs-POs

(Only Ticking)

Program Educational Objectives (PEOs)

Sl. No.	PEOs Name	Program Education Objective Statements
1	PEO - 1	Pursue successful professional career in IT and IT-enabled sectors. [PO's: 1,2,3,4,5,7,8,9,10,11 and 12] [PSO's: 1 and 2]
2	PEO – 2	Pursue lifelong learning skills to solve complex problems through multidisciplinary-research. [PO's: 1,2,3,4,5,6,7,8,9,10 and 12] [PSO's: 1, 2]
3	PEO – 3	Exhibits professionalism, ethics and inter-personal skills to develop leadership qualities. [PO's: 1,2,3,4,5,6,7,8,9,10,11 and 12] [PSO's: 2]

Program Outcomes (POs)

PO Name	Graduate Attributes	PO Statements
PO1	Engineering knowledge	Apply mathematics, science, engineering fundamentals to solve complex engineering problems. [PEO's: 1,2 and 3] [PSO's: 1,2]
PO 2	Problem analysis	Identify, formulate and analyze complex engineering problems to reach substantiated conclusions. [PEO's: 1,2 and 3] [PSO's: 1,2]
PO 3	Design/development of solutions	Design and develop a component/system/process to solve complex societal engineering problems. [PEO's: 1,2 and 3] [PSO's: 1,2]
PO 4	Conduct investigations of complex problems	Design and conduct experiments to analyze, interpret and synthesize data for valid conclusions. [PEO's: 1,2 and 3] [PSO's: 1,2]
PO 5	Modern tool usage	Create, select and apply modern tools, skills, resources to solve complex engineering problems. [PEO's: 1,2 and 3] [PSO's: 1,2]
PO 6	The engineer and society	Apply contextual engineering knowledge to solve societal issues. [PEO's: 2 and 3]
PO 7	Environment and sustainability	Adapt modern engineering practices with environmental safety and sustainable development. [PEO's: 1,2 and 3]
PO 8	Ethics	Apply professional code of ethics, responsibilities and norms in engineering practices. [PEO's: 1,2 and 3] [PSO's: 2]
PO 9	Individual and team work	Compete as an individual and/or as a leader in collaborative cross cultural teams. [PEO's: 1,2 and 3]
PO 10	Communication	Communicate effectively through technical reports, designs, documentations and presentations. [PEO's: 1,2 and 3] [PSO's: 2]
PO 11	Project management and finance	Endorse cognitive management skills to prepare project report using modern tools and finance. [PEO's: 1 and 3] [PSO's: 2]
PO 12	Life-long learning	Engage in independent and life-long learning in the broad context of technological changes. [PEO's: 1,2 and 3] [PSO's: 1,2]

6.Mapping Of Course Outcomes With POs

No	Course Outcomes	P0 ₁	P0 ₂	P0 ₃	P0 ₄	P0 ₅	P0 ₆	P0 ₇	P0 ₈	P0 ₉	P0 ₁₀	P0 ₁₁	P0 ₁₂	Avg
1	CO - 1				3	3				3				
2	CO – 2				3	3				3				
3	CO – 3				3	3				3				
4	CO – 4				3	3				3				
5	CO – 5				3	3				3				
	Avg				3	3				3				

7.Direct Course Assessment

(As mentioned in following table of 10 parameters, of which consider only the parameters required for this courses)

No	Description	Targeted Performance	Actual Performance	Remarks	Course Attainment
1	Internal Marks(25)	80% of Students(182 Students) should Secure 60% of Internal Marks i.e., 15 Marks			
2	External Marks(50)	80% of Students(182 Students) should Secure 70% of External Marks i.e., 35 Marks			
3	Clearing of Subject	A minimum of 95% of Students(216 Students) should clear this course in first attempt			
4	Getting First Class	90% of Students(205 Students) should Secure I Class Marks i.e., 45 Marks in my course			
5	Distinction	80% of Students (182 Students) should secure First Class With Distinction i.e., 53 Marks in my course			
6	Outstanding Performance	60% of Students (137 Students) should secure 80% and above Marks. i.e., 60 Marks in my course			

8. Indirect Course Assessment

(As mentioned-strong (3), moderate (2), weak (1) & no comment (0))

Mission Statement of CSE

- Impart fundamentals through state of art technologies for research and career in Computer Science & Engineering(DS).
- Create value-based, socially committed professionals for anticipating and satisfying fast changing societal requirements.
- Foster continuous self learning abilities through regular interaction with various stake holders for holistic development.

Correlation of Mission Elements with Mission Statement of CSE(DS) Department related to the Course
(only Ticking given by faculty)

No	Mission Elements	Strong	Moderate	Weak	No Comment
M-1	Impart Fundamentals	√			
M-2	State Of Art Technologies	√			
M-3	Research & Career Development	√			
M-4	Value based Socially Committed Professional	√			
M-5	Anticipating & Satisfying Industry Trends		√		
M-6	Changing Societal Requirements			√	
M-7	Foster Continuous Learning	√			
M-8	Self Learning Abilities	√			
M-9	Interaction with stakeholders	√			
M-10	Holistic Development		√		

Indirect Course Assessment through Student Satisfaction Survey

(Note for *): Parameters used for course teaching like

a: Classroom teaching b: Simulations c: labs d: Mini_Projects
 e: Major Projects f: Conferences g: professional activities
 h: Technical Clubs i: Guest Lectures j: Workshops k: Technical Fests l: Tutorials
 m: NPTLs n: Digital Library o: Industrial Visits p: software Tools q:
 Internship/training r: Technical Seminars
 s: NSS t: NSS u: sports etc.

No	Question Based on PEO/ PO/PSO/CO	Parameters (a /b /c.../)*	Strong (3)	Moderate (2)	Weak (1)	No comment (0)
1	Did the course impart fundamentals through interactive learning and contribute to core competence?					
2	Did the course provide the required knowledge to foster continuous learning?					
3	Whether the syllabus content anticipates & satisfies the industry and societal needs?					
4	Whether the course focuses on value based education to be a socially committed professional?					
5	Rate the role of the facilitator in mentoring and promoting the self learning abilities to excel academically and professionally?					
6	Rate the methodology adopted and techniques used in teaching learning processes?					
7	Rate the course in applying sciences & engineering fundamentals in providing research based conclusions with the help of modern tools?					
8	Did the course have any scope to design, develop and test a system or component?					
9	Rate the scope of this course in addressing cultural, legal, health, environment and safety issues?					
10	Scope of applying management fundamentals to demonstrate effective technical project presentations & report writing?					
	Total					
	Average					
	Total Average					

9.Overall Course Assessment

(80% Direct + 20% Indirect, if any)

No	Assessment Type	Weightage	Attainment Level
1	Direct-Assignment, Quiz, Subjective, University Exams, Results, Bench Marks	0.8	
2	Indirect-Surveys-Questionnaire	0.2	
	Overall		

DEVOPS LAB Course Attainment level:

10.Pi diagrams, Bar charts, Histograms

(For representing previous results, if any)

ICS Pass % for Last 4 Academic Years	Appeared	Passed	Pass%

11.Lesson/Course Plan

Week No.	Name of the Program	Week	Text Books	Mode of Assessment
1	Start DevOps with a workflow that includes four phases: to do, in progress, code review, and done	1	R1	By observations, lab records, viva-voice
2	Setup Eclipse for DevOps.	2	R1	By observations, lab records, viva
3	Jenkins Setup on AWS.	3	R1	By observations, lab records, viva
4	Build WAR file in DevOps.	4	R1	By observations, lab records, viva
5	Ansible Setup and SSH keys.	5	R1	By observations, lab records, viva
6	Deploy the artifact on the Test Server.	6	R1	By observations, lab records, viva
7	Perform automation using Jenkins.	7	R1	By observations, lab records, viva
8	Build and deploy a grid for Chrome and Firefox based testing.	8	R1	By observations, lab records, viva
9	Create deployment resource using Kubernetes.	9	R1	By observations, lab records, viva
10	Create a docker image for any application using Docker file and push it to Docker hub.	10	R1	By observations, lab records, viva
11	Setup Grafana for Devops.	11	R1	By observations, lab records, viva
12	Setup Prometheus for Devops.	12	R1	By observations, lab records, viva

12.Programs

Experiment-1

Aim: Start DevOps with a workflow that includes four phases: to do, in progress, code review, and done.

Require Software& Tools: JIRA, KANBAN.

Procedure:

Phase 1: To Do

- **Objective:** Identify and prioritize tasks or features to be developed.
- **Key Actions:**
 - Define tasks clearly in a backlog.
 - Prioritize tasks based on impact, urgency, and dependencies.
 - Assign owners or teams to each task.

Tools:

Jira, Trello, GitHub Issues, or Asana.

Phase 2: In Progress

- **Objective:** Actively work on tasks selected from the "To Do" phase.
- **Key Actions:**
 - Begin coding or configuring based on task requirements.
 - Update the task status to reflect ongoing work.
 - Ensure team members collaborate effectively (e.g., stand-ups, pair programming).

Best Practices:

- Use branches in version control systems for individual tasks (e.g., Git feature branches).
- Write unit tests alongside development.

Phase 3: Code Review

- **Objective:** Validate the quality, functionality, and security of the code.
- **Key Actions:**
 - Submit pull requests for peer review.
 - Review code for adherence to standards, logic, and potential issues.
 - Approve or request changes.

Tools:

GitHub Pull Requests, GitLab Merge Requests, Bitbucket.

Automation:

Integrate CI/CD pipelines to run tests automatically during reviews.

Phase 4: Done

- **Objective:** Mark tasks as completed and deploy changes if necessary.
- **Key Actions:**
 - Merge the approved code into the main branch.
 - Deploy to staging or production environments.
 - Monitor deployment and validate functionality.

Post-Completion:

- Add documentation for the changes.
 - Gather feedback from stakeholders or users.
-

Workflow Visualization

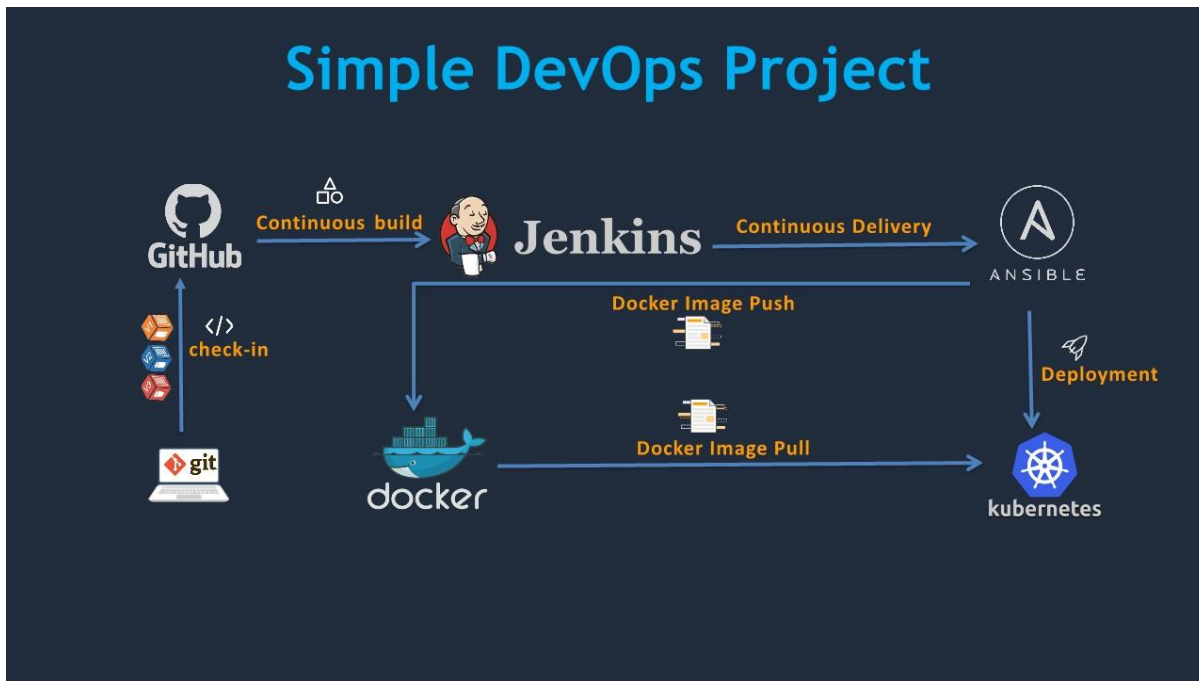
A Kanban board or similar visual representation can help track the status of tasks across these phases. For example:

1. **To Do:** Contains all pending tasks.
2. **In Progress:** Tasks currently being worked on.
3. **Code Review:** Tasks awaiting review or approval.
4. **Done:** Completed and deployed tasks.

Tools:

Trello, Jira, Azure.

Output:-



Experiment-2

Aim: Setups Eclipse for Devops

Note: Write the given steps in your lab manual. The provided image is just for your better understanding.

Require Software& Tools: Eclipse, Java jdk-17, Tomcat v.9, TestNG and Dependencies.

Procedure:

Step-1: Install Jdk-17 and set the java path in System environment

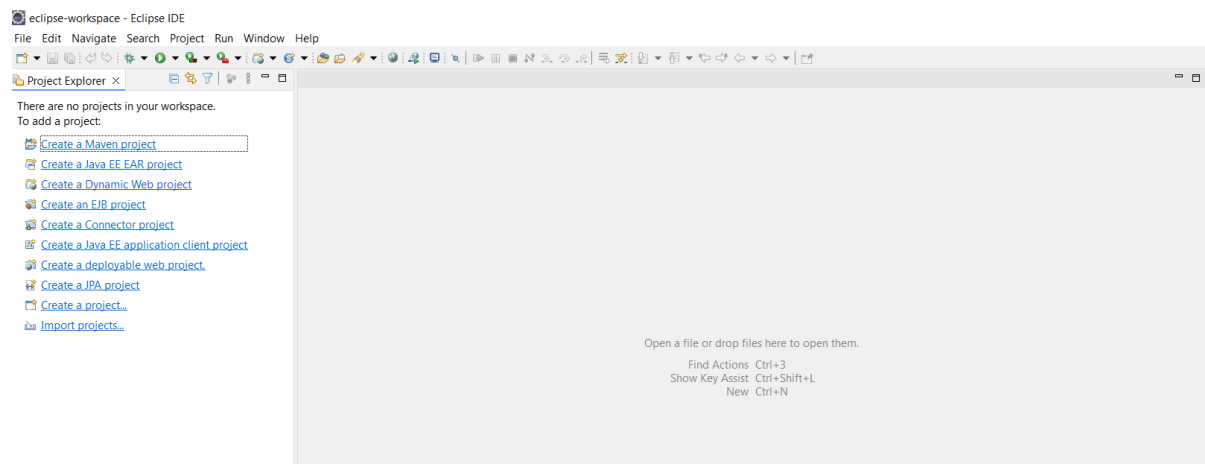
Step-2: Download eclipse zipfile and extract the contents the all eclipse file

Step-3: Create a Maven Project from eclipse as:

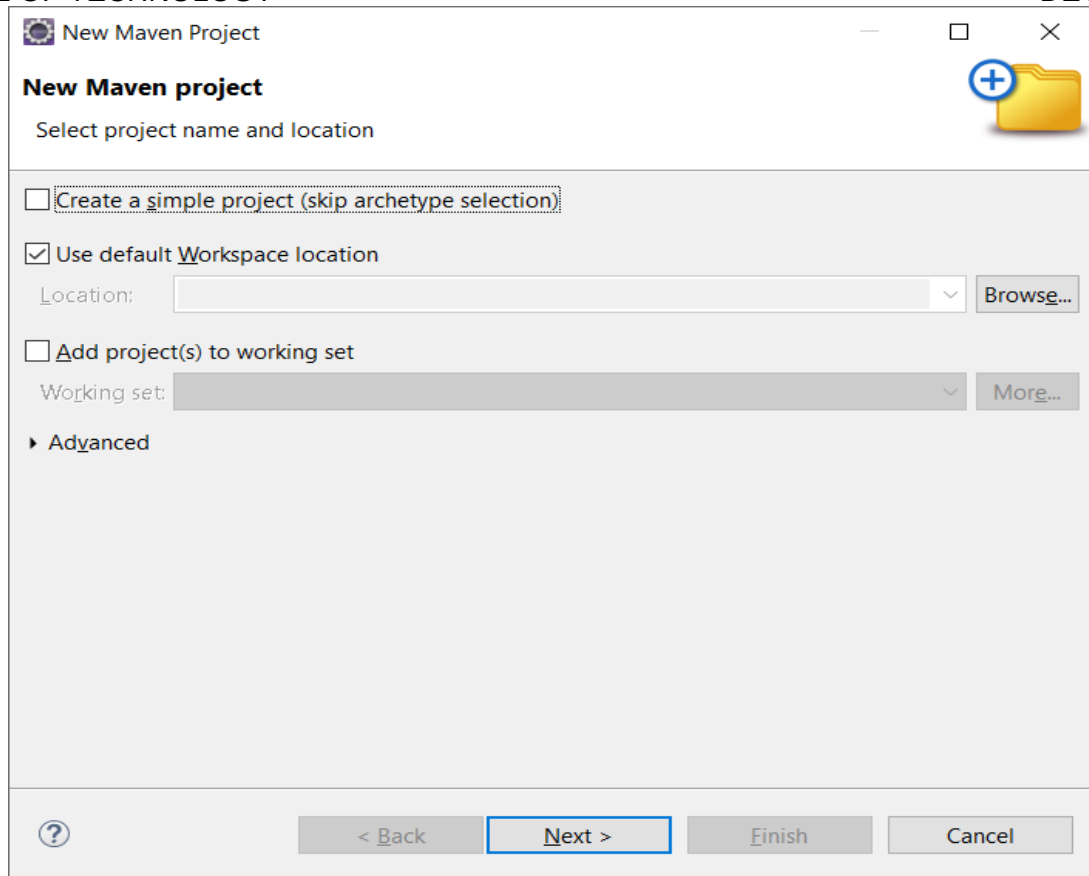
OR

Click on File in left corner -> Click on new -> click on Maven Project and follow the given image steps.

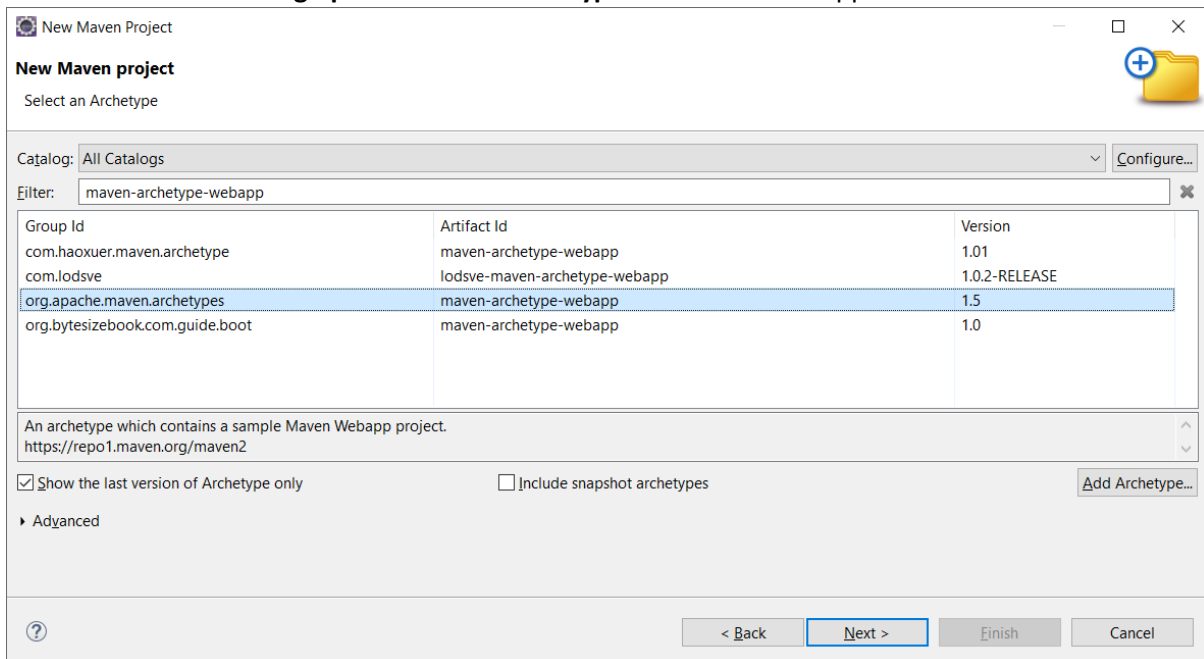
a.



b.



- b. Click Next and Search **org.apache.maven.archetypes** and select webapp file



- c. In a **group id** you can type anything like name and in **aartefact id**: you can type anything like your roll number

New Maven Project

Specify Archetype parameters

Group Id: DevOpsExp2

Artifact Id: MyDevOpsPipeline

Version: 0.0.1-SNAPSHOT

Package: war

☒ run archetype generation interactively

Properties available from archetype:

Name	Value

► Advanced

< Back Next > Finish Cancel

- d. Click Finish

Open a file or drop files here to open them.

Find Actions Ctrl+3
Show Key Assist Ctrl+Shift+L
New Ctrl+N

Problems Servers Terminal Data Source Explorer Properties Console X

C:\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.5.v20241023-1957\jre\bin\javaw.exe (26-Dec-2024, 7:07:36 pm) [pid: 5840]

```
Progress (1): 16/16 MB
Progress (1): 16/16 MB
Progress (1): 16/16 MB
Progress (1): 16/16 MB
Progress (1): 16/16 MB
Progress (1): 16 MB

Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (16 MB at 3.3 MB/s)
[INFO] Archetype repository not defined. Using the one from [org.apache.maven.archetypes:maven-archetype-webapp:1.5] found in catalog remote
[INFO] Using property: groupId = DevOpsExp2
[INFO] Using property: artifactId = MyDevOpsPipeline
[INFO] Using property: version = 0.0.1-SNAPSHOT
[INFO] Using property: package = war
Confirm properties configuration:
groupId: DevOpsExp2
artifactId: MyDevOpsPipeline
version: 0.0.1-SNAPSHOT
package: war
Y: Y
```

- e. Type Y and Press enter, You should see a Build Success message as below:

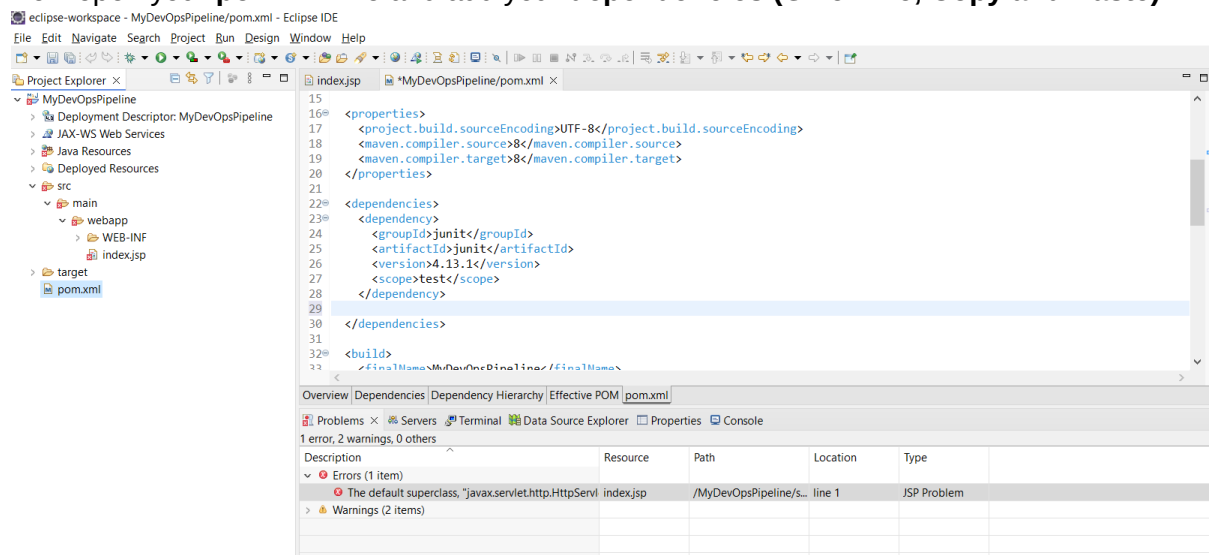
```

Show Key Assist Ctrl+Shift+L
New Ctrl+N

[terminated] C:\eclipse\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64_21.0.5.v20241023-1957\jre\bin\javaw.exe (26-Dec-2024, 7:07:36 pm) [pid: 5840]
[INFO] Parameter: groupId, Value: DevOpsExp2
[INFO] Parameter: artifactId, Value: MyDevOpsPipeline
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[INFO] Parameter: package, Value: war
[INFO] Parameter: packageInPathFormat, Value: war
[INFO] Parameter: package, Value: war
[INFO] Parameter: groupId, Value: DevOpsExp2
[INFO] Parameter: artifactId, Value: MyDevOpsPipeline
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[WARNING] CP Don't override file C:\Users\Dell\eclipse-workspace\MyDevOpsPipeline\src\main\webapp
[WARNING] CP Don't override file C:\Users\Dell\eclipse-workspace\MyDevOpsPipeline\src\main\resources
[INFO] Project created from Archetype in dir: C:\Users\Dell\eclipse-workspace\MyDevOpsPipeline
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:22 min
[INFO] Finished at: 2024-12-26T19:09:00+05:30
[INFO] -----

```

Step-4: now open your pom.xml file and add your dependencies (Given file, Copy and Paste)



Step-5: Update your project once (Right click on Project -> click on Maven -> click on Update Project)

Step-6: Download Apache tomcat v9 from Official website.

The Apache Tomcat® software is an open source implementation of the [Jakarta Servlet](#), [Jakarta Pages](#), [Jakarta Expression Language](#), [Jakarta WebSocket](#), [Jakarta Annotations](#) and [Jakarta Authentication](#) specifications. These specifications are part of the [Jakarta EE platform](#).

The Jakarta EE platform is the evolution of the Java EE platform. Tomcat 10 and later implement specifications developed as part of Jakarta EE. Tomcat 9 and earlier implement specifications developed as part of Java EE.

The Apache Tomcat software is developed in an open and participatory environment and released under the [Apache License version 2](#). The Apache Tomcat project is intended to be a collaboration of the best-of-breed developers from around the world. We invite you to participate in this open development project. To learn more about getting involved, [click here](#).

Apache Tomcat software powers numerous large-scale, mission-critical web applications across a diverse range of industries and organizations. Some of these users and their stories are listed on the [PoweredBy](#) wiki page.

Apache Tomcat, Tomcat, Apache, the Apache feather, and the Apache Tomcat project logo are trademarks of the Apache Software Foundation.

Tomcat Migration Tool for Jakarta EE 1.0.9 Released

2025-01-21

The Apache Tomcat Project is proud to announce the release of 1.0.9 of the Apache Tomcat Migration Tool for Jakarta EE. This release contains a number of bug fixes and improvements compared to [version 1.0.8](#).

The notable changes in this release are:

- Fix an issue that `matchExcludesAgainstPathName` didn't work for files. Based on a pull request by Semiao Marco.
- Added a new profile, `SERVLET` that only migrates the `javax.servlet` package and sub-packages. Provided by Ralf Wiebicke.
- Update dependencies

Full details of these changes, and all the other changes, are available in the [changelog](#).

Mirrors

You are currently using <https://d1cdn.apache.org/>. If you encounter a problem with this mirror, please select another mirror. If all mirrors are failing, there are *backup* mirrors (at the mirrors list) that should be available.

Other mirrors:

9.0.98

Please see the [README](#) file for packaging information. It explains what every distribution contains.

Binary Distributions

- Core:
 - [zip \(pgp, sha512\)](#)
 - [tar.gz \(pgp, sha512\)](#)
 - [32-bit Windows zip \(pgp, sha512\)](#)
 - [64-bit Windows zip \(pgp, sha512\)](#)
 - [32-bit/64-bit Windows Service Installer \(pgp, sha512\)](#)
- Full documentation:
 - [tar.gz \(pgp, sha512\)](#)
- Deployer:
 - [zip \(pgp, sha512\)](#)
 - [tar.gz \(pgp, sha512\)](#)
- Embedded:
 - [tar.gz \(pgp, sha512\)](#)
 - [zip \(pgp, sha512\)](#)

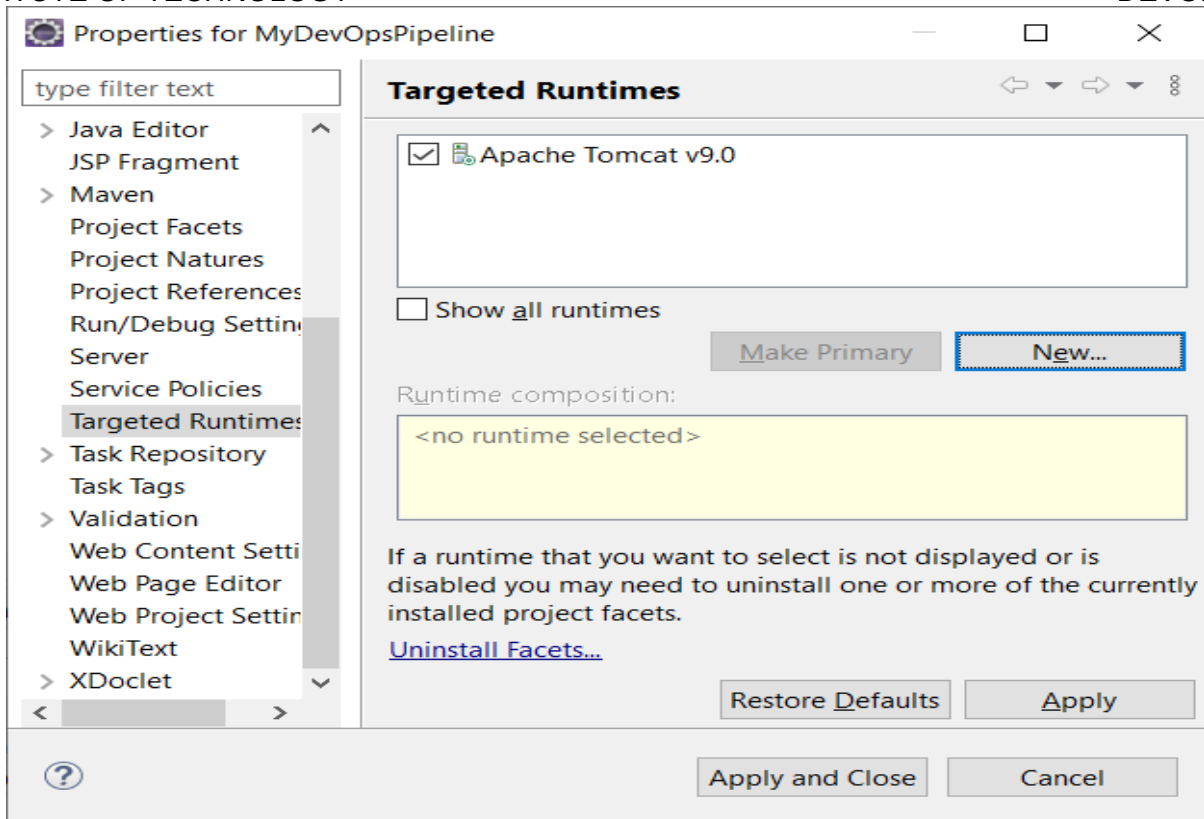
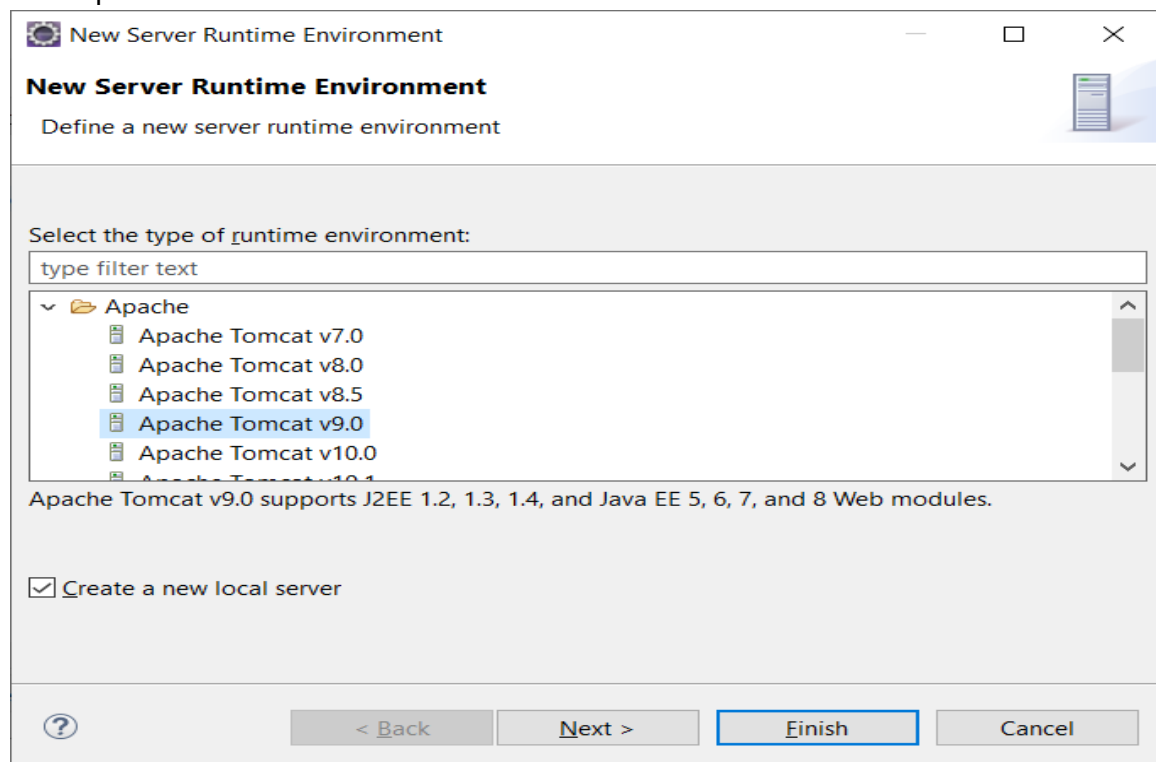
Source Code Distributions

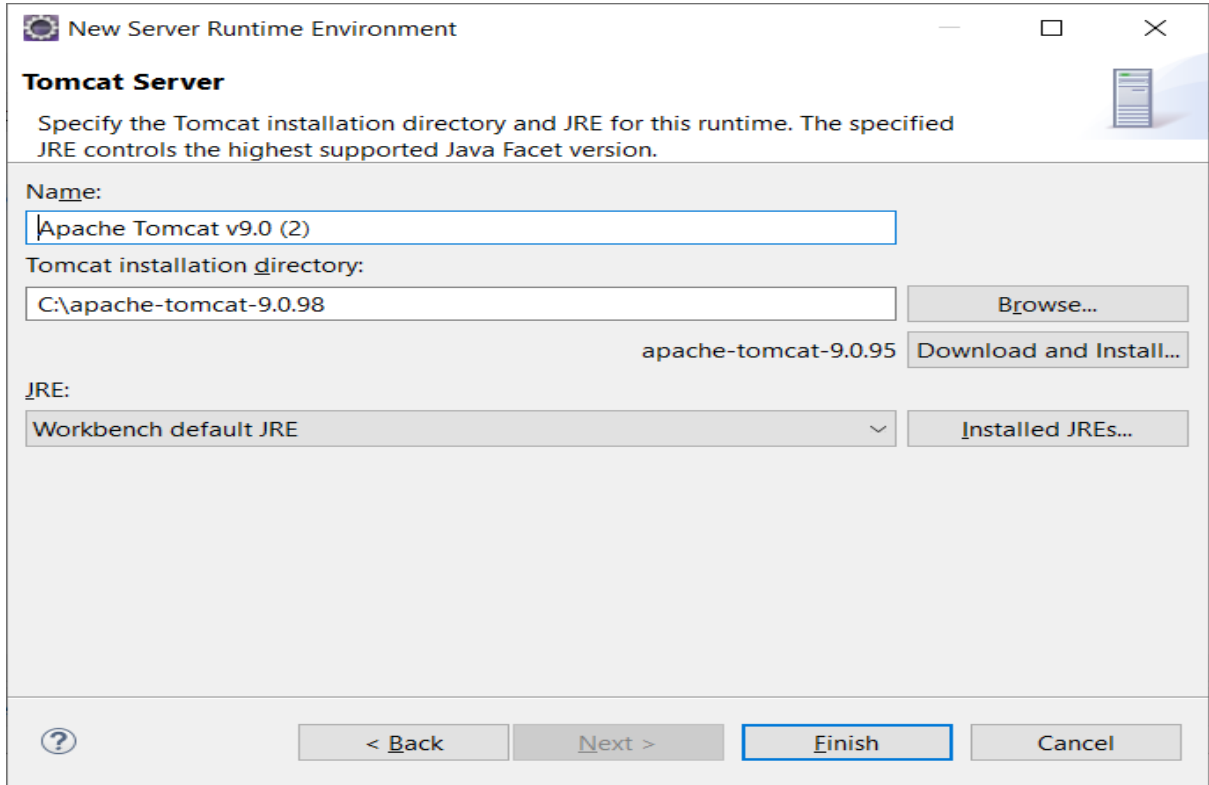
- [tar.gz \(pgp, sha512\)](#)
- [zip \(pgp, sha512\)](#)

Step-7: After Download the Apache tomcat, Extract the .zip file and paste your apache-tomcat-9.0.98 folder in your folder

Step-8: Now click on your **project option in Menu** -> **Click on Properties** -> **Click on Targeted Runtime**

Step-9: Click on **new** or follow the given image

**Step-10:** Select Apache Tomcat v9.0**Step-11:** Click on Browse and Select your Extracted file and then click on finish as given image

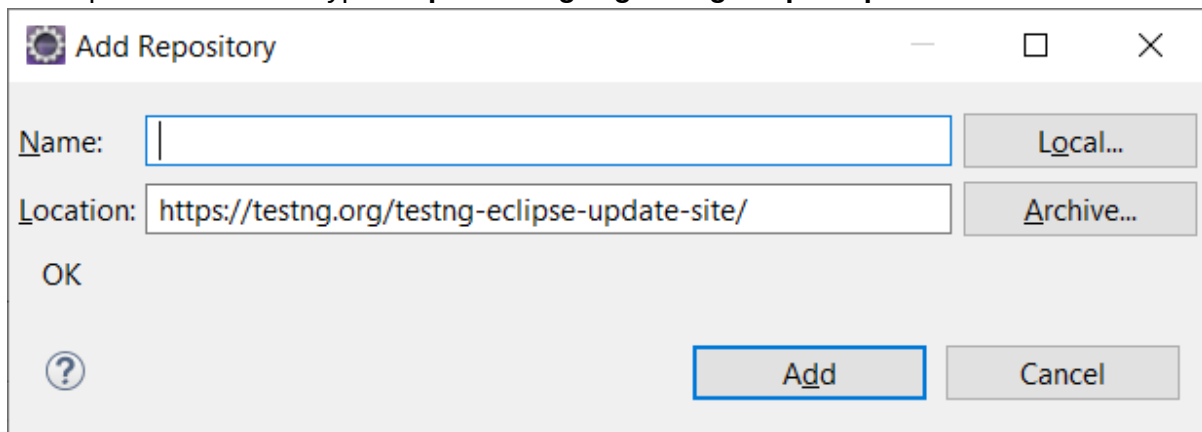


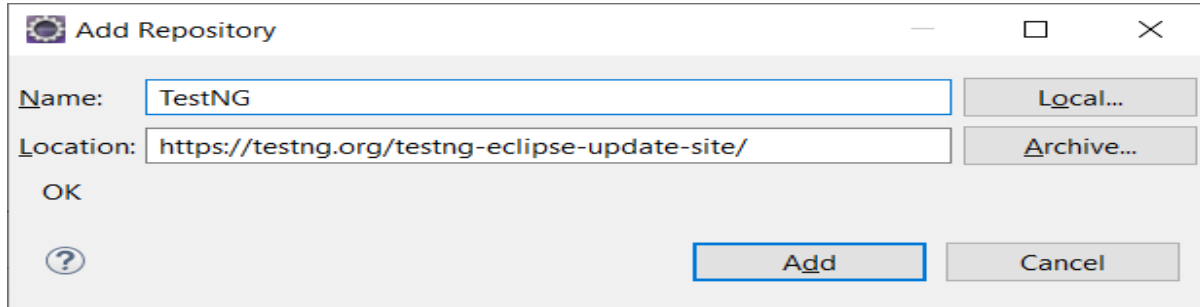
Step-12: Now Click on **help Menu** ->click on **Install new Software**.

Step-13: Click on **Add** and it will show a **popup dialog box** like given image

In the place of Name type: **TestNG**

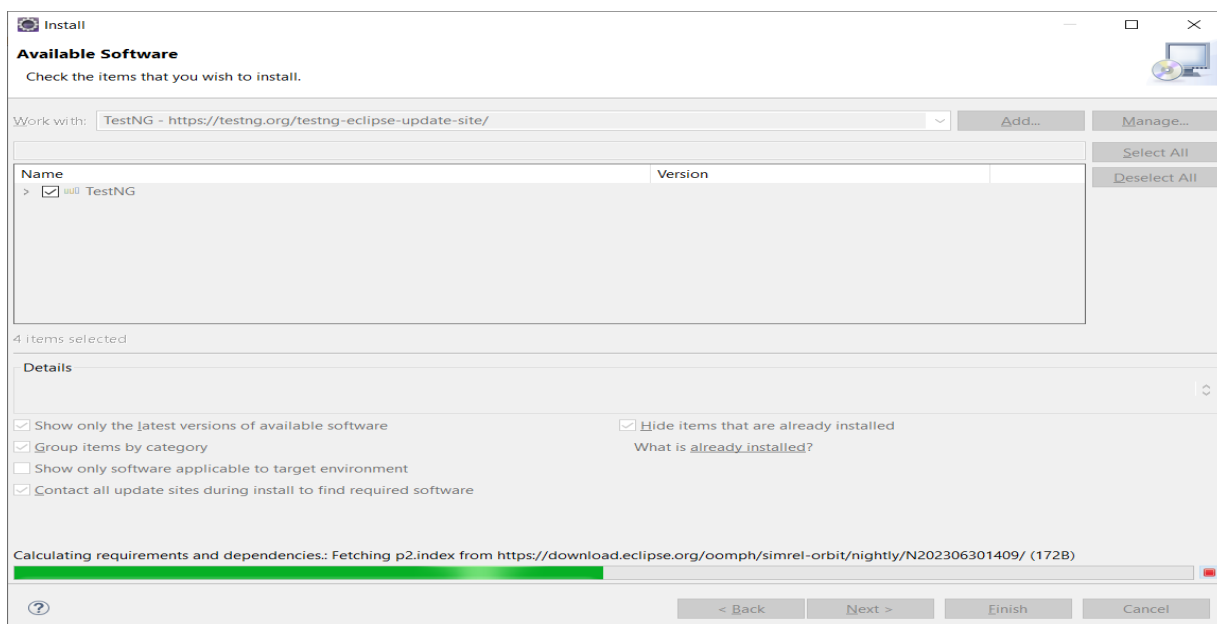
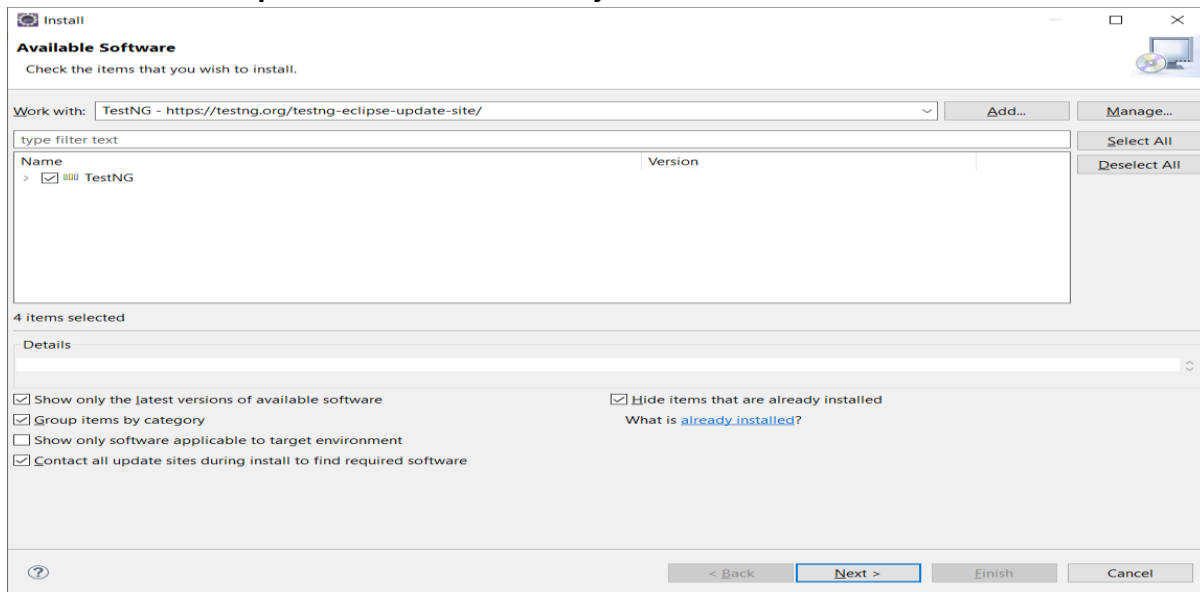
In the place of Location type: **<https://testng.org/testng-eclipse-update-site/>**



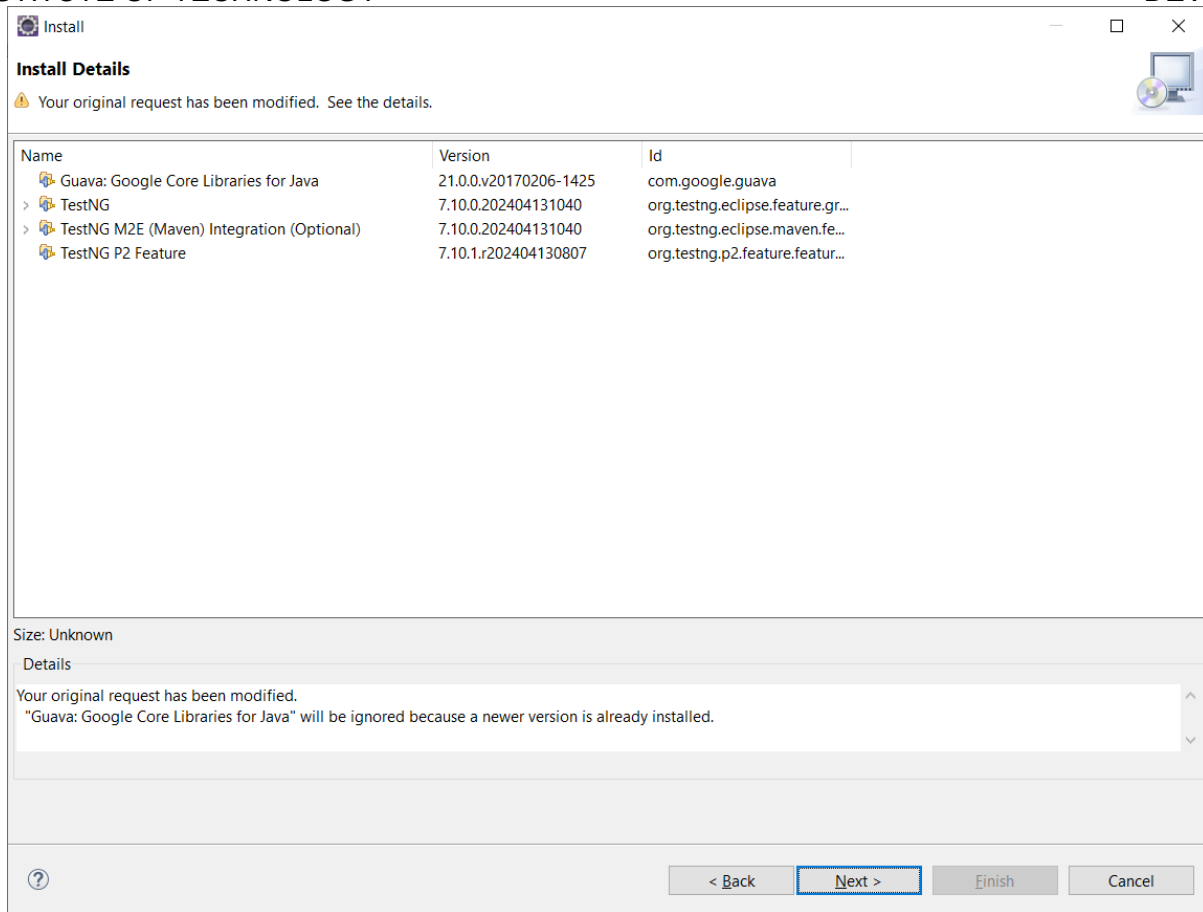


Step-14: Click on **Add** -> It will load **at**estNG **Dependencies** -> Select **TestNg** like given Image and then click **Next**.

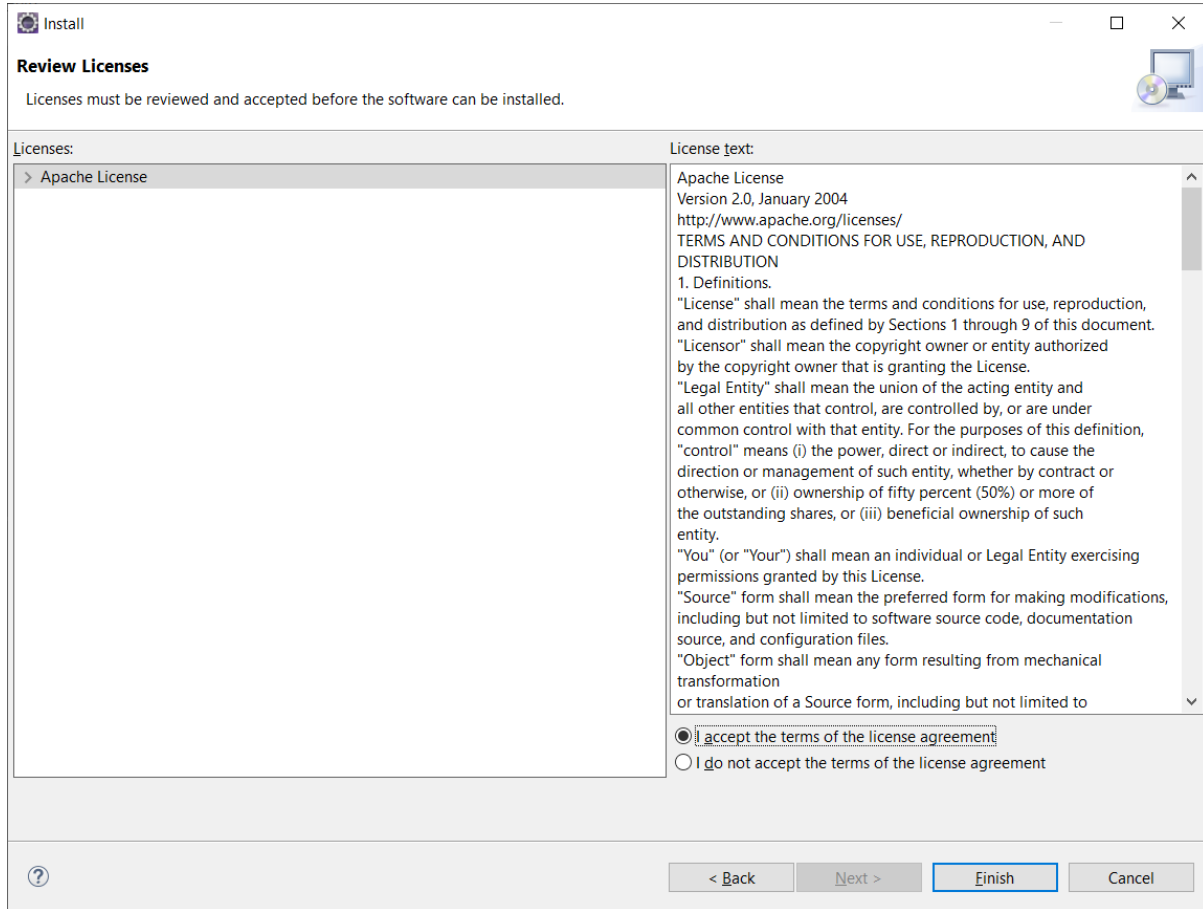
It will take 10 minute to update TestNG in our Project



Step-15: After downloading the all dependencies it will show some file select all and click on next.



Step-16: Accept Terms and condition and click on finish

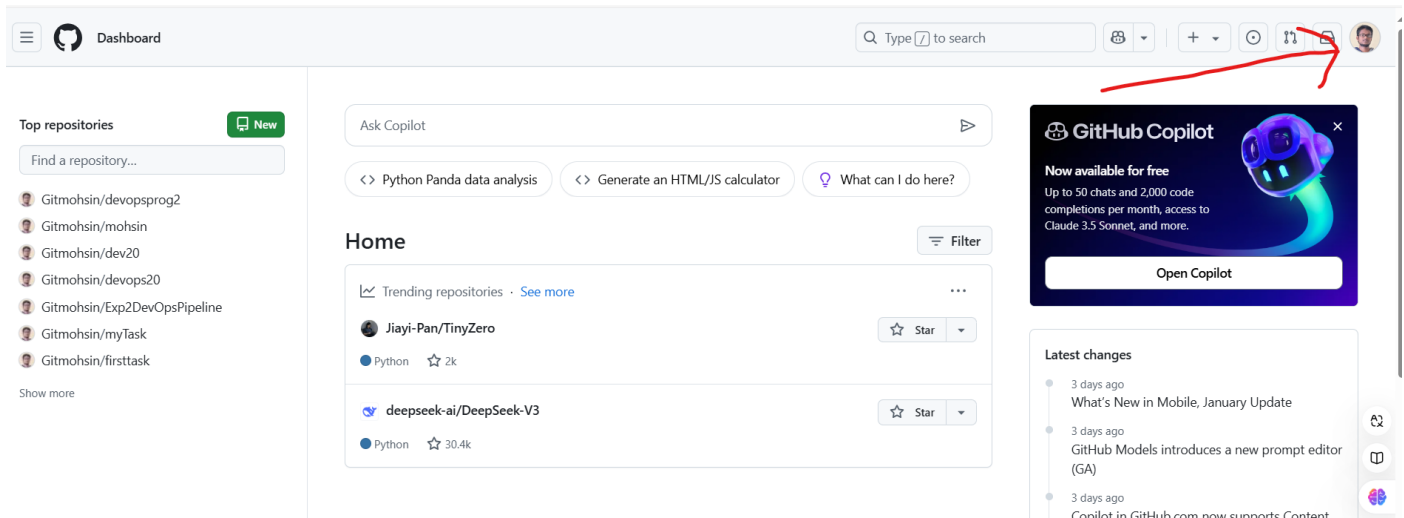


Step-17: After finish it will show **restart option (Restart the Project)** otherwise just **update** once of your project.

Step-18: Now Login your **GitHub Account**.

Step-19: Create a New **Repository** and Copy your **Repository** and paste in notepad

Step-20: After that **Click** on your **Profile** in Right corner -> **Click** on **Setting**.



Step-21: It will show a new page, scroll down and select the **developer setting** -> click on **personal access token** -> select **Token(Classic)** -> click on **Generate new token** and select **Generate new token(Classic)** -

>write your **token name** and select **repo option** and scroll down and click on **Generate Token**. (Follow the given Image)

The image shows the GitHub Developer Settings page. A red arrow points to the user profile icon in the top right corner. Another red arrow points to the 'Developer settings' link in the left sidebar. A third red arrow points to the 'Tokens (classic)' option under 'Personal access tokens'.

GitHub Apps

- OAuth Apps
- Personal access tokens
 - Fine-grained tokens
 - Tokens (classic)

GitHub Apps

Want to build something that integrates with and extends GitHub? Register a new GitHub App to get started developing on the GitHub API.

[New GitHub App](#)

[View documentation](#)



No personal access token created

Need an API token for scripts or testing? Generate a personal access token for quick access to the GitHub API.

[Generate new token](#)

Generate new token Beta
Fine-grained, repo-scoped

Generate new token (classic)
For general use

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

[Preview](#)

Note

What's this token for?

Expiration *

[30 days](#) The token will expire on Wed, Feb 26 2025

Select scopes

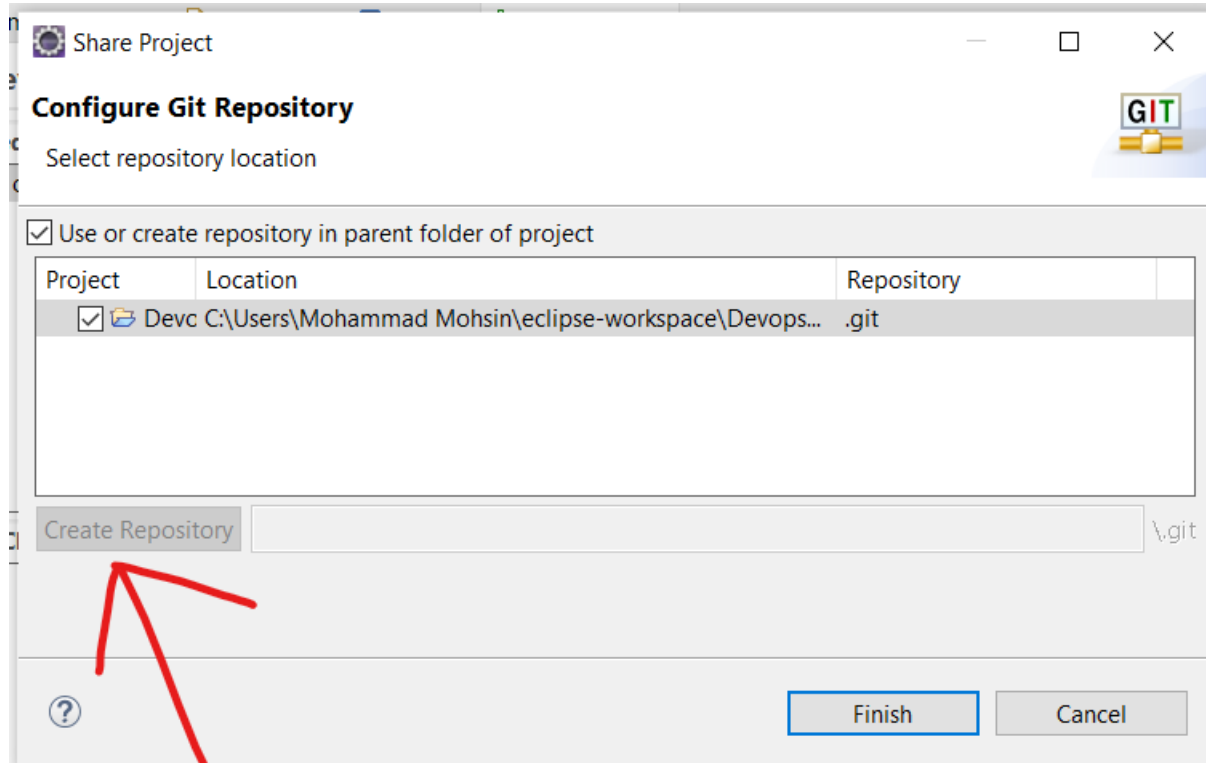
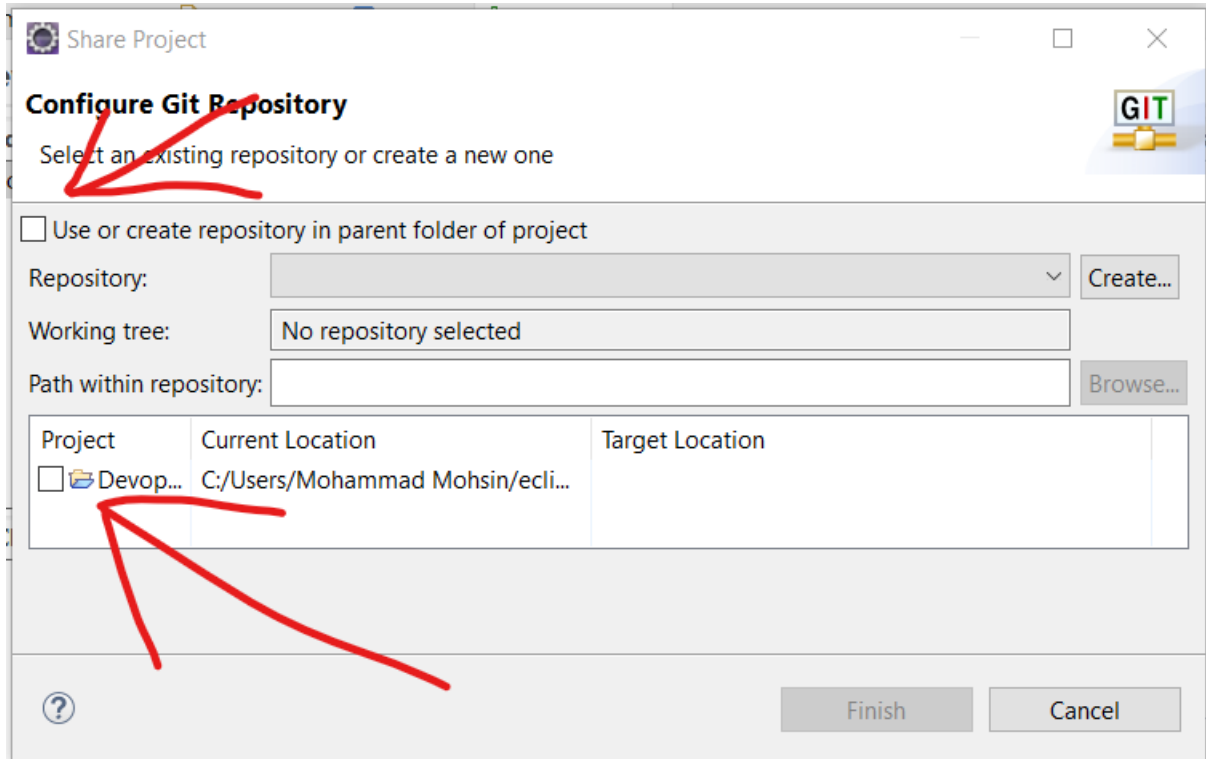
Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☐ repo	Full control of private repositories
☐ repo:status	Access commit status
☐ repo_deployment	Access deployment status
☐ public_repo	Access public repositories
☐ repo:invite	Access repository invitations
☐ security_events	Read and write security events

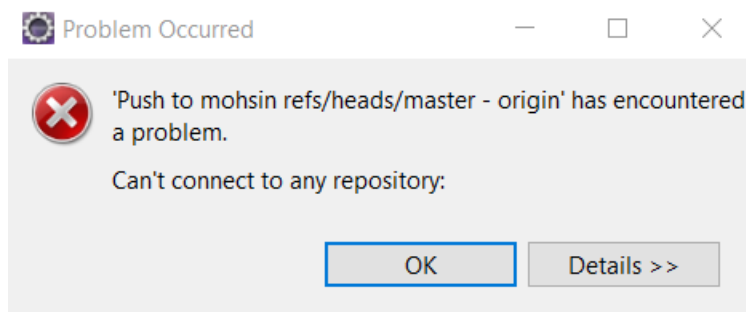
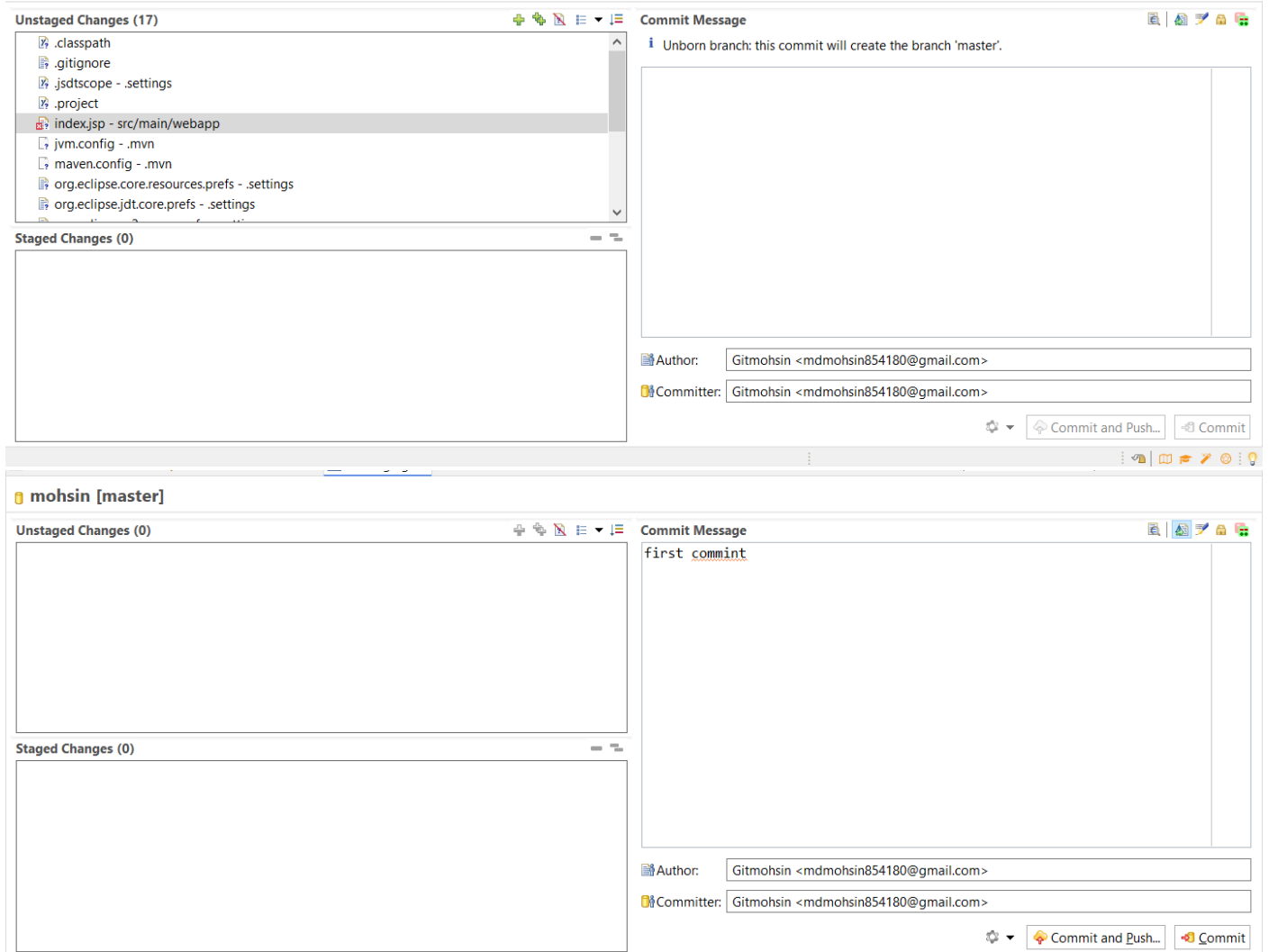
Step-22: After Generating the **token** copy the token id and paste in a**NotePad**.

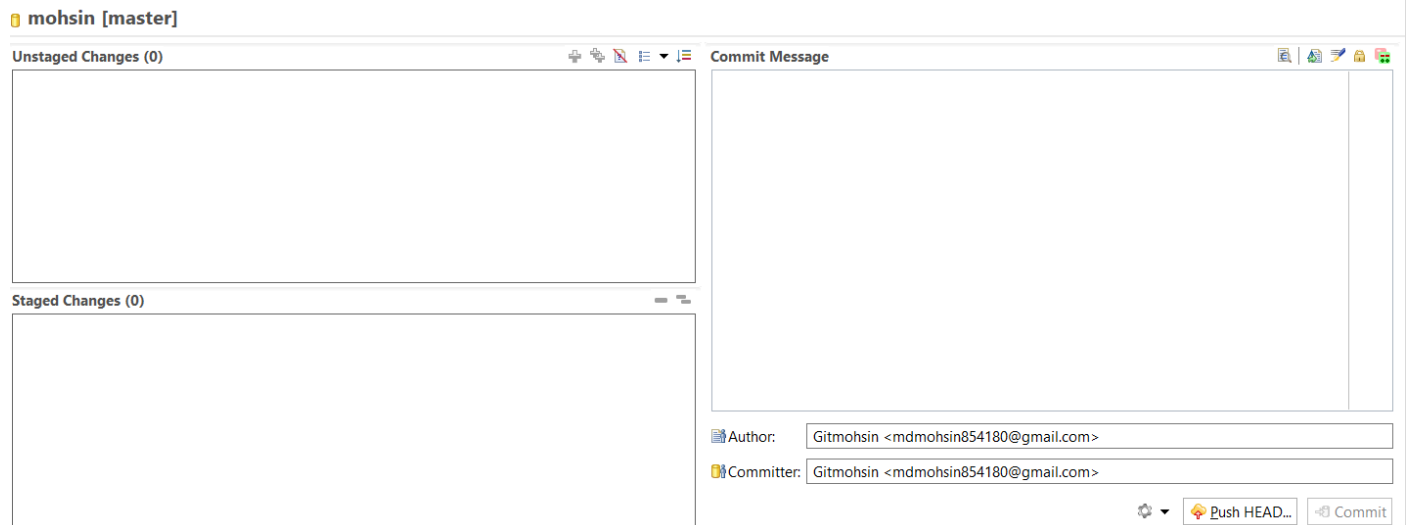
Step-23: Now come on your project and right click on your**project** ->Click on **Team** ->Click on **Share Project**.

Step- 24: It will open a Dialog Box for**GitHubSetup**, select the **option Use or create repository in parent folder of project** - >Select your Project and Click on **Create Repository** and click on **Finish**.

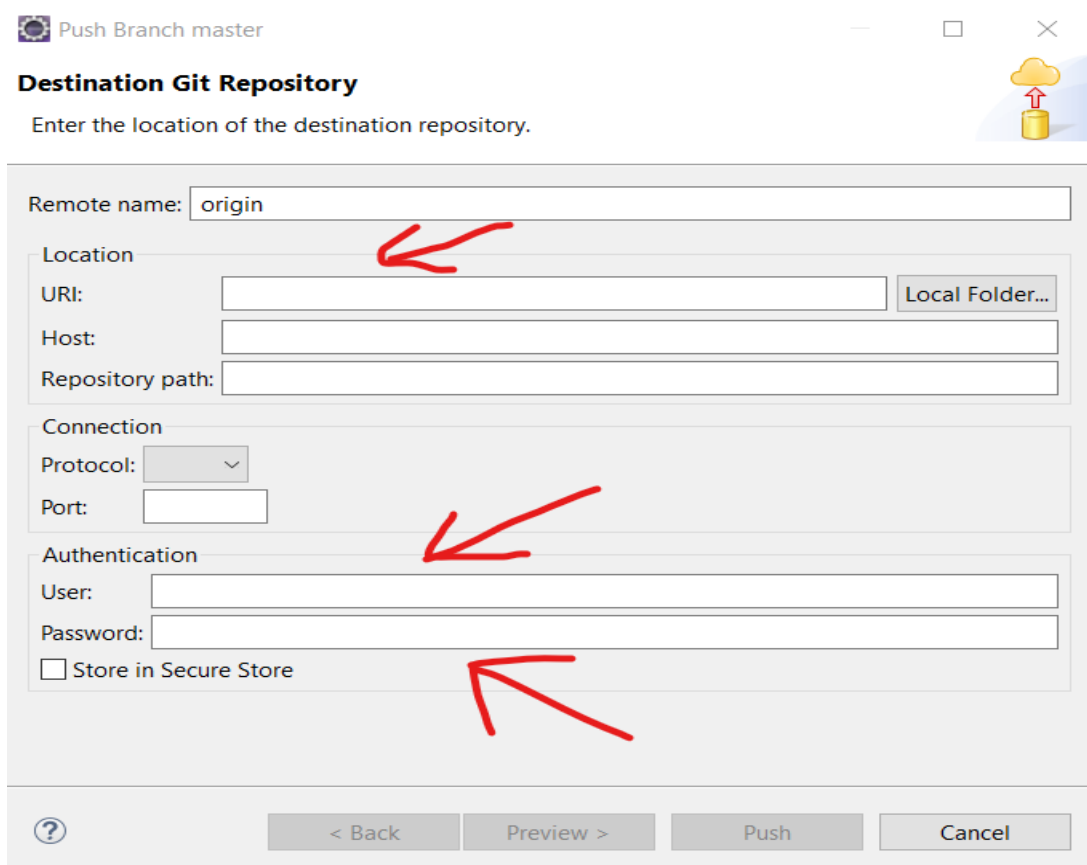


Step-25: After that again **Right click on yourProject** and select the **Team** -> click on **Commit**-> and **stage your all file** -> and Write a comment (i.e. First Commit) and click on **Commit and push**-> after that it show an **error dialog** ->click **OK** ->now again click on **Push Head** Button



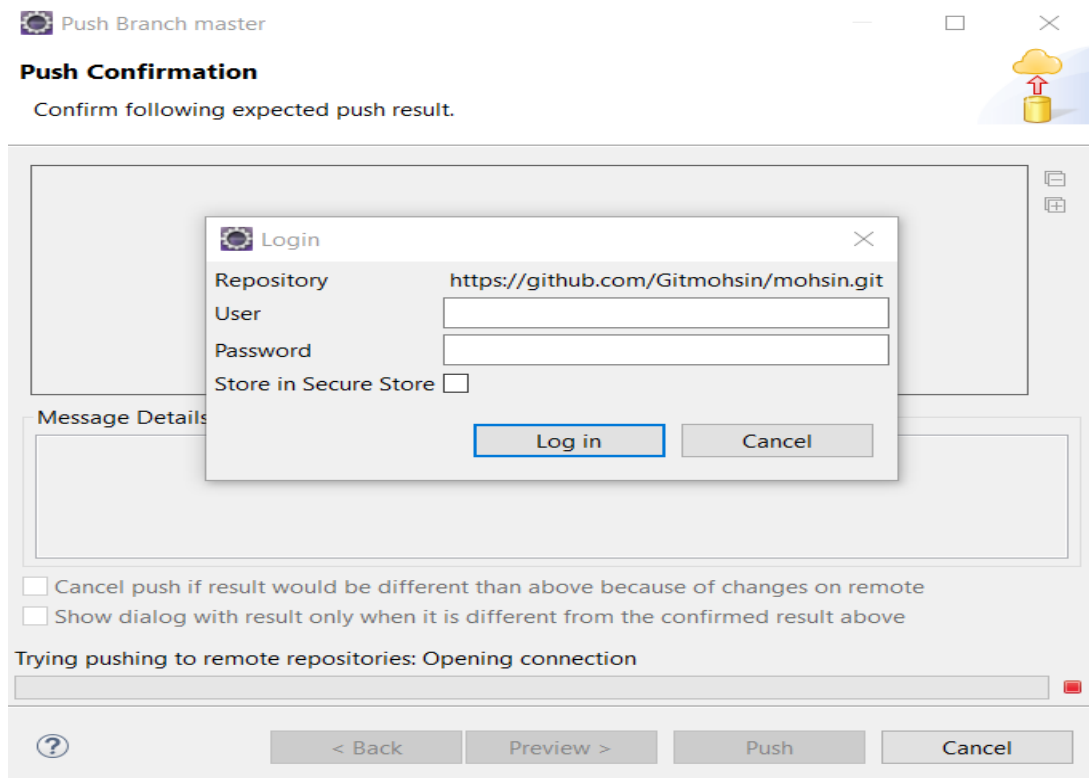


Step-26: After that again click on **Push Head**, it will show a **dialog** like given image, paste your **Repository URL** in **URL** section and type your **GitHub User Id and Password** in User, password section -> Click on **preview** -> Again click on **Preview**.



Step-27: After that it will again show a **user Id** and **Password** option -> just type your **Github id** in user section and paste your **Token id** in Password section -> click on **push** -> one more time it will ask **user id** and

password just repeat your last step with **user id and token id** ->now check your **repository on github**, your file is **uploaded or not**



Step-28: Now you have to create a simple java code in SRC File, so first open your project from **file manager** -> open **SRC** -> Create two folder in **SRC** -> **first name: java, second name: test** -> now open **test folder** and create two more folder in **test folder** -> now come on your **eclipse IDE** and **Update** your project once -> After that create a java class file with a Statement **"Hello World"** in your **SRC/TEST/java** folder.

Step-29: Now Push again your **all unstage files** in your **GitHubRepository** with **different version** or **Comment** (Its just for Version Control).

Step-30: Now Check again your Repository your recent file is uploaded or not with different version.

Output:-

The screenshot displays the Eclipse IDE interface with the console output of a Maven build and the Project Explorer showing the project structure. Below the IDE, a browser window shows the GitHub repository page for 'devisar / Exp2.2'.

Eclipse Console Output:

```
<terminated> C:\eclipse\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.21.0.5.v20241023-1957\jre\bin\javaw.exe (26-Dec-2024, 7:07:36 pm) [pid: 5840]
[INFO] Parameter: groupId, Value: DevOpsExp2
[INFO] Parameter: artifactId, Value: MyDevOpsPipeline
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[INFO] Parameter: package, Value: war
[INFO] Parameter: packageInPathFormat, Value: war
[INFO] Parameter: package, Value: war
[INFO] Parameter: groupId, Value: DevOpsExp2
[INFO] Parameter: artifactId, Value: MyDevOpsPipeline
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[WARNING] CP Don't override file C:\Users\De11\workspace\MyDevOpsPipeline\src\main\webapp
[WARNING] CP Don't override file C:\Users\De11\workspace\MyDevOpsPipeline\src\main\resources
[INFO] Project created from Archetype in dir: C:\Users\De11\workspace\MyDevOpsPipeline
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 01:22 min
[INFO] Finished at: 2024-12-26T19:09:00+05:30
[INFO] -----
```

Eclipse Project Explorer:

- MyDevOpsPipeline
 - Deployment Descriptor: MyDevOpsPipeline
 - JAX-WS Web Services
 - Java Resources
 - Deployed Resources
 - src
 - main
 - webapp
 - WEB-INF
 - index.jsp
 - target
 - pom.xml

Eclipse IDE View:

The IDE shows the 'pom.xml' file with the following content:

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.devops</groupId>
  <artifactId>MyDevOpsPipeline</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <packaging>war</packaging>
  <name>MyDevOpsPipeline</name>
  <properties>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
  </properties>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>4.13.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
  <build>
    <finalName>MyDevOpsPipeline</finalName>
  </build>
</project>
```

GitHub Repository Page:

The browser shows the GitHub repository page for 'devisar / Exp2.2'. The page includes a navigation bar with links to Product, Solutions, Resources, Open Source, Enterprise, and Pricing. The repository is public and has 1 branch and 0 tags. The file list shows the following files:

File	Commit	Last Commit
.mvn	First123 commit	last month
.settings	First123 commit	last month
src/main/webapp	Update index.jsp	last week
.classpath	First123 commit	last month
.gitignore	First123 commit	last month
.project	First123 commit	last month
pom.xml	First123 commit	last month

The right sidebar shows the repository's activity, including 0 stars, 1 watching, 0 forks, and 0 releases.

Experiment-3

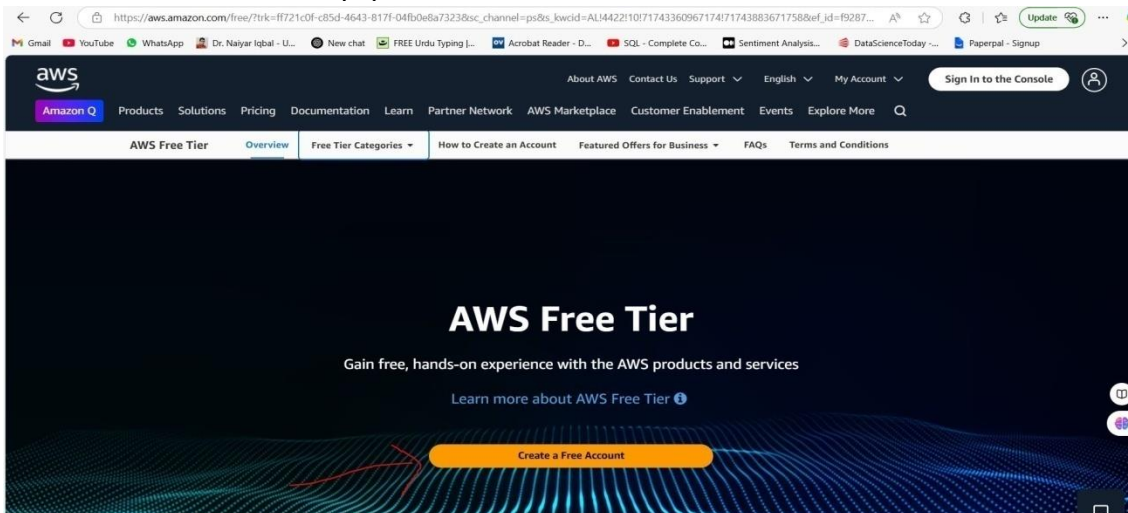
Aim: Jenkins Setup on AWS.

Note: Write the given steps in your lab manual. The provided image is just for your better understanding.

Require Software & Tools: AWS, MobaXterm Software.

Procedure:

Step-1: Search AWS Free Tier Account in any Browser and Create an Account. (if you have already an account then leave the steps)



Sign in

☒ **Root user**
Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**
User within an account that performs daily tasks. [Learn more](#)

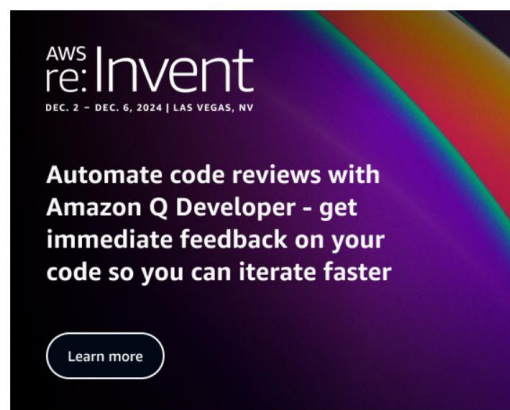
Root user email address

Next

By continuing, you agree to the [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

☐ New to AWS?

Create a new AWS account





Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.

Sign up for AWS

Root user email address
Used for account recovery and some administrative functions

AWS account name
Choose a name for your account. You can change this name in your account settings after you sign up.

Verify email address

OR

Step-2: After Creating the Account, Just Login as a **ROOT User**

Step-3: After Login It will show your account -> Click on **EC2** option -> Click on **Launch Instance**

The screenshot shows the AWS Management Console interface. The top navigation bar includes the AWS logo, a search bar, and the user's profile. The main content area is divided into several sections:

- Console Home:** Includes a "Recently visited" section with links to "Billing and Cost Management", "EC2", and "EC2 Global View". A red arrow points to the "EC2" link.
- Applications (0):** A section for managing applications, currently showing "No applications".
- Resources:** A section showing the user's current resources in the "US East (N. Virginia)" region. It includes a table with columns for "Instances (running)", "Dedicated Hosts", "Key pairs", "Security groups", "Auto Scaling Groups", "Elastic IPs", "Load balancers", "Snapshots", "Capacity Reservations", "Instances", "Placement groups", and "Volumes".
- Launch instance:** A section with a "Launch instance" button and a "Migrate a server" button. A red arrow points to the "Launch instance" button.
- Service health:** A section showing the status of AWS services, currently indicating "This service is operating normally".
- Account attributes:** A section showing account details, including the "Default VPC" and "Settings".
- Explore AWS:** A section with a "10 Things You Can Do Today to Reduce AWS Costs" link.

Step-4: After Clicking on Launch Instance It will ask Instance Name and Other things -> Write Instance Name as **Jenkins** -> Select **Application and IOS Image** as **Ubuntu** -> Scroll Down and Come on **Instance Type** Option and Select **t2.Medium** -> Now Come in **Key Pair(Login)** Section and Click on **Create New Key Pair** -> Write your Key Pair Name as **Exp3** -> Select **RSA** -> Select **.pem**-> Click on **Create Key Pair**.

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or Browse for AMIs if you don't see what you are looking for below

Q Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux aws	macOS Mac	Ubuntu ubuntu	Windows Microsoft	Red Hat Red Hat	SUSE Linux SUSE	Debian debian
---------------------	--------------	------------------	----------------------	--------------------	--------------------	------------------

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-04b4f1a9cf54c11d0 (64-bit (x86)) / ami-0a7a4e87939439934 (64-bit (Arm))
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Jenkins

[Add additional tags](#)

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t2.medium

Family: t2 2 vCPU 4 GiB Memory Current generation: true
On-Demand Ubuntu Pro base pricing: 0.0499 USD per Hour
On-Demand Linux base pricing: 0.0464 USD per Hour On-Demand RHEL base pricing: 0.0752 USD per Hour
On-Demand Windows base pricing: 0.0644 USD per Hour On-Demand SUSE base pricing: 0.1464 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations

[Compare instance types](#)

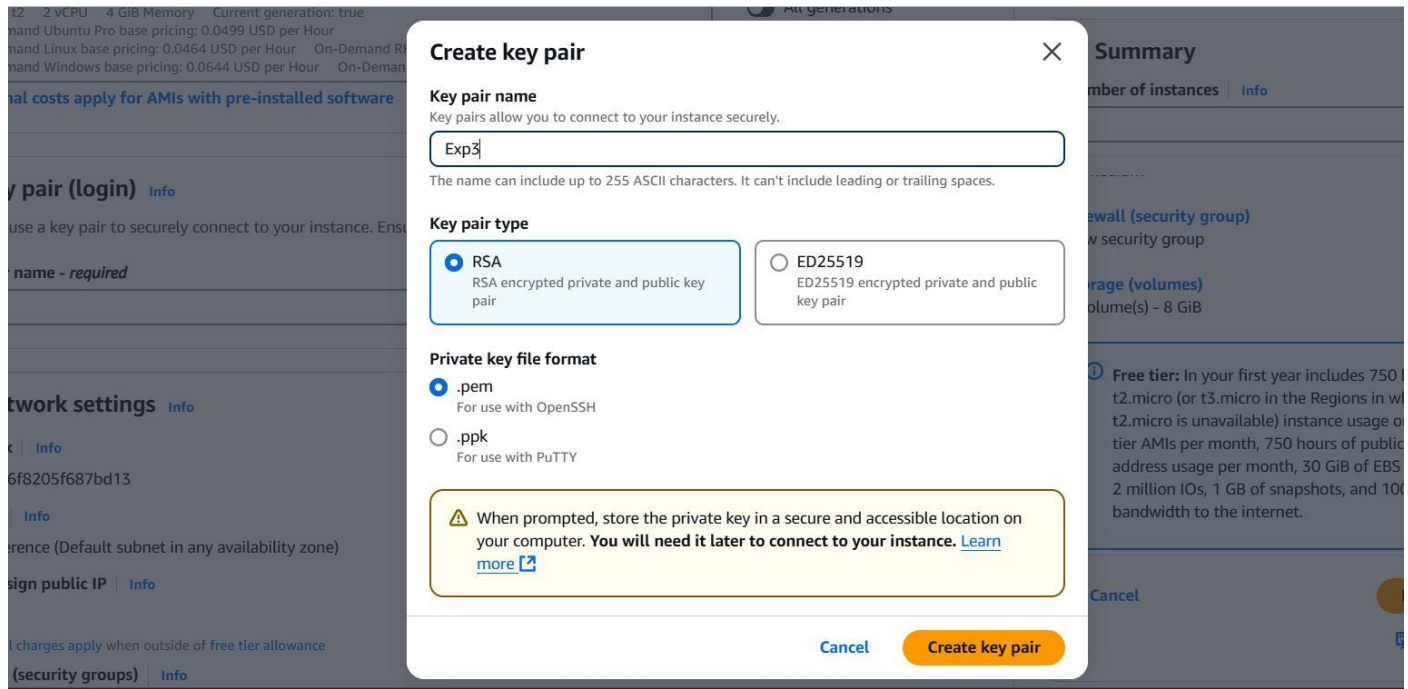
▼ **Key pair (login)** [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

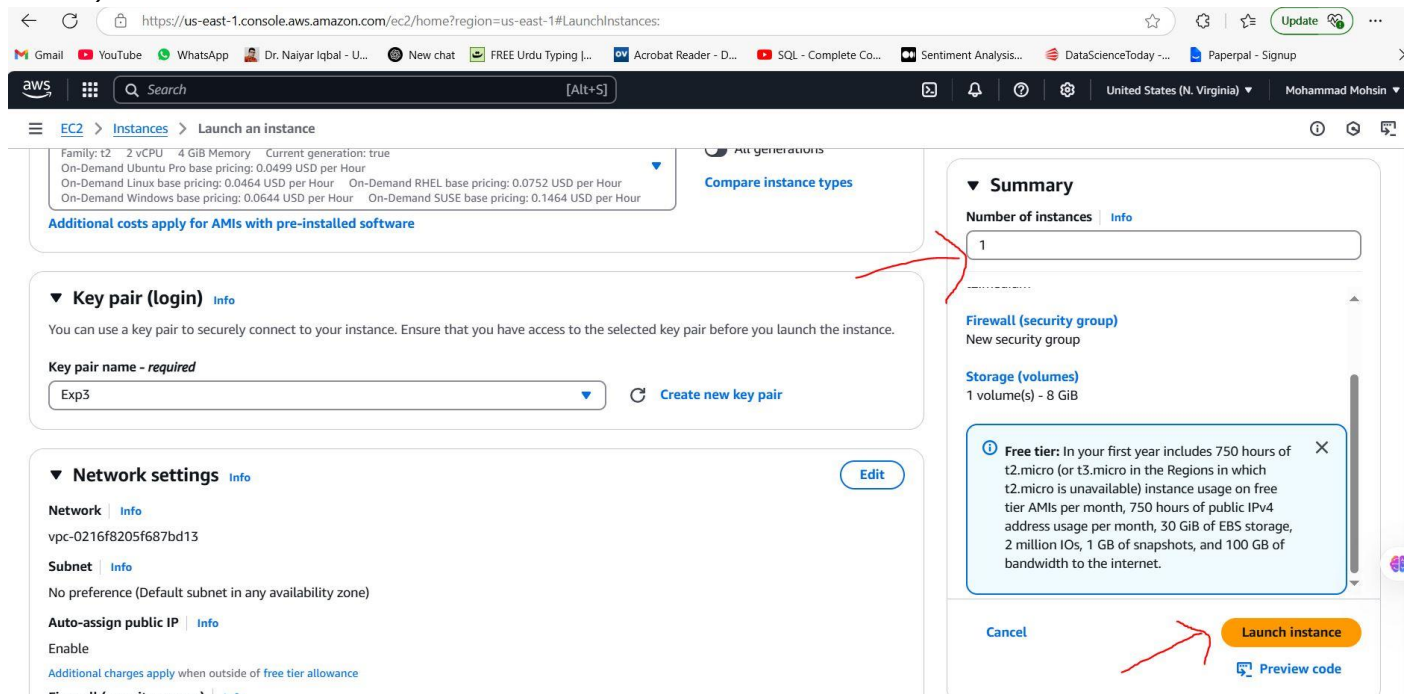
Key pair name - required

Select

[Create new key pair](#)



Step-5: After Click on **Create Key Pair** it will download a **Exp3.pem** file -> After that in Right Side we have a Option **Number of Instances** Select **1** -> Click on **Launch Instances** (It will take some time and will create an instance).



Step-6: After that Refresh the page -> Select the **Created Instance** -> Click on **Security**.

The screenshot shows the AWS Management Console Dashboard for the US East (N. Virginia) region. The left sidebar contains navigation links for Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, and Network & Security. The main content area is divided into several sections:

- Resources:** A table showing the usage of various Amazon EC2 resources in the US East (N. Virginia) Region. A red arrow points to the 'Instances (running)' row.
- Launch instance:** A section with a 'Launch instance' button and a 'Migrate a server' button. A note states: 'Note: Your instances will launch in the US East (N. Virginia) Region.'
- Service health:** A section showing the status of AWS services. The 'AWS Health Dashboard' link is highlighted. The status is 'This service is operating normally.'
- Account attributes:** A section showing account details, including the Default VPC, Settings, and Explore AWS.

Resource	Count
Instances (running)	1
Auto Scaling Groups	0
Capacity Reservations	0
Dedicated Hosts	0
Elastic IPs	0
Instances	1
Key pairs	1
Load balancers	0
Placement groups	0
Security groups	2
Snapshots	0
Volumes	1

The screenshot shows the AWS Management Console Instances page. The left sidebar contains navigation links for Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, and Network & Security. The main content area shows a list of instances. A red arrow points to the 'Jenkins' instance.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
Jenkins	i-09fa49fc85615f9d2	Running	t2.medium	Initializing	View alarms +	us-east-1b	ec2-34-...

The screenshot shows the AWS Management Console Instance details page for the 'Jenkins' instance. The left sidebar contains navigation links for Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, and Network & Security. The main content area shows the instance details, including the instance ID, name, state, type, status check, alarm status, availability zone, and public IP. A red arrow points to the 'Security' tab.

i-09fa49fc85615f9d2 (Jenkins)

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

Instance summary Info

Instance ID	Public IP address	Private IP address
i-09fa49fc85615f9d2	ec2-34-...	10.0.1.10

Step-7: After Clicking on Security -> It Will Show a Blue Link name as [Security Group](#) -> Click on that [blue Link](#) it will Open a Page -> Click On [Edit Inbounds rule](#) -> Click on [Add Rule](#) -> Add one Rule [HTTP](#) and Source type [Anywhere IPv4](#) -> Addone More Rule [Custom TCP](#), port range [8080](#) and Source Anywhere [IPv4](#) -> Click on [Save Rule](#)

EC2 > Security Groups > sg-02a4089f784882382 - launch-wizard-1

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive)

Instance state = running

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
Jenkins	i-09fa49fc85615f9d2	Running	t2.medium	Initializing	View alarms +	us-east-1b	ec2-34-...

i-09fa49fc85615f9d2 (Jenkins)

Security details

IAM Role: -

Owner ID: 102250941657

Launch time: Wed Feb 05 2025 13:45:56 GMT+0530 (India Standard Time)

Security groups

sg-02a4089f784882382 (launch-wizard-1)

Inbound rules

Details

Security group name launch-wizard-1	Security group ID sg-02a4089f784882382	Description launch-wizard-1 created 2025-02-05T07:51:46.636Z	VPC ID vpc-0216f8205f687bd13
Owner 102250941657	Inbound rules count 1 Permission entry	Outbound rules count 1 Permission entry	

Inbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-075fd1ed42717eadd	IPv4	SSH	TCP	22

Manage tags Edit inbound rules

EC2 > Security Groups > sg-02a4089f784882382 - launch-wizard-1 > Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sgr-075fd1ed42717eadd	SSH	TCP	22	Custom 0.0.0.0/0	

Add rule

Step-7: After that click on Dashboard -> click on instance Running -> select Jenkins -> click on connect.

The screenshot shows the AWS Management Console interface. On the left, the navigation menu includes 'EC2', 'Dashboard', 'EC2 Global View', 'Events', 'Instances', 'Instance Types', 'Launch Templates', 'Spot Requests', 'Savings Plans', and 'Reserved Instances'. The main content area displays the 'Resources' section for the Asia Pacific (Mumbai) Region, listing various EC2 resources and their counts. Below this, the 'Instances (1/1)' page is shown, featuring a search bar, filters, and a table of instances. The 'jenkins' instance is selected, and the 'Connect' button is highlighted with a red arrow.

Resource	Count
Instances (running)	1
Auto Scaling Groups	0
Capacity Reservations	0
Dedicated Hosts	0
Elastic IPs	0
Instances	1
Key pairs	1
Load balancers	0
Placement groups	0
Security groups	2
Snapshots	0
Volumes	1

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
jenkins	i-03bbdd67796a22a	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-3-108-217-235.ap-...	3.108.217.235	-

Step-8: After that click on connect it will open a Linux terminal -> just type `clear` command and follow the **setp-9** documents.

```
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:      https://ubuntu.com/pro

System information as of Sat Feb 15 04:20:11 UTC 2025

System load: 0.0      Processes:    101
Usage of /:  21.7% of 7.57GB   Users logged in: 0
Memory usage: 21%      IPv4 address for eth0: 172.31.1.31
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-1-31:~$
```

Step-9: Now You have to Setup Jenkins -> Search [Jenkins Document](#) in Any Browser -> Click on first Jenkins link -> Click on [Installing Jenkins](#) -> click on [Linux](#) -> Click on [Debian/Ubuntu](#) -> After that it will show the [Linux Command](#)

The screenshot shows a Microsoft Bing search for "jenkins Documents". The search results include a link to "Jenkins User Documentation" with a red arrow pointing to it. Below the search results, there is a sidebar with a "User Handbook" section containing a list of links: "User Handbook Overview", "Installing Jenkins", "Platform Information", "Using Jenkins", "Pipeline", "Blue Ocean", "Managing Jenkins", and "Securing Jenkins". A red arrow points to "Installing Jenkins". Below this, there is a "User Handbook" section with a list of links: "User Handbook Overview", "Installing Jenkins", "Docker", "Kubernetes", "Linux", "macOS", "Windows", and "Other Systems". A red arrow points to "Linux".

The main content area shows the "Jenkins User Documentation" page. It has a header with "Jenkins User Documentation" and a sub-header "Welcome to the Jenkins user documentation - for people wanting to use Jenkins's existing functionality and plugin features." Below this, there is a section "What is Jenkins?" which states: "Jenkins is a self-contained, open source automation server which can be used to automate all sorts of tasks related to building, testing and delivering or deploying software." Below this, there is a section "Installing Jenkins" which states: "The procedures in this chapter are for new installations of Jenkins. Jenkins is typically run as a standalone application in its own process. The Jenkins WAR file bundles Winstone, a Jetty servlet container wrapper, and can be started on any operating system or platform with a version of Java supported by Jenkins." Below this, there is a section "Chapter Sub-Sections" with links: "Docker", "Kubernetes", "Linux", "macOS", "Windows", and "Other Systems".

User Handbook

- User Handbook Overview
- Installing Jenkins
 - Docker
 - Kubernetes
 - Linux
 - macOS
 - Windows
 - Other Systems
 - WAR file

User Handbook

- User Handbook Overview
- Installing Jenkins
 - Docker
 - Kubernetes
 - Linux
 - macOS
 - Windows
 - Other Systems
 - WAR file
 - Other Servlet Containers
 - Offline Installations
 - Initial Settings
- Platform Information
- Using Jenkins
 - Pipeline
 - Blue Ocean
 - Managing Jenkins

Linux

Jenkins installers are available for several Linux distributions.

- Debian/Ubuntu
- Fedora
- Red Hat Enterprise Linux and derivatives

Table of Contents

- Prerequisites
- Debian/Ubuntu
 - Long Term Support release
 - Weekly release
 - Installation of Java

On Debian and Debian-based distributions like Ubuntu you can install Jenkins through `apt`.

Long Term Support release

A LTS (Long-Term Support) release is chosen every 12 weeks from the stream of regular releases as the stable release for that time period. It can be installed from the `debian-stable apt repository`.

```
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt-get update
sudo apt-get install jenkins
```

Weekly release

A new release is produced weekly to deliver bug fixes and features to users and plugin developers. It can be installed from the `debian apt repository`.

Step-10: Now Come on your **MobaXtrem Application** -> Type **Clear** and Hit **Enter key** -> it will clear your Command Screen -> Now copy the the **Long Term Support release** code till `/dev/null` -> Paste in your **Linux Terminal** and hit the **Enter Key** -> it will Download the Jenkins Support File

Code:

```
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
echo "deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
```

Step-11: Clear your Screen -> Now Update your Project Using Given Code.

Code:

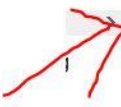
```
sudo apt-get update
```

Step-12: clear your screen -> Now Check Java version Using Command: `java --version`

Step-13: if It Show **Java Not Found** then it show the option **install Java** -> so install java using given Command -> it will ask **Y/N** just type **Y** and hit **Enter** -> It will take some time to download java.

Code:

```
sudo apt install fontconfig openjdk-17-jre
```

```

Reading package lists... Done
ubuntu@ip-172-31-18-41:~$ java --version
Command 'java' not found, but can be installed with:
sudo apt install openjdk-17-jre-headless # version 17.0.12+7-1ubuntu2~24.04, or
sudo apt install openjdk-21-jre-headless # version 21.0.4+7-1ubuntu2~24.04
sudo apt install default-jre # version 2:1.17-75
sudo apt install openjdk-11-jre-headless # version 11.0.24+8-1ubuntu3~24.04.1
sudo apt install openjdk-8-jre-headless # version 8u422-b05-1~24.04
sudo apt install openjdk-19-jre-headless # version 19.0.2+7-4
sudo apt install openjdk-20-jre-headless # version 20.0.2+9-1
sudo apt install openjdk-22-jre-headless # version 22~22ea-1
ubuntu@ip-172-31-18-41:~$

libwayland-client0 libwayland-server0 libwebp7 libx11-xcb1 libxaw7 libxcb-dri2-0 libxcb-dr
libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xf86vm0 libxcomposite1 libxcursor1 libxdamag
libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt6t64 libxtst6 libxv1 libxxf86dga1 libxxf
openjdk-17-jre-headless session-migration ubuntu-mono x11-common x11-utils
The following packages will be upgraded:
libdrm-common libdrm2
2 upgraded, 120 newly installed, 0 to remove and 83 not upgraded.
Need to get 123 MB of archives.
After this operation, 543 MB of additional disk space will be used.
Do you want to continue? [Y/n]

```

Step-14: After Downloading the Java, Clear your Screen using **clear** command -> Once again Update your Project Using **Update Command** (follow Step-14 command) -> now Install Jenkins Using Jenkins Command, i.e given code (It will ask again **Y/N** just type Y and Hit Enter) -> After Installing the Jenkins Once again Update your Project using **Update Command** Just Check Java and Jenkins Version Using (**java -version** and **Jenkins -version** Command)

Code: `Sudo apt-get install jenkins`

Step-15: Now **Enable** the **Jenkins Server** Using Given Command

Command: `Sudo systemctl enable jenkins`

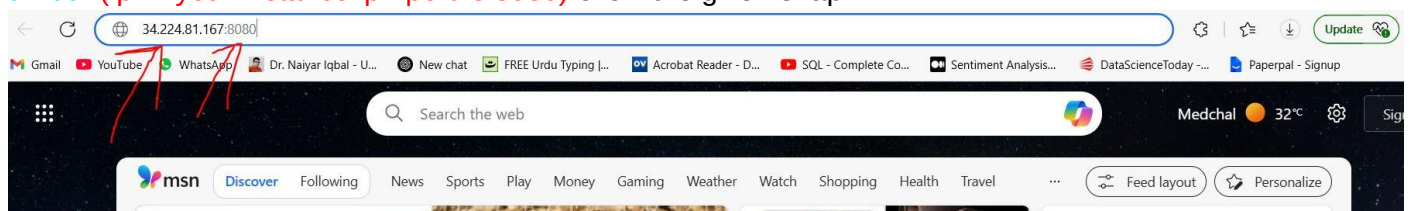
Step-16: After **Enable** the **Jenkins Server** Start **Jenkins Server** Using Given Command

Command: `Sudo systemctl start jenkins`

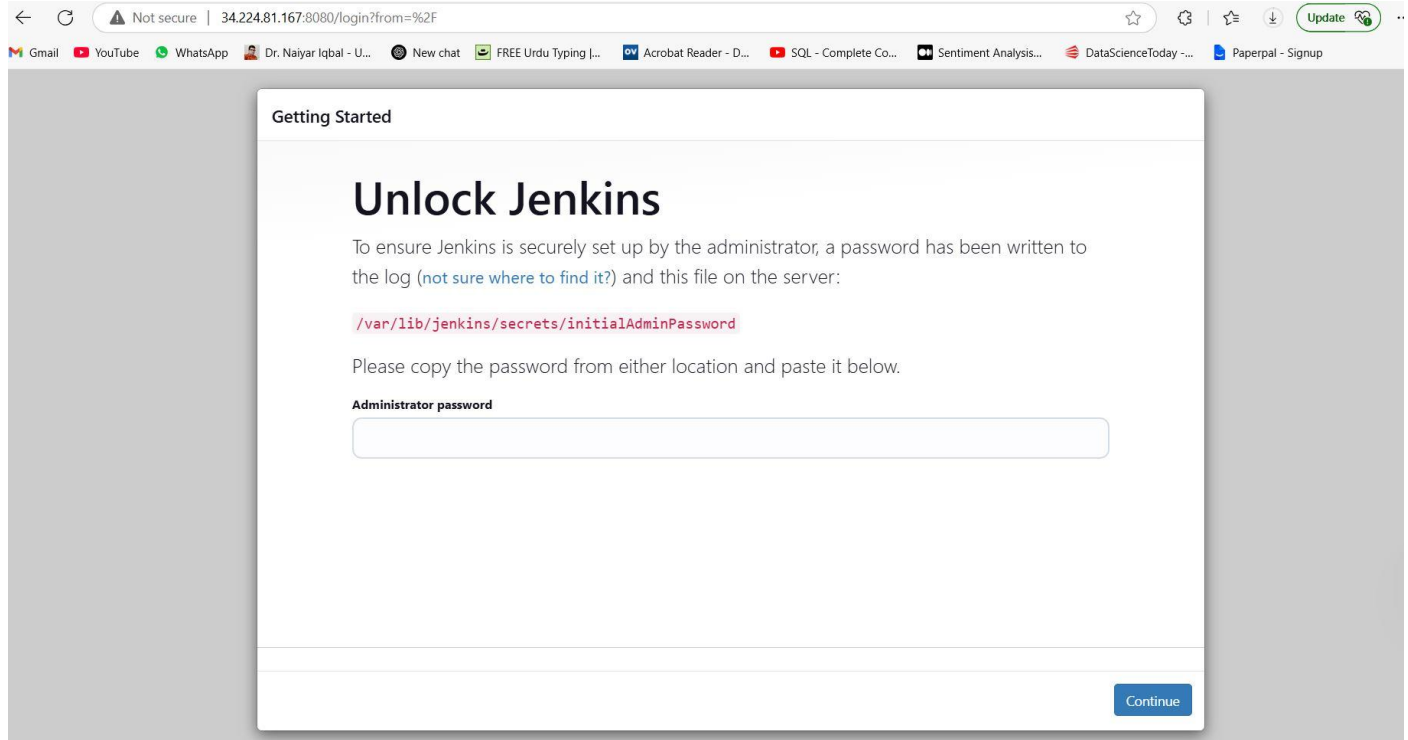
Step-17: After **Starting** the **Jenkins Server** Check **Jenkins Server** Status using Given Command. It will show the Status Active running and show the status of memory, CPU etc.

Command: `Sudo systemctl status jenkins`

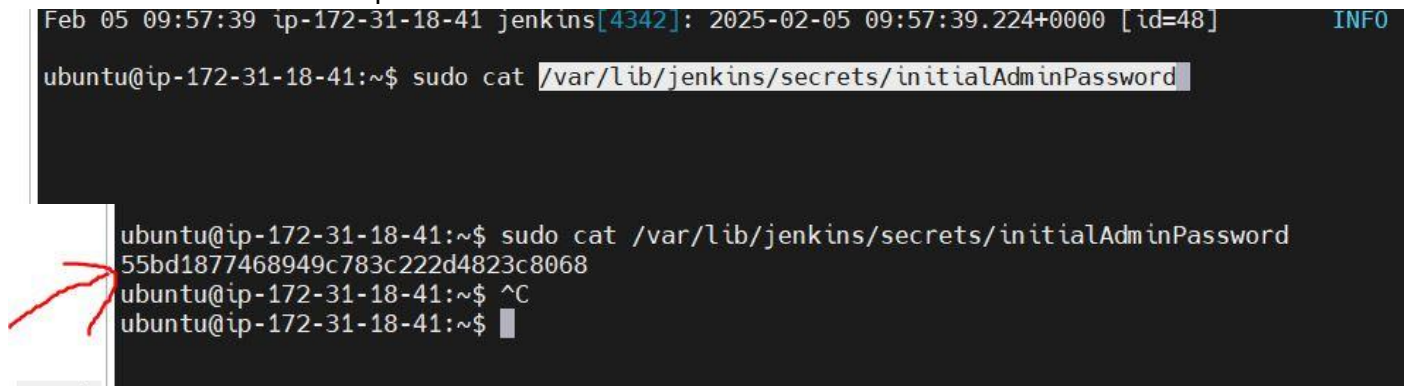
Step-18: Now Open your Any Browser and Search the Jenkins Server using **Instance IP** and **Custom Port Number** (ip= <your instance ip> port is 8080) follow the given snap



Step-19: It will start the **Jenkins Server**, show some path with red color and Ask the Administrator Password like Given Image.



Step-20: Copy the **RED color path** and Paste in your **Linux Terminal** followed by **sudo cat** command and hit the Enter it will show **64-bit Alpha-Numeric code** -> Copy that code and paste in the place of **password** in your browser -> click on continue option.



Step-21: After clicking on **continue** option it will show **two option** for **installing the plugging**-> Just click on **installed suggested plugin** -> it will take some time for install all plugins -> After installing all plugins, it will ask Creating an account using **user ID** and **password** so create an account using user id and password (fill all related field). -> Click on **Save and Continue**

Getting Started

Create First Admin User

Username

Password

Confirm password

Full name

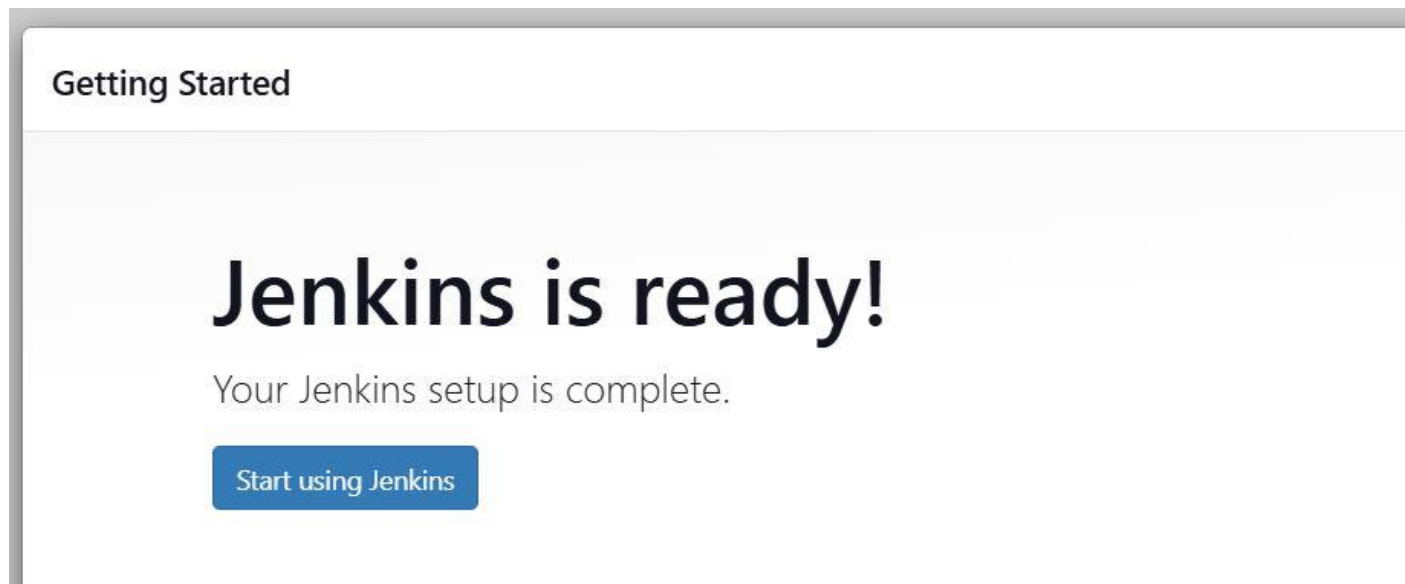
E-mail address

Jenkins 2.479.3

[Skip and continue as admin](#)[Save and Continue](#)

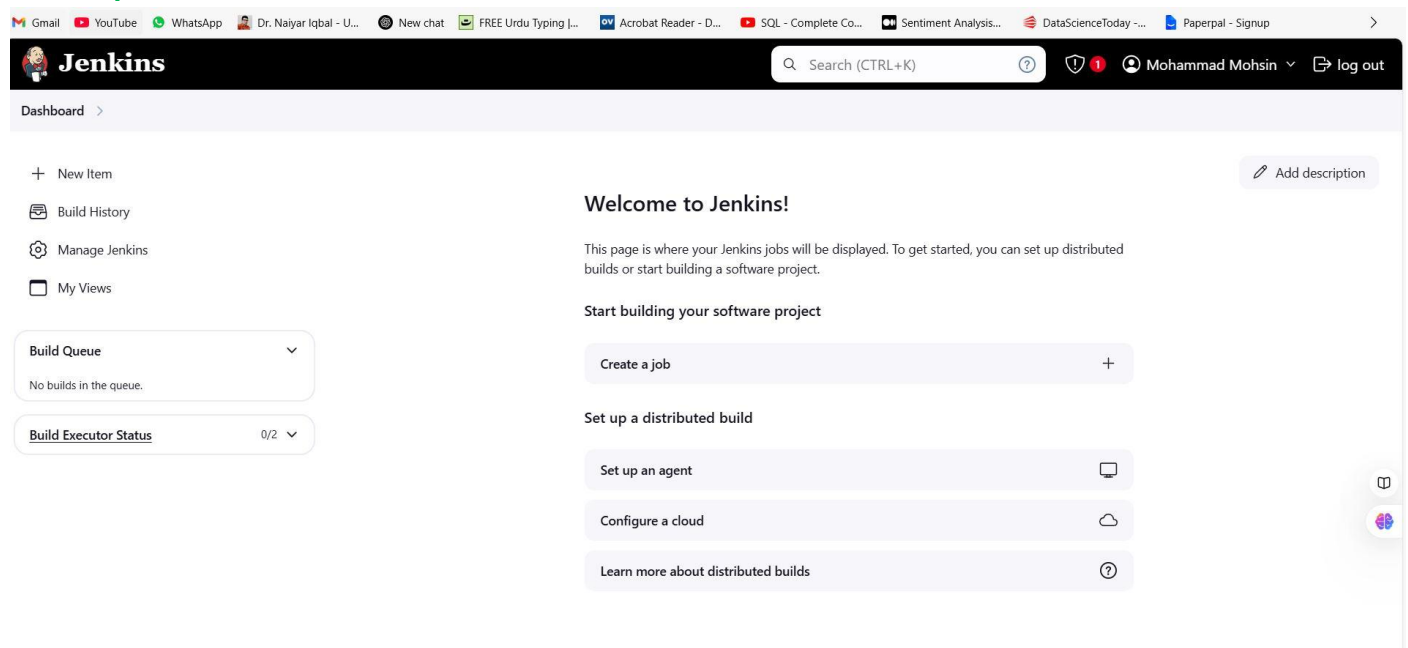
Step-22: After click on [Save and continue](#) -> it show some links and show the option [Save & Finish](#) -> Click on [Save & Finish](#) -> After that it show Jenkins in Ready and show a option [Start using Jenkins](#) -> just click on it -> It will open a Jenkins Dashboard.

Output:-



OUTPUT

Your Output is the Jenkins Dashboard



Experiment-4

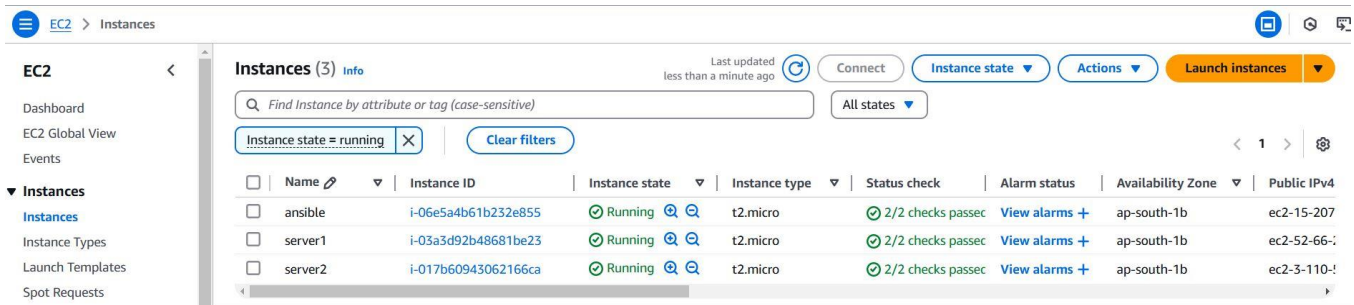
Aim: Ansible Setup and SSH key

(Note: All Blue Words is a command, just copy and paste)

Require Software & Tools: AWS Account, Puttygen software, Wincp Software.

Procedure:

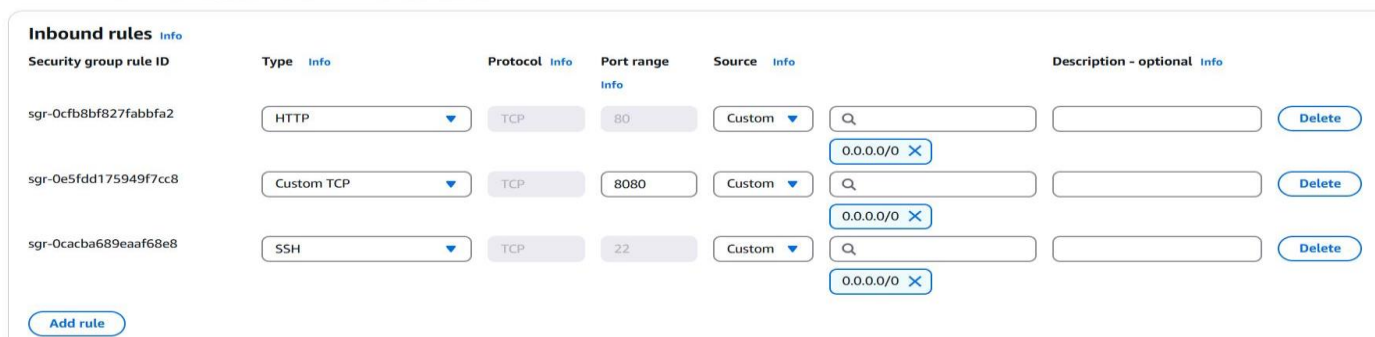
Step-1: Login AWS and Create three Instances on AWS account using ubuntu OS and save key pair also.



Step-2: change the 2nd and 3rd instance name as a server1 and server2 (you can named anything) -> Edit inbound rules of 1st instance (ansible). Edit inbound rules like given details.

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.



Step-3: Now come on instance and select 1st instance (ansible) and click on connect. It will open a linux terminal -> type **Clear** command.

```

Usage of /: 24.9% of 6.71GB  Users logged in: 0
Memory usage: 20%          IPv4 address for enX0: 172.31.13.107
Swap usage: 0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-13-107:~$

```

i-06e5a4b61b232e855 (ansible)

PublicIPs: 15.207.222.240 PrivateIPs: 172.31.13.107

Step-4: type command `sudo apt update` and hit enter.

Step-5: type command `sudo apt install ansible` and hit enter -> it will ask yes/no type yes and hit enter.

Step-6: type command `sudo apt install openssh-client` and hit enter.

Step-7: type command `sudo apt install openssh-server` and hit enter.

Step-8: type command `cd ~/.ssh/` and hit enter.

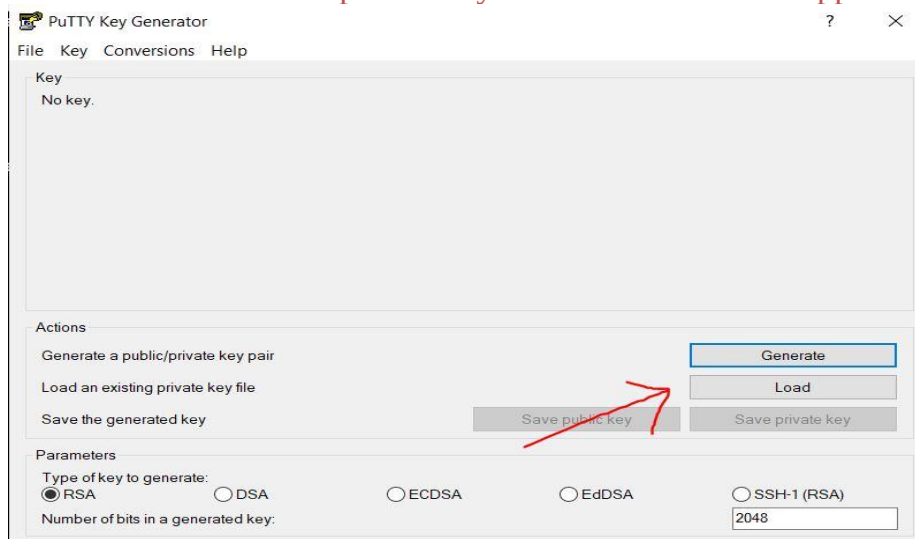
Step-9: type command `ls` and hit enter.

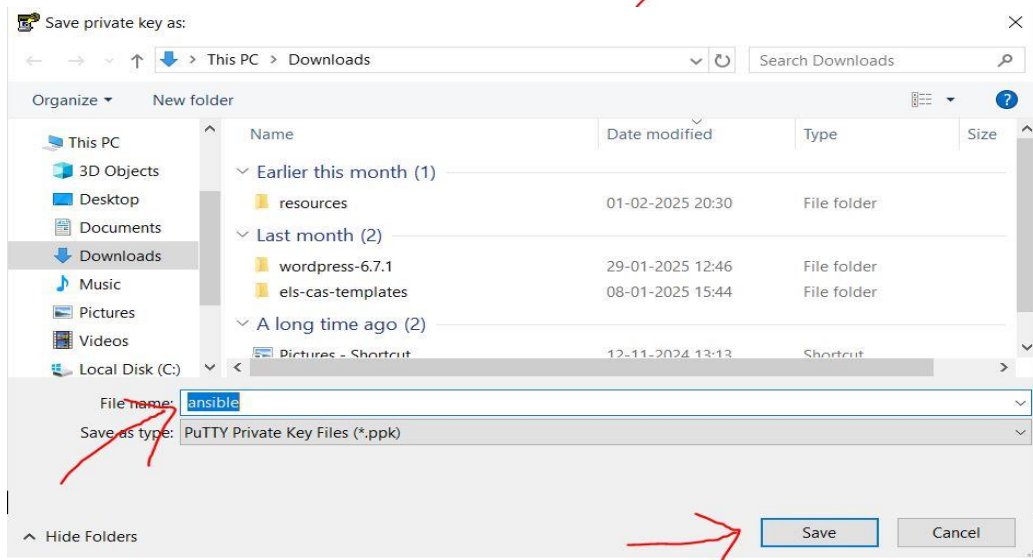
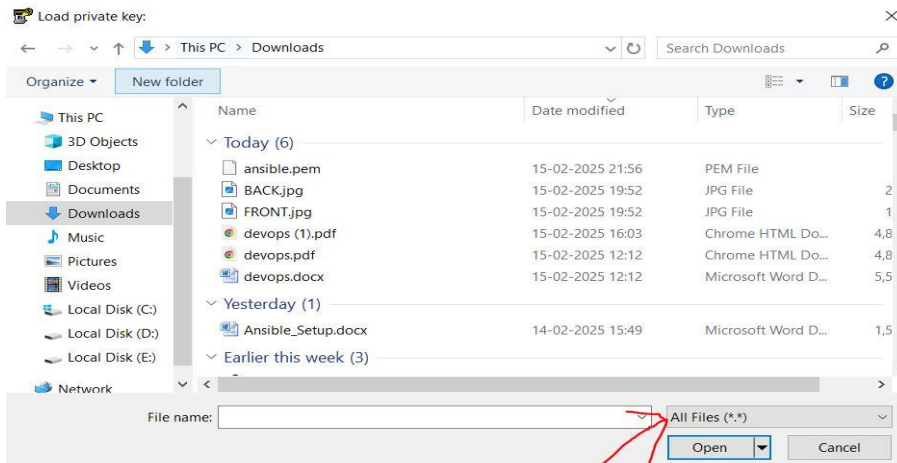
Step-10: type command `cd` and hit enter.

Step-11: type command `cd -` and hit enter. (it will show ssh path)

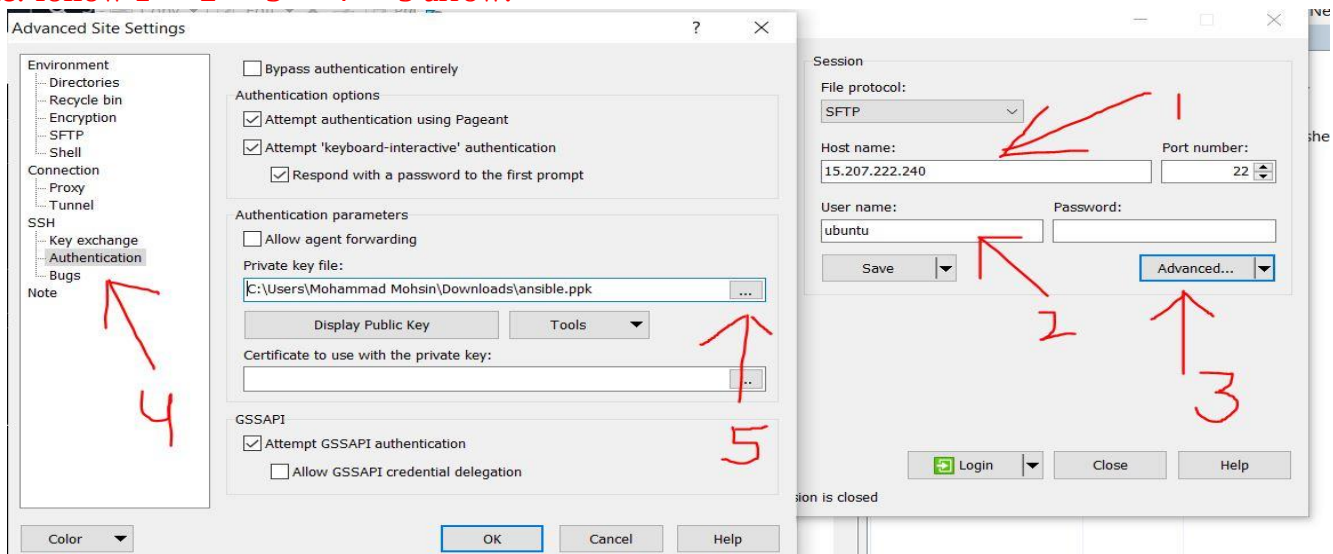
Step-12: now again type command `cd` and hit enter.

Step-13: open **puttygen software** -> click on **load** -> select as **all files** -> select **ansible.pem** (your key pair) -> click on **save** and -> click on **save private key** and save the file with **.ppk extension**.





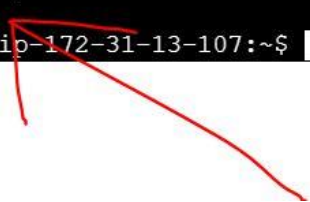
Step-14: Now open **winCP software** -> it will ask for **host name** just copy and paste your **ansibleinstance Ip**-> In the place of **user name** type **Ubuntu** -> click on **advance** -> select **authentication** -> in the place of **private key** select your downloaded **.ppk file** - > click on **Ok** -> click on **Login**. (it will connect to your server and show a **Ubuntu wincp window**)
Note: follow 1 -> 2 -> 3 -> 4 -> 5 arrow.



Step-15: after connecting to your server it will show **two directory** (one is your local disk, another is ssh directory) -> right click in **ssh directory** -> click on **new** -> select **directory** -> create a **folder** name as **key** -> now select your **.pemfile** from left side local disk directory and **drag in your key folder**.

Step-16: Now come on in your terminal -> type **cd** command and hit enter -> type **ls** command hit enter, you will see key.

```
ubuntu@ip-172-31-13-107:~/.ssh$ cd
ubuntu@ip-172-31-13-107:~$ ls
ubuntu@ip-172-31-13-107:~$ cd
ubuntu@ip-172-31-13-107:~$ ls
key
ubuntu@ip-172-31-13-107:~$
```



Step-17: Now type **cd key/** and hit enter, you will see ansible.pem

Step-18: Now type **cp ansible.pem ~/.ssh/** command and hit enter -> after that type **cd** hit enter -> after that **ls ~/.ssh/** and hit enter, you will see ansible.pem and authorized_key

Step-19: Now create one more directory using **mkdir ansible-script** command and hit enter -> select directory using **cd ansible-script/** command and hit enter.

Step-20: Now write **vi inventory.ini** command and hit enter -> it will open your Vim Editor -> copy the given command and paste in your Vim editor -> after that in the place of [server1-ip] copy and paste server1 instance public ip and in the place of [server2-ip] copy and paste server2 instance public ip -> after that save this vim editor using **:wq!** Command and hit enter

Command:

```
[webservers]
server1 ansible_host=[server1-ip] ansible_user=ubuntu ansible_ssh_private_key_file=
/home/ubuntu/.ssh/ansible.pem
server2 ansible_host=[server2-ip] ansible_user=ubuntu ansible_ssh_private_key_file=
/home/ubuntu/.ssh/ansible.pem
```

Step-21: Now write the command **vi playbook.yml** and create a yml file -> it will again open your Vim editor -> copy and paste the given command in your vim editor after save and exit using command **:wq!** and hit enter.

Command:

```
-- name: Install Nginx on AWS EC2
hosts: webservers
become: yes
tasks:
  - name: Install Nginx
    apt:
      name: nginx
      state: present
      when: ansible_os_family == "Debian"

  - name: start and enable nginx server
    systemd:
      name: nginx
      state: started
```


Step-22: Now you can check your .ini file and .yml file using `cat inventory.ini` command and `cat playbook.yml` command.

Step-23: Now type `ansible-playbook -i playbook.yml` command and hit enter.

Step-24: Now type `ansible-playbook -i inventory.ini playbook.yml` command and hit enter -> it will ask for type `yes` and again `yes` and again `yes` or may be it will show server Unreachable.

Step-25: Now type `cd` and hit enter -> after that `cd ~/.ssh/` hit enter, you will see `ansible.pem` authorized_key and known_hosts.

Step-26: Now again run the command `ansible-playbook -i inventory.ini playbook.yml`. it will run your server or may be it will show again an error server unreachable.

Step-27: If server is reachable is then check your server using server ip otherwise -> write command `chmod 400 ~/.ssh/ansible.pem` hit enter -> after that `ls -l ~/.ssh/` and hit enter, it will show Ubuntu file.

Step-28: Now again run the command `ansible-playbook -i inventory.ini playbook.yml`. now it will run your server properly. After that you can check your server in any browser using server ip.

OUTPUT:-

```
ubuntu@ip-172-31-4-180:~$ sudo apt-get install -y ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

```

ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml --syntax-check
playbook: playbook1.yml
ubuntu@ip-172-31-4-180:~$ vim playbook1.yml
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml --syntax-check
playbook: playbook1.yml
ubuntu@ip-172-31-4-180:~$ ansible-playbook playbook1.yml -b

PLAY [Configure tomcat9] *****
TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 172.31.7.139 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.7.139]
[WARNING]: Platform linux on host 172.31.5.200 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter
could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [172.31.5.200]
TASK [Install tomcat9] *****
ok: [172.31.7.139]
ok: [172.31.5.200]
TASK [Change port no of tomcat 8080 to 9090] *****
ok: [172.31.7.139]
ok: [172.31.5.200]
PLAY RECAP *****
172.31.5.200 : ok=3  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0
172.31.7.139 : ok=3  changed=0  unreachable=0  failed=0  skipped=0  rescued=0  ignored=0

```

⚠ Not secure 43.204.218.193

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

⚠ Not secure 13.233.103.243

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working. Further configuration is required.

For online documentation and support please refer to nginx.org.
Commercial support is available at nginx.com.

Thank you for using nginx.

Experiment-5

(Note: All Blue Words is a command, just copy and paste)

Aim: Build War file in Devops

Require Software & Tools: Jenkins, Jdk-17 or jdk-21, Git bash.

Procedure:

Step-1: First open any browser and Search **download Jenkins** -> click on **first link** -> scroll down and click on **windows logo** -> **download will be started.**

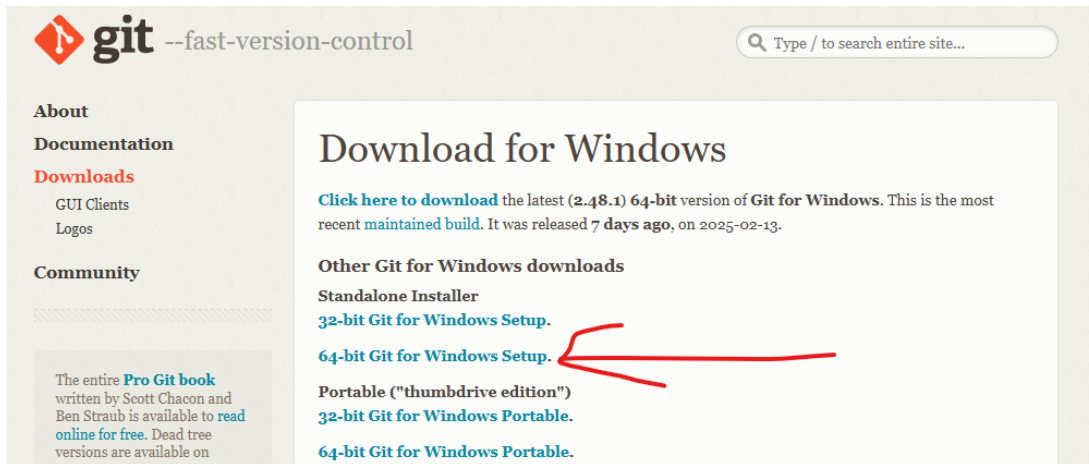
The screenshot shows a Bing search results page for 'download jenkins'. The top result is the Jenkins project page, which includes buttons for 'Overview', 'Download', and 'Documentation'. Below this, there's a 'Copilot Answer' section. The main result is titled 'Download and deploy - Jenkins' with a link to 'https://www.jenkins.io/download'. A sidebar on the right shows 'Jenkins - Wikipedia'. The main content area lists various operating systems for which Jenkins can be downloaded: Docker, Kubernetes, Ubuntu/Debian, Red Hat Enterprise Linux and derivatives, Fedora, Windows, and openSUSE. A red arrow points to the 'Windows' option in the list.

Step2: First check java jdk in your system, if java is available check the java version using cmd. otherwise download and install jdk-17.

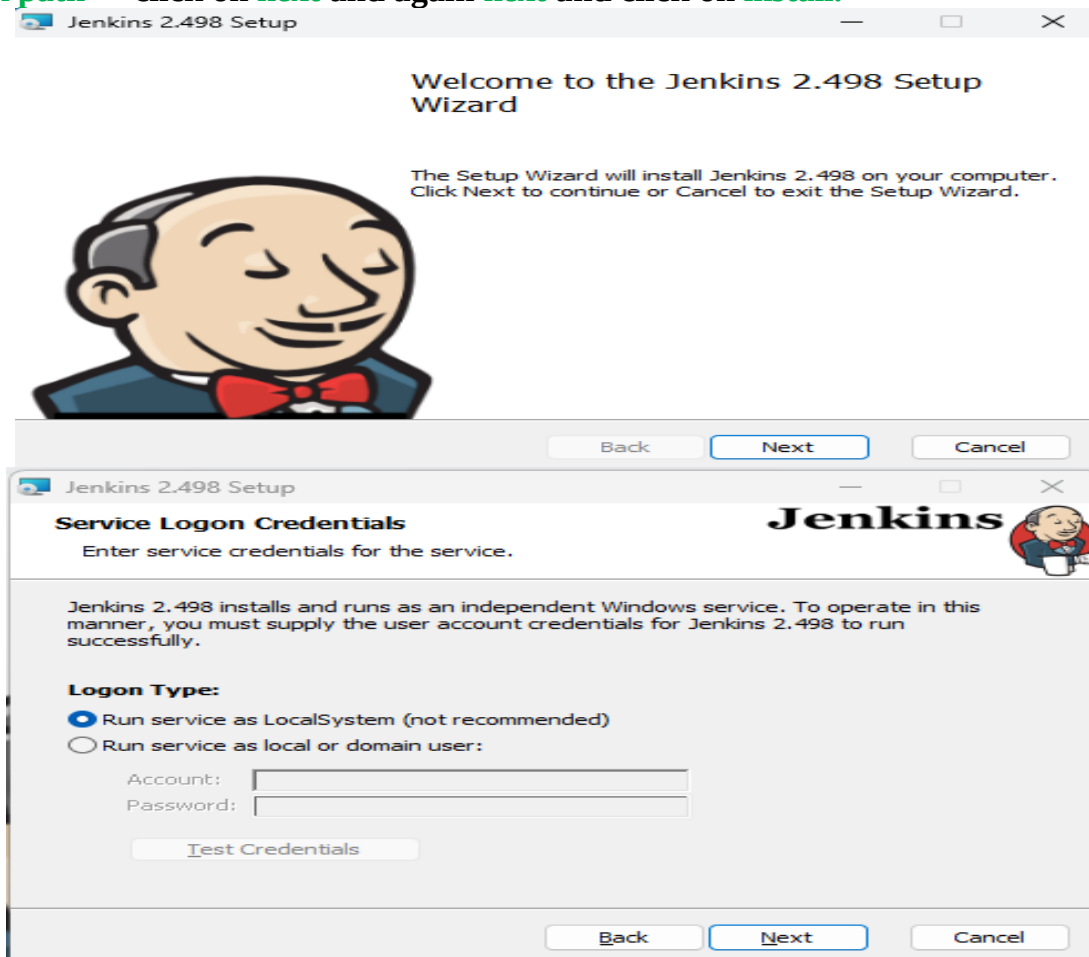
```
C:\Users\student>java --version
java 17.0.12 2024-07-16 LTS
Java(TM) SE Runtime Environment (build 17.0.12+8-LTS-286)
Java HotSpot(TM) 64-Bit Server VM (build 17.0.12+8-LTS-286, mixed mode, sharing)

C:\Users\student>
```

Step-3: Download Git Bash 64-bit Standalone Installer, and install git bash in your system. After installation set the **git.exe** path in your **runtime environment**.



Step-4: Now install Jenkins in your system -> click on **next and again **next** -> select **run service as a local system** -> click on **next** and click on **Test port** -> click on **next** and click on **change** and set the **java jdk path** -> click on **next** and again **next** and click on **install**.**

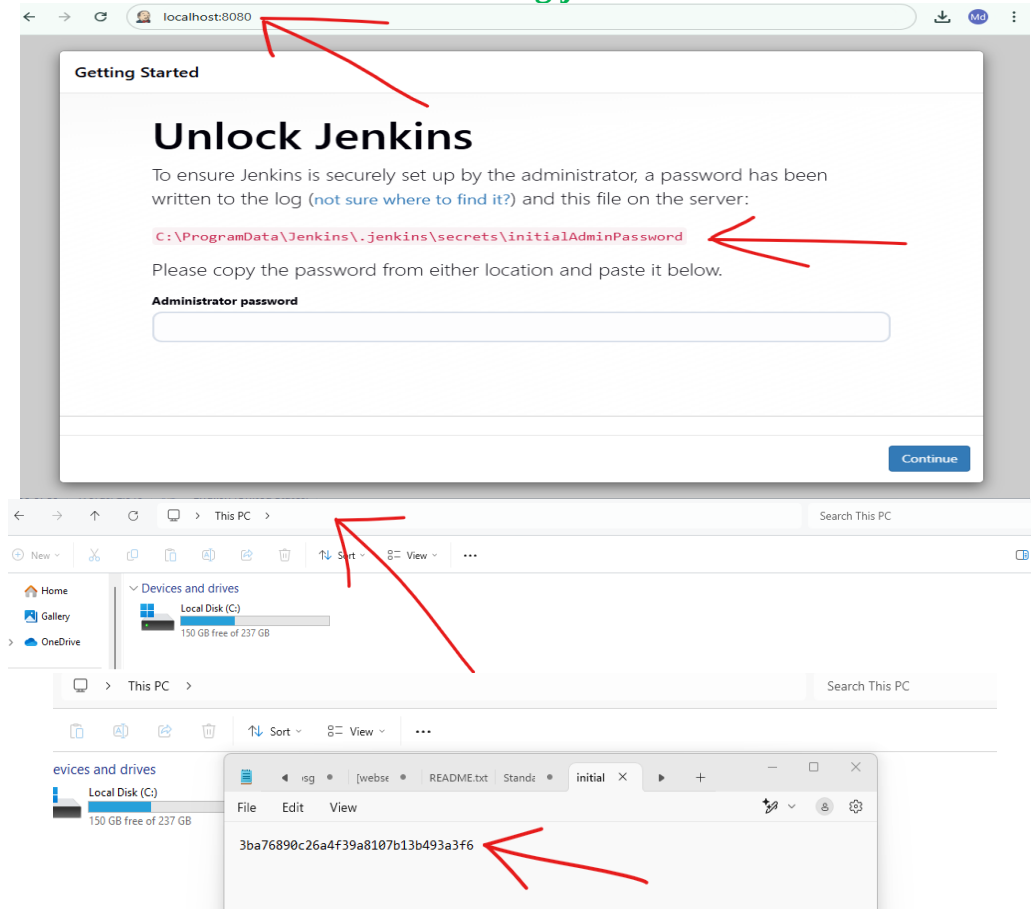


Step-5: After that install Git Bash in your system (no need to change anything just click on next next and next till last).

Step-6: Now open a browser -> search `localhost:8080`

Step-7: It will open your **Jenkins server -> Copy **red line location** and search it to in your **File manager**.**

Step-7: It will show **open with option just select **notepad** -> after that **copy the password** and paste in **Jenkins server** -> click on **continue** -> click on **suggested plugin** -> it will take some time for installation -> create **admin user** using given details and click on **save and continue** -> after that click on **save and finish** -> Click on **start using jenkins** It will show **Jenkins Dashboard**.**



Getting Started

Getting Started

✓ Folders	✓ OWASP Markup Formatter	⌚ Build Timeout	🔑 Credentials Binding	** Ionicons API
⌚ Timestamp	🔄 Workspace Cleanup	⌚ Ant	🔄 Gradle	Folders
🔄 Pipeline	🔄 GitHub Branch Source	🔄 Pipeline: GitHub Groovy Libraries	🔄 Pipeline Graph View	OWASP Markup Formatter
⌚ Git	🔄 SSH Build Agents	🔄 Matrix Authorization Strategy	⌚ PAM Authentication	** ASM API
⌚ LDAP	🔄 Email Extension	⌚ Mailer	⌚ Dark Theme	** JSON Path API
				** Struts
				** Pipeline: Step API

** - required dependency

Jenkins 2.498

Step-8: After that click on **manage Jenkins -> click on **plugin** -> click on **available plugin** -> search **maven** -> select **maven integration** -> click on **install** -> scroll down and check **maven integration** and **loading plugin** is green and success or not -> After success click on **go back to the top page****

Dashboard > Manage Jenkins

Manage Jenkins

Building on the built-in node can be a security issue. You should set up distributed builds. See the [documentation](#). [Set up agent](#) [Set up cloud](#) [Dismiss](#)

+ New Item

📅 Build History

⚙️ **Manage Jenkins**

📁 My Views

Build Queue

No builds in the queue.

Build Executor Status 0/2

⚙️ System

Configure global settings and paths.

🔧 Tools

Configure tools, their locations and automatic installers.

⚙️ **Plugins**

Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

💻 Nodes

Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

☁️ Clouds

Add, remove, and configure cloud instances to provision agents on-demand.

🔧 Appearance

Configure the look and feel of Jenkins

Dashboard > Manage Jenkins > Plugins

Plugins

Search: Install Refresh

Available plugins

Install	Name	Released
☒	Maven Integration 3.25 Build Tools This plugin provides a deep integration between Jenkins and Maven. It adds support for automatic triggers between projects depending on SNAPSHOTS as well as the automated configuration of various Jenkins publishers such as Junit.	13 days ago
☒	Config File Provider 982.vb_a_e458a_37021 Groovy-related External Site/Tool Integrations Maven Ability to provide configuration files (e.g. settings.xml for maven, XML groovy, custom files...) loaded through the UI which will be copied to the job workspace.	26 days ago
☐	Jira 3.13 External Site/Tool Integrations Maven jira This plugin integrates Jenkins to Atlassian Jira.	11 mo ago

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

PAM Authentication	Success
LDAP	Success
Email Extension	Success
Mailer	Success
Theme Manager	Success
Dark Theme	Success
Loading plugin extensions	Success
Javadoc	Success
Dev Tools Symbols API	Success
JSch dependency	Success
Maven Integration	Success
Loading plugin extensions	Success

[Go back to the top page](#)
 (you can start using the installed plugins right away)

☐ Restart Jenkins when installation is complete and no jobs are running

Step-9: Now again click on **Manage Jenkins** -> click on **tools** -> scroll down and check **Git path** -> after that scroll down and in a **Maven installation section** -> click on **add maven** -> write maven name and click on **apply** -> click on **save**.

The screenshot shows the Jenkins 'Manage Jenkins' page. On the left sidebar, 'Manage Jenkins' is selected. The main area has a 'System Configuration' section with icons for 'System', 'Tools', 'Nodes', 'Clouds', 'Plugins', and 'Appearance'. Red arrows point to the 'Tools' icon and the 'Path to Git executable' field. Below this, the 'Git' installation configuration is shown with 'Name' set to 'Default' and 'Path to Git executable' set to 'git.exe'. The 'Install automatically' checkbox is checked. Below the Git configuration, the 'Maven installations' section is visible, showing 'Add Maven' and 'Install from Apache' with 'Version' set to '3.9.9'. At the bottom, there are 'Save' and 'Apply' buttons.

Step-10: now click on **dashboard** -> click on **create a job** -> type your **buildwar name** (i.e. myWar) -> select **Maven Project** -> click on **OK**

The screenshot shows the Jenkins Dashboard. The top header includes the Jenkins logo, a search icon, a shield icon with a red exclamation mark, the user name 'Mohsin', and a 'log out' link. The left sidebar has a 'Dashboard' link and a list of items: 'New Item', 'Build History', 'Project Relationship', 'Check File Fingerprint', 'Manage Jenkins', and 'My Views'. The main area displays a 'Welcome to Jenkins!' message with instructions on how to get started. Below the message, there is a 'Start building your software project' section with a 'Create a job' button. A red arrow points to the 'Create a job' button.

Dashboard > New Item

New Item

Enter an item name

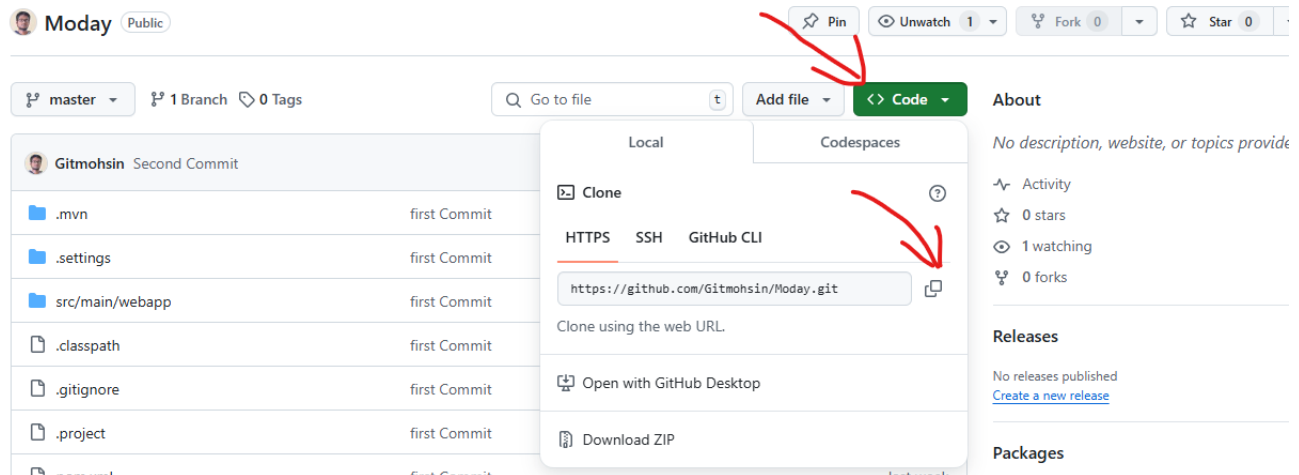
buildwar

Select an item type

-  **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
-  **Maven project**
Build a maven project. Jenkins takes advantage of your POM files and drastically reduces the configuration.
-  **Pipeline**
Orchestrates long-running activities that can span multiple build agents. Suitable for building pipelines (formerly known as workflows) and/or organizing complex activities that do not easily fit in free-style job type.

OK

Step-10: Now take a new page -> login your **GitHub -> check second experiment repository like **Devops_pipeline** -> copy the git link of **second experiment**.**



Step-11: scroll down and go to **Source code management section -> select **git** -> paste **git link** -> scroll down go to **build option** and type **clean install package** in **goals** option -> click on **apply** -> click on **save**.**

Configure

Source Code Management

Connect and manage your code repository to automatically pull the latest code for your builds.

☐ None

☒ **Git** ?

Repositories ?

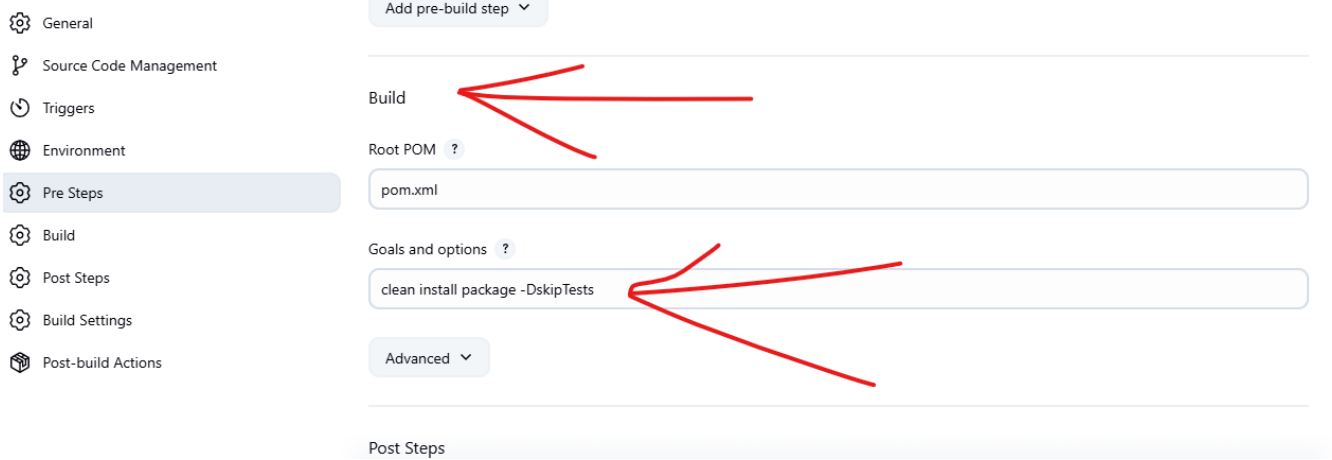
Repository URL ?

https://github.com/Gitmohsin/Moday.git

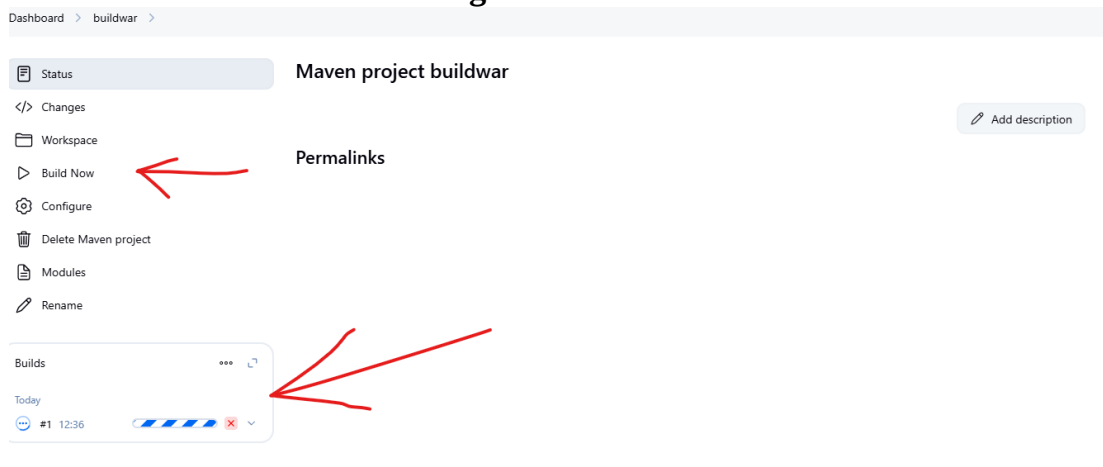
Credentials ?

- none -

+ Add



Step-12: After save the file -> it will show an option in left side **build now -> click on **build now** -> It will take two to five minute for creating build file.**



Step-12: After building it will show success with **green mark -> click on **that section** -> it will open new section -> click on **console output** -> scroll down and check **status**, **build success** or not.**

Dashboard > buildwar >

Status

Maven project buildwar

</> Changes

Workspace

Build Now

Configure

Delete Maven project

Modules

Rename

Ac

Permalinks

- Last build (#1), 6 min 42 sec ago
- Last stable build (#1), 6 min 42 sec ago
- Last successful build (#1), 6 min 42 sec ago
- Last completed build (#1), 6 min 42 sec ago

Builds

Filter

Today

✓ #1 12:36

Status



#1 (20 Feb 2025, 12:36:30)

</> Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

Git Build Data

Redeploy Artifacts

See Fingerprints

Started by user [Mohsin](#)

This run spent:

- 5 ms waiting;
- 3.8 sec build duration;
- 3.8 sec total from scheduled to completion.

**Revision:** d530f6158be5ef19c6fc8070bfd34a12ce3fec53**Repository:** <https://github.com/Gitmohsin/Moday.git>

- refs/remotes/origin/master

Module Builds

Output:-

Dashboard > warjar >

Maven project warjar

Status

Changes

Workspace

Build Now

Configure

Delete Maven project

Modules

Rename

Add description

Permalinks

Builds

Today

#1 1:06 AM

```

utils/4.0.1/plexus-utils-4.0.1.jar (193 kB at 10 MB/s)
[INFO] Installing /home/ubuntu/.jenkins/workspace/warjar/pom.xml to
/home/ubuntu/.m2/repository/DevOpsCourse/MyDevPipeline/0.0.1-SNAPSHOT/MyDevPipeline-0.0.1-SNAPSHOT.pom
[INFO] Installing /home/ubuntu/.jenkins/workspace/warjar/target/MyDevPipeline.war to
/home/ubuntu/.m2/repository/DevOpsCourse/MyDevPipeline/0.0.1-SNAPSHOT/MyDevPipeline-0.0.1-SNAPSHOT.war
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 13.732 s
[INFO] Finished at: 2025-02-15T01:07:19Z
[INFO] -----
Waiting for Jenkins to finish collecting data
[JENKINS] Archiving /home/ubuntu/.jenkins/workspace/warjar/pom.xml to DevOpsCourse/MyDevPipeline/0.0.1-SNAPSHOT/MyDevPipeline-0.0.1-SNAPSHOT.pom
[JENKINS] Archiving /home/ubuntu/.jenkins/workspace/warjar/target/MyDevPipeline.war to
DevOpsCourse/MyDevPipeline/0.0.1-SNAPSHOT/MyDevPipeline-0.0.1-SNAPSHOT.war
channel stopped
Finished: SUCCESS
  
```

Jenkins

admin log out

Dashboard >

+ New Item

Build History

Project Relationship

Check File Fingerprint

Manage Jenkins

My Views

Add description

All +

S	W	Name 1	Last Success	Last Failure	Last Duration
✓	☀	warjar	8 min 40 sec #1	N/A	28 sec

Experiment-6

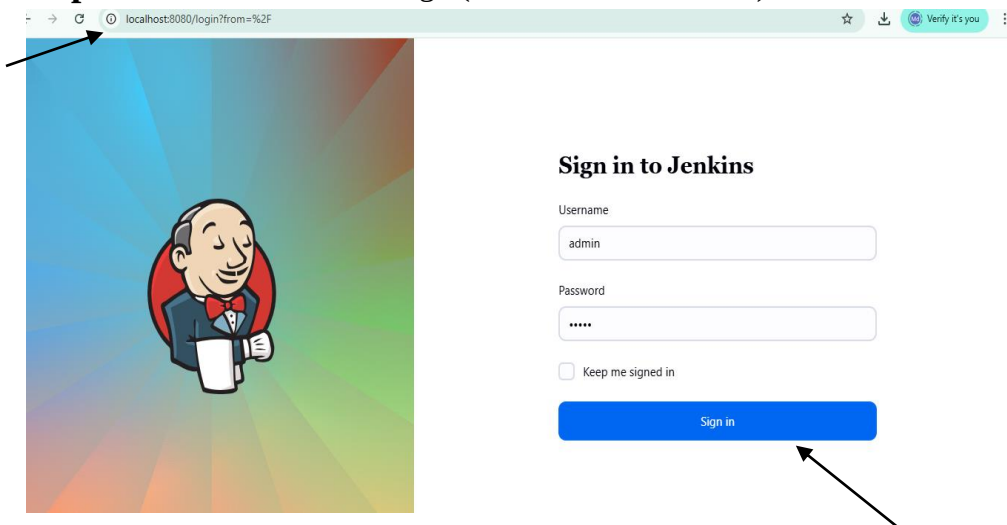
AIM:-Deploy the artifact on the Test Server

Softwares:Jenkins(port:8080),JDK 17 or 21,Tomcat 9 (port:9090),Gitbash

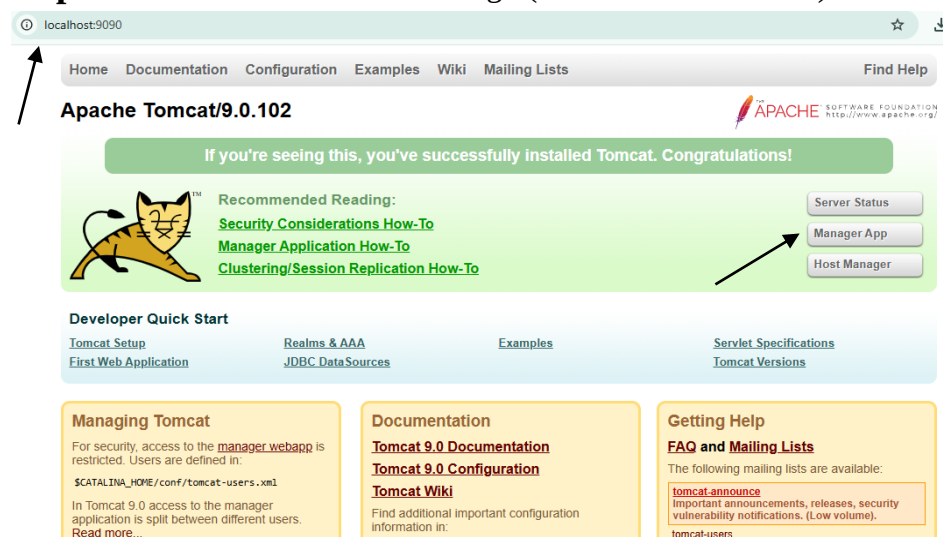
(Here we are creating one artifact war file. That war file we are deploy to test server(Tomcat server) using Jenkins Server)

(Note:-Images are purpose of understanding only don't paste it in records but output images paste it in output area)

Step1:-Lunch Jenkins and Login(Use AWS or Software)



Step2:-Lunch Tomcat Server and login(use AWS or Software)



Open Jenkins do following steps

Step 3:-Install plugins

Go to Manage Jenkins→Plugins→Install Plugin select verify Git→Go to Available plugins install Maven integration and Deploy to container install.

The screenshot shows the Jenkins Dashboard with the 'Manage Jenkins' page selected. The left sidebar contains navigation links: 'New Item', 'Build History', 'Project Relationship', 'Check File Fingerprint', 'Manage Jenkins' (highlighted with an arrow), and 'My Views'. The main content area displays a table of builds and a warning message about the EDDSA API Plugin. Below the warning, the 'System Configuration' section is visible, with the 'Plugins' link highlighted by an arrow. The 'Plugins' page shows a search bar with 'git' entered and a list of installed plugins, including 'Git client plugin 6.1.2' and 'Git plugin 5.7.0'. Arrows point to the 'Available plugins', 'Installed plugins', and 'Advanced settings' links in the left sidebar.

Dashboard >

+ New Item

Build History

Project Relationship

Check File Fingerprint

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

Dashboard > Manage Jenkins

Check File Fingerprint

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

(0 of 2 executors busy)

Warnings have been published for the following currently installed components:

EDDSA API Plugin 0.3.0-13.v7cb_69ed68f00:
EdDSA implementation exhibits signature malleability
A fix for this issue is available. Go to the [plugin manager](#) to update the plugin.

Go to plugin manager

Configure which of these warnings are shown

System Configuration

System
Configure global settings and paths.

Tools
Configure tools, their locations and automatic installers.

Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

Clouds
Add, remove, and configure cloud instances to provision agents on-demand.

Appearance
Configure the look and feel of Jenkins

Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

Dashboard > Manage Jenkins > Plugins

Plugins

Updates

Available plugins

Installed plugins

Advanced settings

Download progress

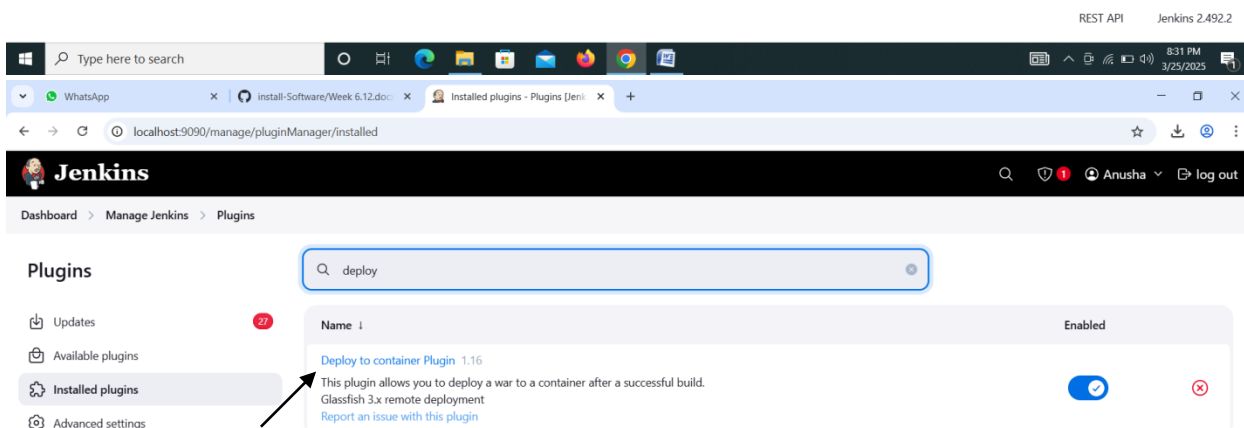
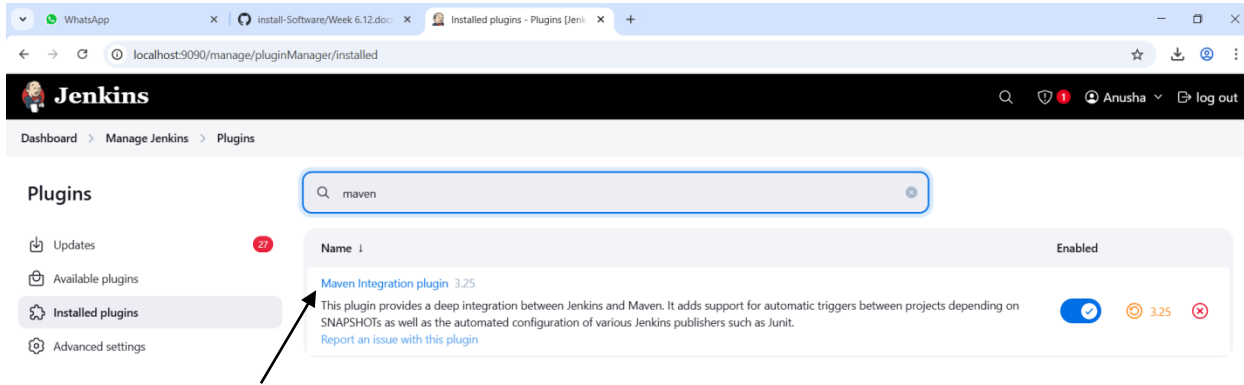
git

Name ↓

Git client plugin 6.1.2
Utility plugin for Git support in Jenkins
[Report an issue with this plugin](#)

Git plugin 5.7.0
This plugin integrates [Git](#) with Jenkins.
[Report an issue with this plugin](#)

Above picture three plugins are installed , If not showing please go to available plugins install it.



Step 4:-set up git path and maven installation in Tools

Go to manage Jenkins→Tools→In git installation select path(Git bash) if your using Jenkins software or AWS leave it →In maven Enter name Maven and default installation→Apply and Save.

The screenshot shows the Jenkins 'Manage Jenkins' interface. The 'System Configuration' section is active, displaying options for System, Tools, Nodes, Clouds, Plugins, and Appearance. An arrow points to the 'Tools' icon. Below this, the 'Tools' configuration page is shown, specifically the 'Git installations' section. A form for adding a new Git installation is visible, with the 'Path to Git executable' field set to 'C:\Program Files\Git\bin\git.exe'. An arrow points to this field. The 'Add Git' button is also visible. Below the Git section, the 'Gradle installations' section is partially visible with an 'Add Gradle' button. At the bottom, 'Save' and 'Apply' buttons are present.

localhost:9090/manage/

Jenkins

Dashboard > Manage Jenkins

+ New Item

- Build History
- Project Relationship
- Check File Fingerprint
- Manage Jenkins
- My Views

Build Queue: No builds in the queue.

Build Executor Status: 0/2

Manage Jenkins

Warnings have been published for the following currently installed components:

EDDSA API Plugin 0.3.0-13.v7cb_69ed68f00:
EdDSA implementation exhibits signature malleability
A fix for this issue is available. Go to the [plugin manager](#) to update the plugin.

Go to plugin manager | Configure which of these warnings are shown

System Configuration

- System**: Configure global settings and paths.
- Tools**: Configure tools, their locations and automatic installers.
- Nodes**: Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Clouds**: Add, remove, and configure cloud instances to provision agents on-demand.
- Plugins**: Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- Appearance**: Configure the look and feel of Jenkins.

Security

- Security
- Credentials
- Credential Providers

localhost:9090/manage/configureTools/

Dashboard > Manage Jenkins > Tools

Git installations

Git

Name: Default

Path to Git executable: C:\Program Files\Git\bin\git.exe

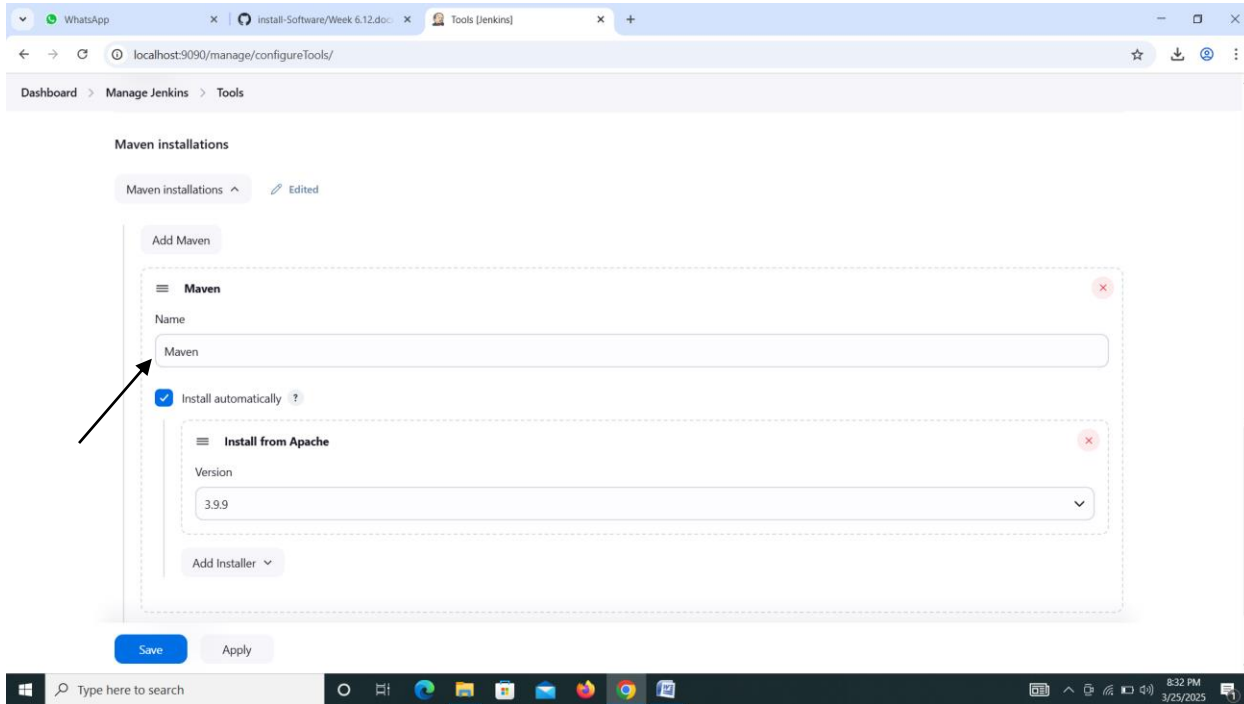
☐ Install automatically

Add Git

Gradle installations

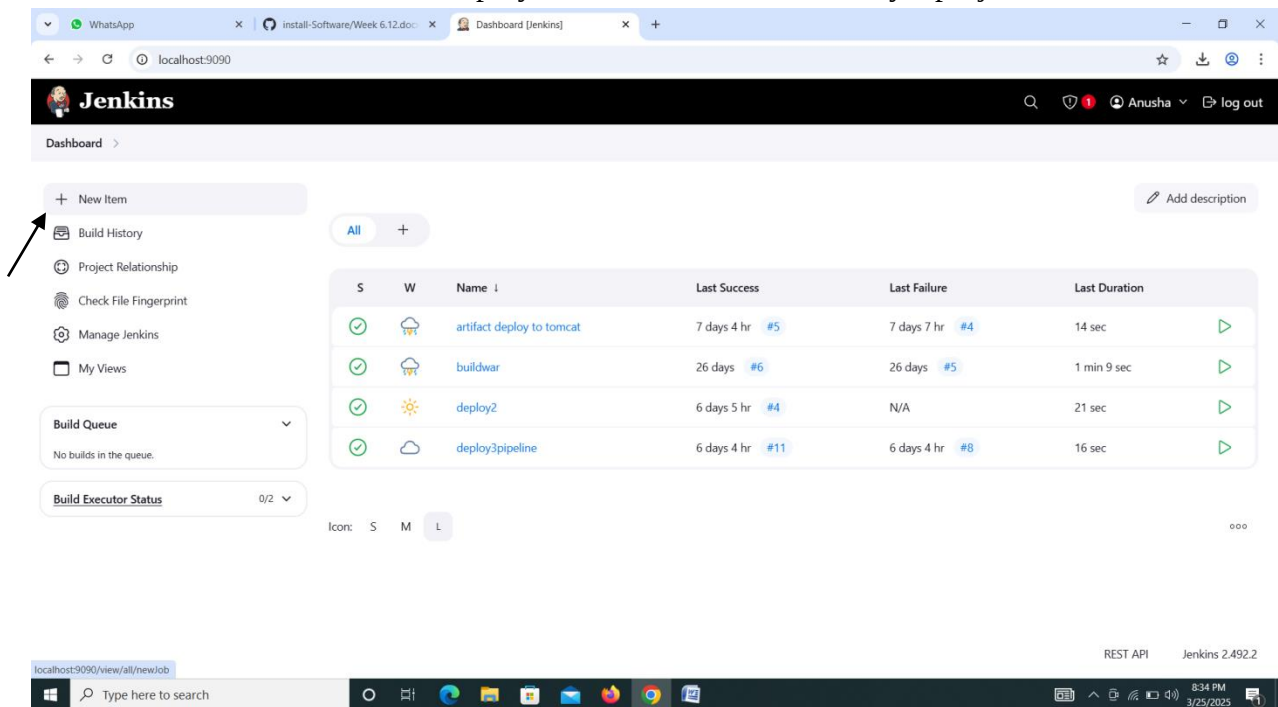
Add Gradle

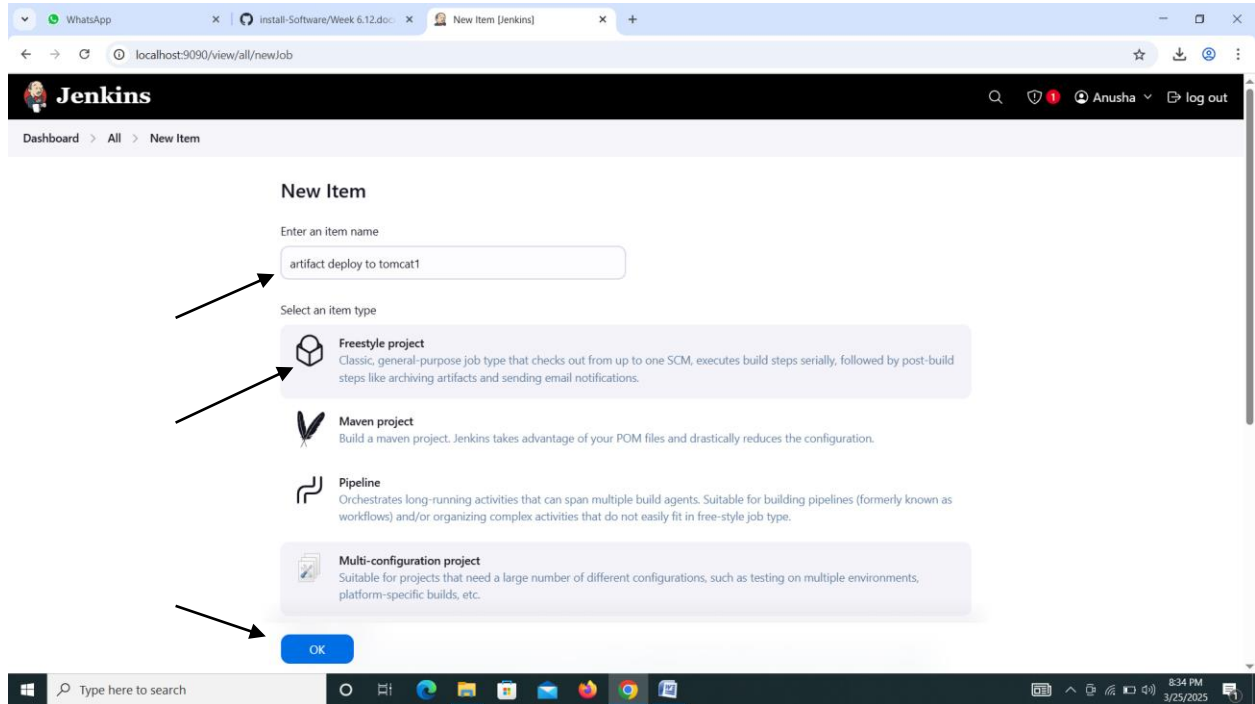
Save | Apply



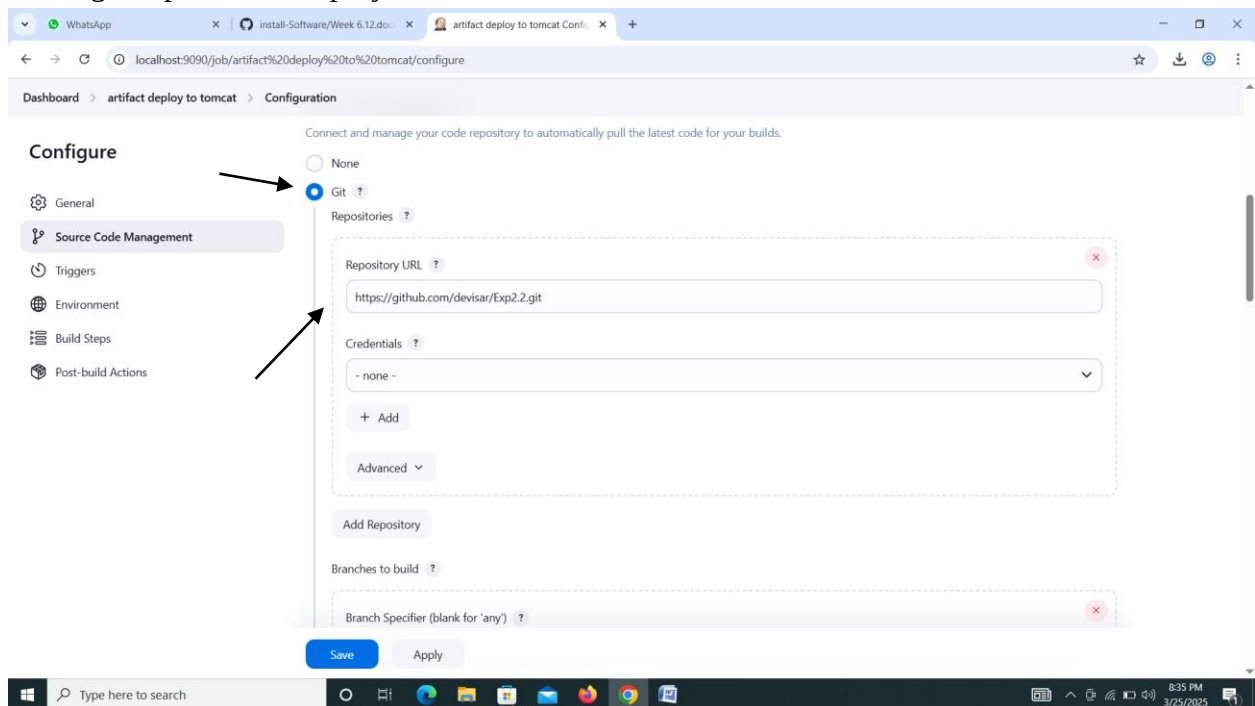
Step 5:-Creating New Job for deploy artifact in tomcat test server

1. Create New Item → Enter name DeploytoTestServer → select freestyle project → click ok

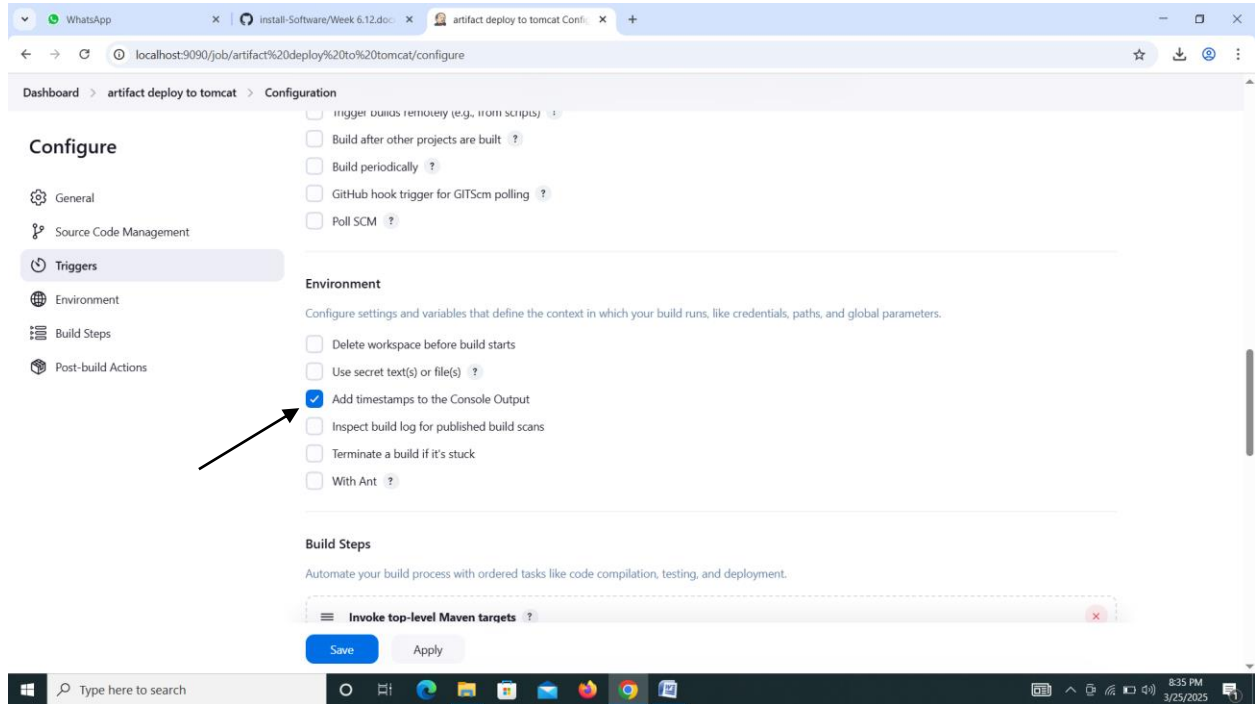




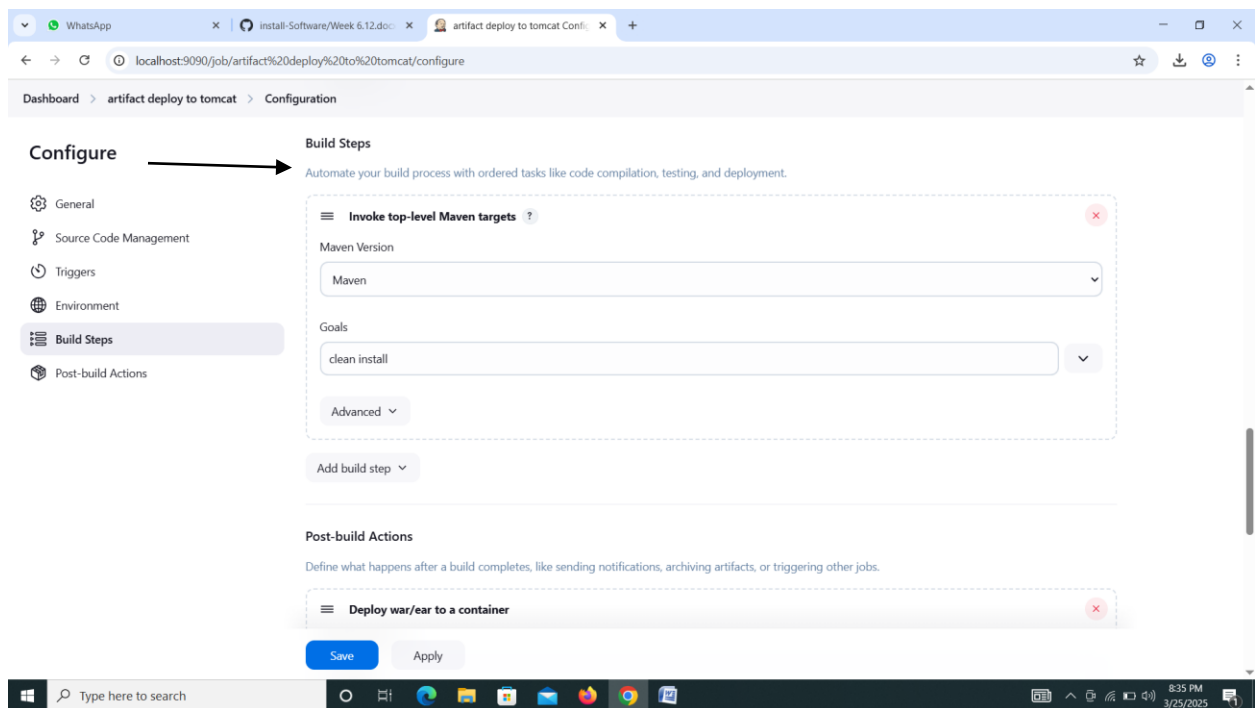
2. Select git → paste maven project url form Github → Must and should match branches



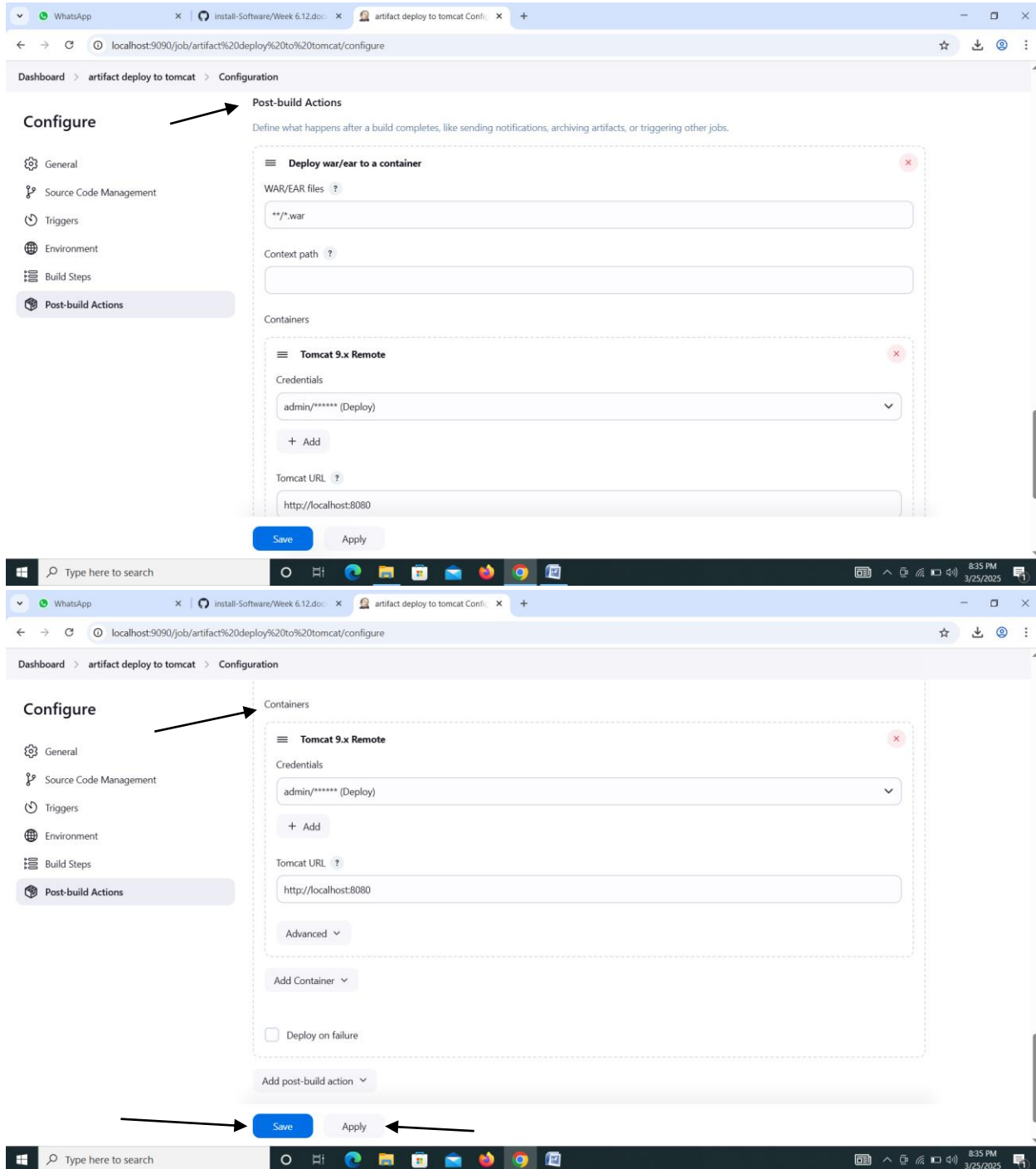
3. Go to Environment → Select Add timestamps to the Console Output



4. Go to Build Setups→Select **Invoke top-level Maven targets**→Maven Version enter maven →Goals **clean install**

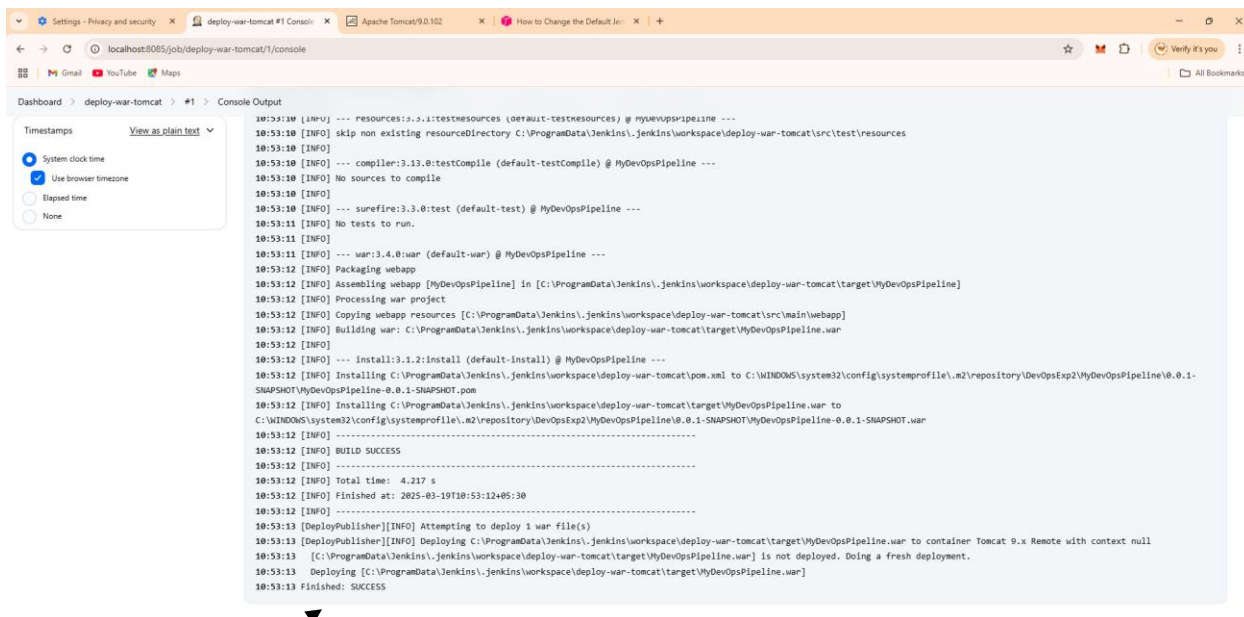
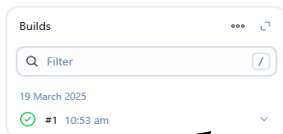
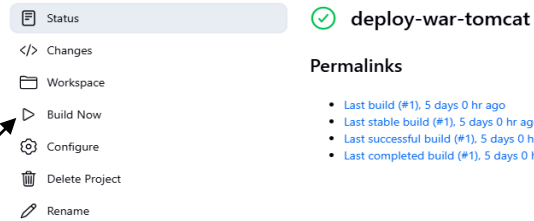
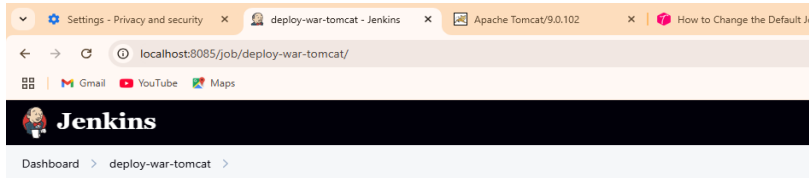


5. Go to Post-build Actions→Select **Deploy war/ear to a container**→WAR/EAR files enter ****/*.war**→select Containers **Tomcat 9.x Remote**→Add credentials like username, password, Id and description.-->Tomcat URL **http://localhost:9090** →Apply and save.

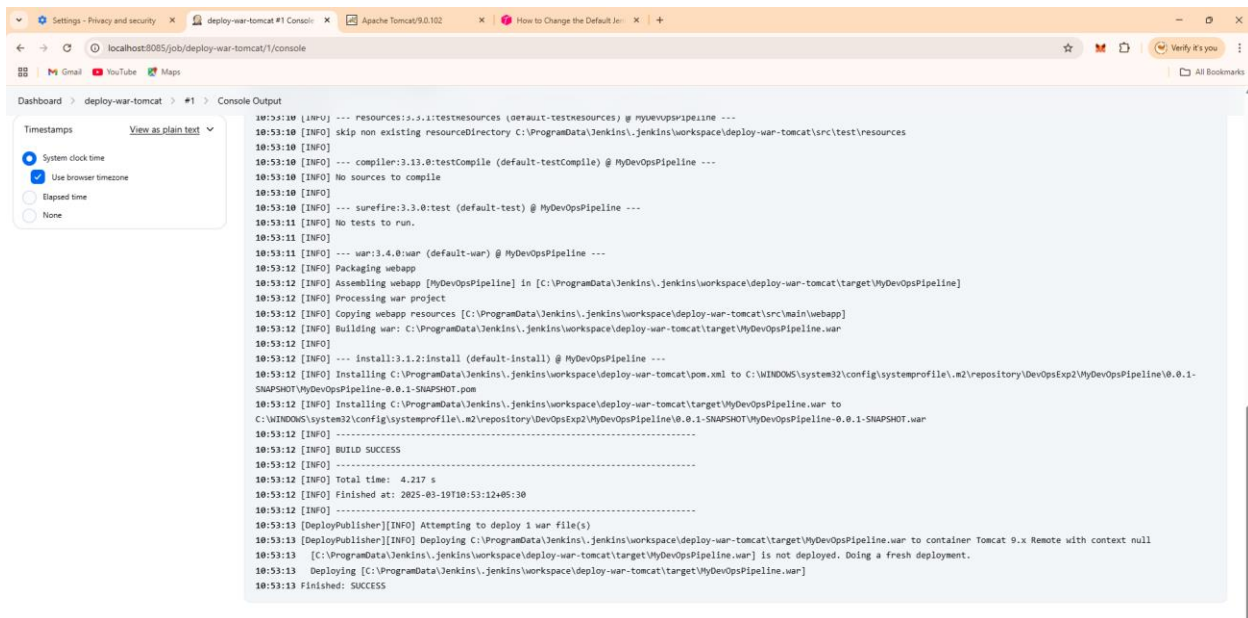
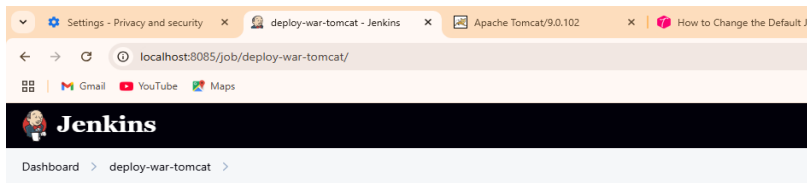


Step 6:- Build war File

1. Click Build now → Go to Build select console output now you got Successfully Finish.



Output:-



Tomcat Web Application Manager

Message: OK

Manager

List Applications HTML Manager Help Manager Help Server Status

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/MyDevOpsPipeline	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes
/manager	None specified	Tomcat Manager Application	true	1	Start Stop Reload Undeploy Expire sessions with idle > 30 minutes

Deploy

Deploy directory or WAR file located on server

Context Path:

Version (for parallel deployment):

XML Configuration file path:

WAR or Directory path:

Deploy

localhost:9090/MyDevOpsPipeline/

Hello Students12345

Experiment-7

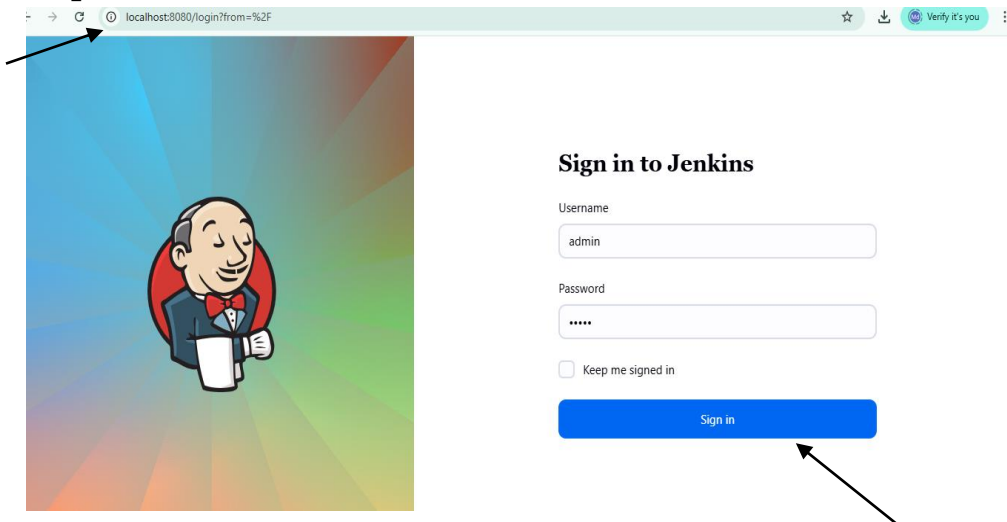
AIM:-Perform Automation using Jenkins.

Softwares:Jenkins(port:8080),JDk 17 or 21,Tomcat 9 (port:9090),Gitbash

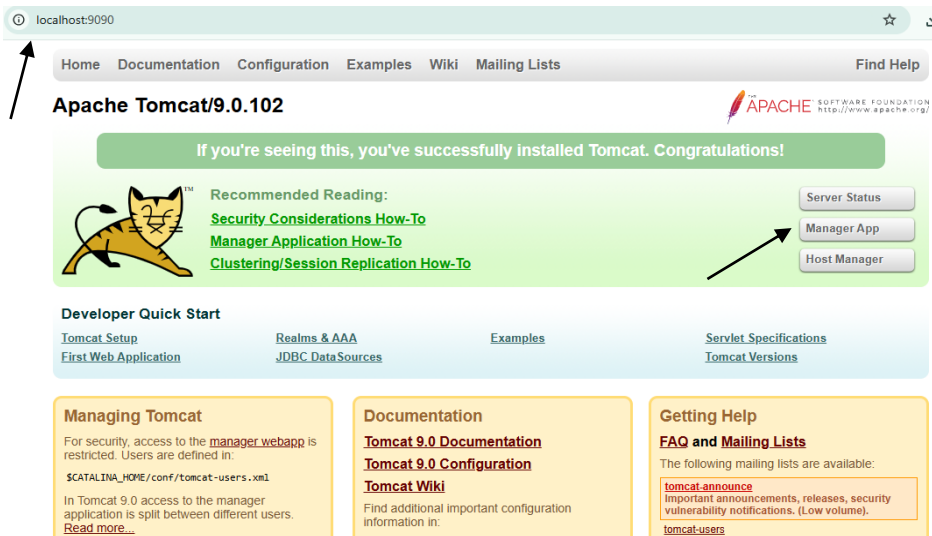
(Here we are creating one artifact war file. That war file we are deploy to test server(Tomcat server) using Jenkins Server)

(Note:-Images are purpose of understanding only don't paste it in records but output images paste it in output area)

Step1:-Lunch Jenkins and Login(Use AWS or Software)



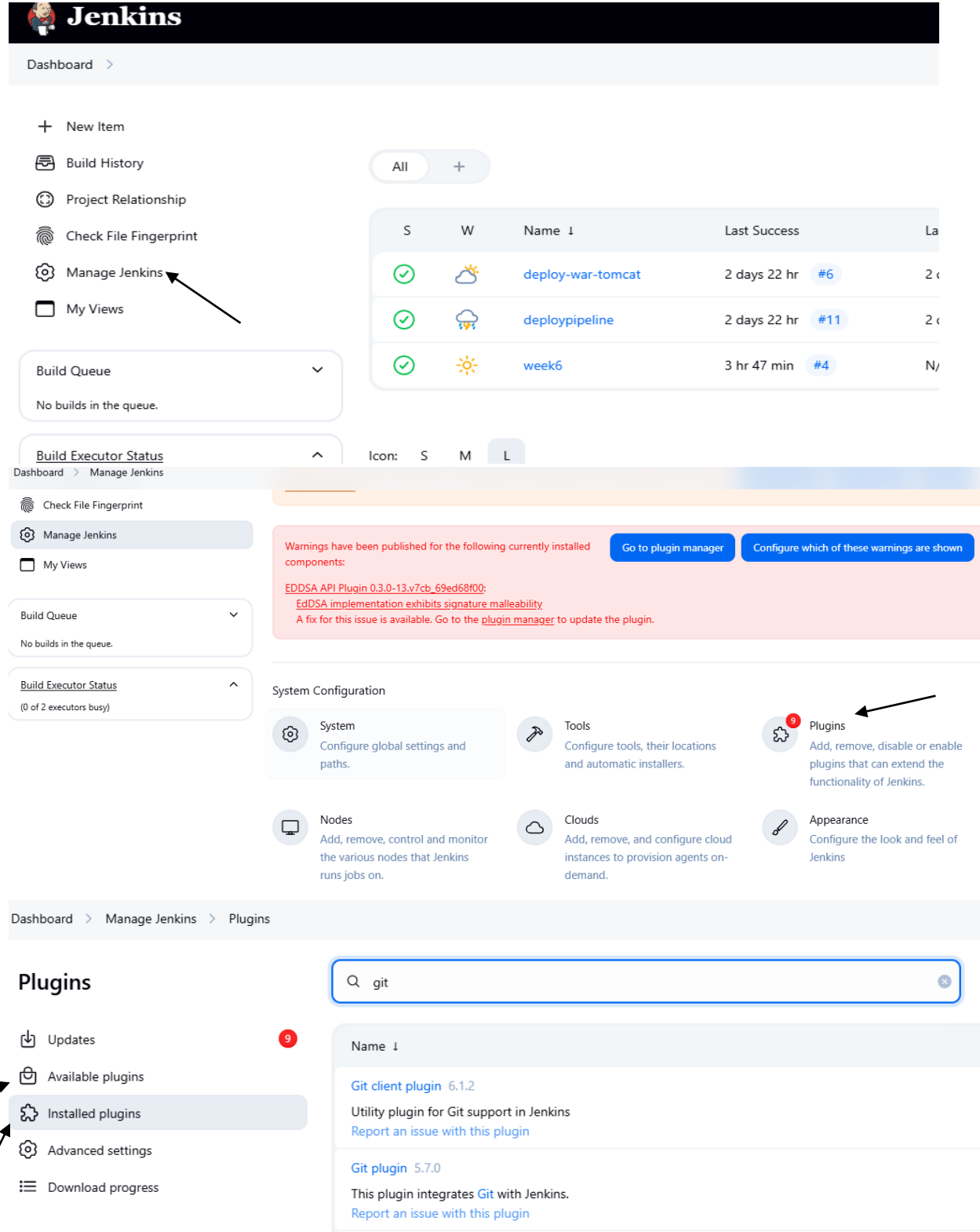
Step2:-Lunch Tomcat Server and login(use AWS or Software)



Open Jenkins do following steps

Step 3:-Install plugins

Go to Manage Jenkins→Plugins→Install Plugin select verify Git→Go to Available plugins install Maven integration and Deploy to container install.



Dashboard >

+ New Item

Build History

Project Relationship

Check File Fingerprint

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

Dashboard > Manage Jenkins

Check File Fingerprint

Manage Jenkins

My Views

Build Queue

No builds in the queue.

Build Executor Status

(0 of 2 executors busy)

Warnings have been published for the following currently installed components:

EDDSA API Plugin 0.3.0-13.v7cb_69ed68f00:
EdDSA implementation exhibits signature malleability
A fix for this issue is available. Go to the [plugin manager](#) to update the plugin.

Go to plugin manager

Configure which of these warnings are shown

System Configuration

System
Configure global settings and paths.

Tools
Configure tools, their locations and automatic installers.

Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.

Clouds
Add, remove, and configure cloud instances to provision agents on-demand.

Appearance
Configure the look and feel of Jenkins

Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.

Dashboard > Manage Jenkins > Plugins

Plugins

Updates

Available plugins

Installed plugins

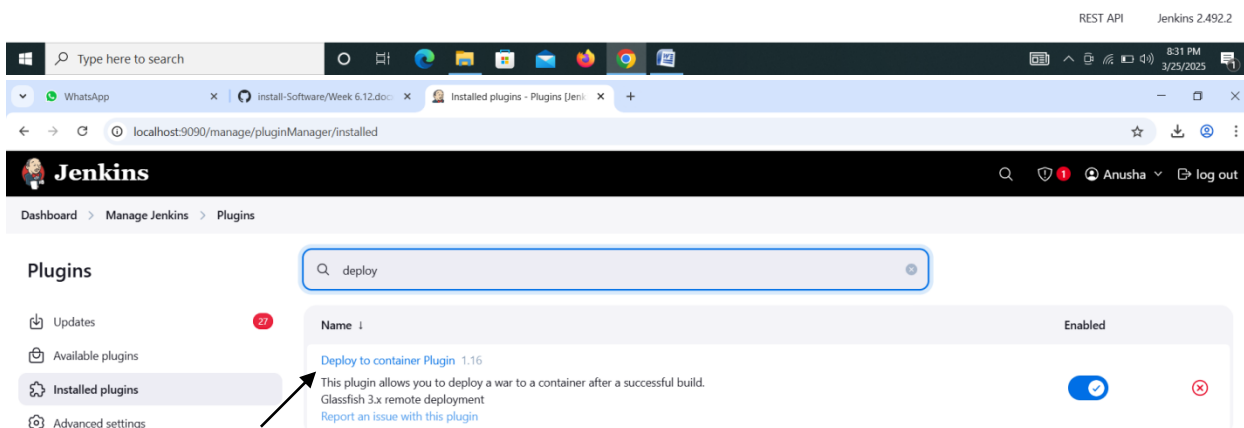
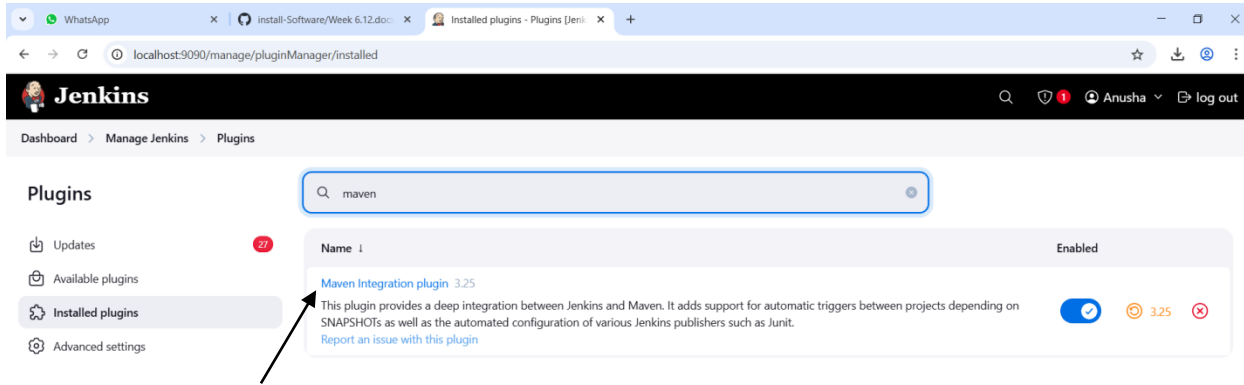
Advanced settings

Download progress

git

Name ↓
Git client plugin 6.1.2 Utility plugin for Git support in Jenkins Report an issue with this plugin
Git plugin 5.7.0 This plugin integrates Git with Jenkins. Report an issue with this plugin

Above picture three plugins are installed , If not showing please go to available plugins install



Step 4:-set up git path and maven installation in Tools

Go to manage Jenkins→Tools→In git installation select path(Git bash) if your using Jenkins software or AWS leave it →In maven Enter name Maven and default installation→Apply and Save.

The screenshot shows the Jenkins 'Manage Jenkins' interface. The 'System Configuration' section is active, displaying options for System, Tools, Nodes, Clouds, Plugins, and Appearance. An arrow points to the 'Tools' icon. Below this, the 'Tools' configuration page is shown, specifically the 'Git installations' section. A form for adding a new Git installation is visible, with the 'Path to Git executable' field containing 'C:\Program Files\Git\bin\git.exe'. An arrow points to this field. The 'Add Git' button is also visible. Below the Git section, the 'Gradle installations' section is partially visible with an 'Add Gradle' button. At the bottom, 'Save' and 'Apply' buttons are present.

localhost:9090/manage/

Jenkins

Dashboard > Manage Jenkins

+ New Item

- Build History
- Project Relationship
- Check File Fingerprint
- Manage Jenkins
- My Views

Build Queue: No builds in the queue.

Build Executor Status: 0/2

Manage Jenkins

Warnings have been published for the following currently installed components:

EDDSA API Plugin 0.3.0-13.v7cb_69ed68f00:
EdDSA implementation exhibits signature malleability
A fix for this issue is available. Go to the [plugin manager](#) to update the plugin.

Go to plugin manager | Configure which of these warnings are shown

System Configuration

- System**: Configure global settings and paths.
- Tools**: Configure tools, their locations and automatic installers.
- Nodes**: Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
- Clouds**: Add, remove, and configure cloud instances to provision agents on-demand.
- Plugins**: Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
- Appearance**: Configure the look and feel of Jenkins.

Security

- Security
- Credentials
- Credential Providers

localhost:9090/manage/configureTools/

Dashboard > Manage Jenkins > Tools

Git installations

Git

Name: Default

Path to Git executable: C:\Program Files\Git\bin\git.exe

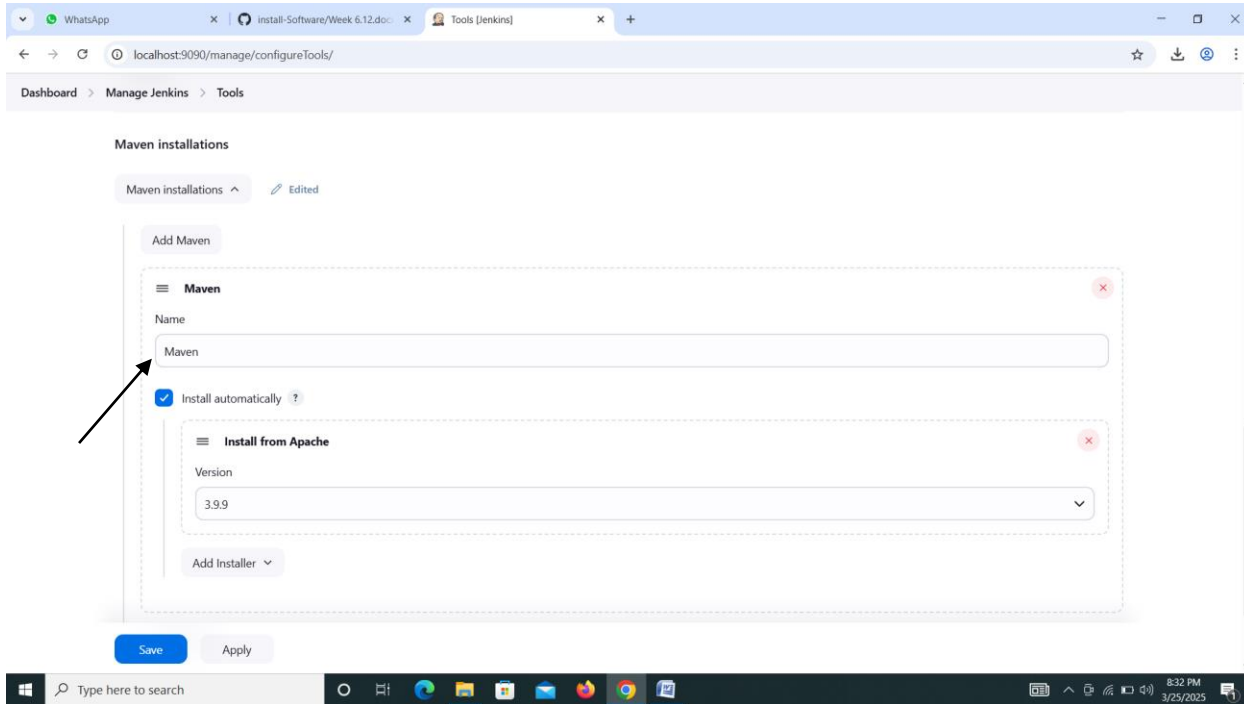
☐ Install automatically

Add Git

Gradle installations

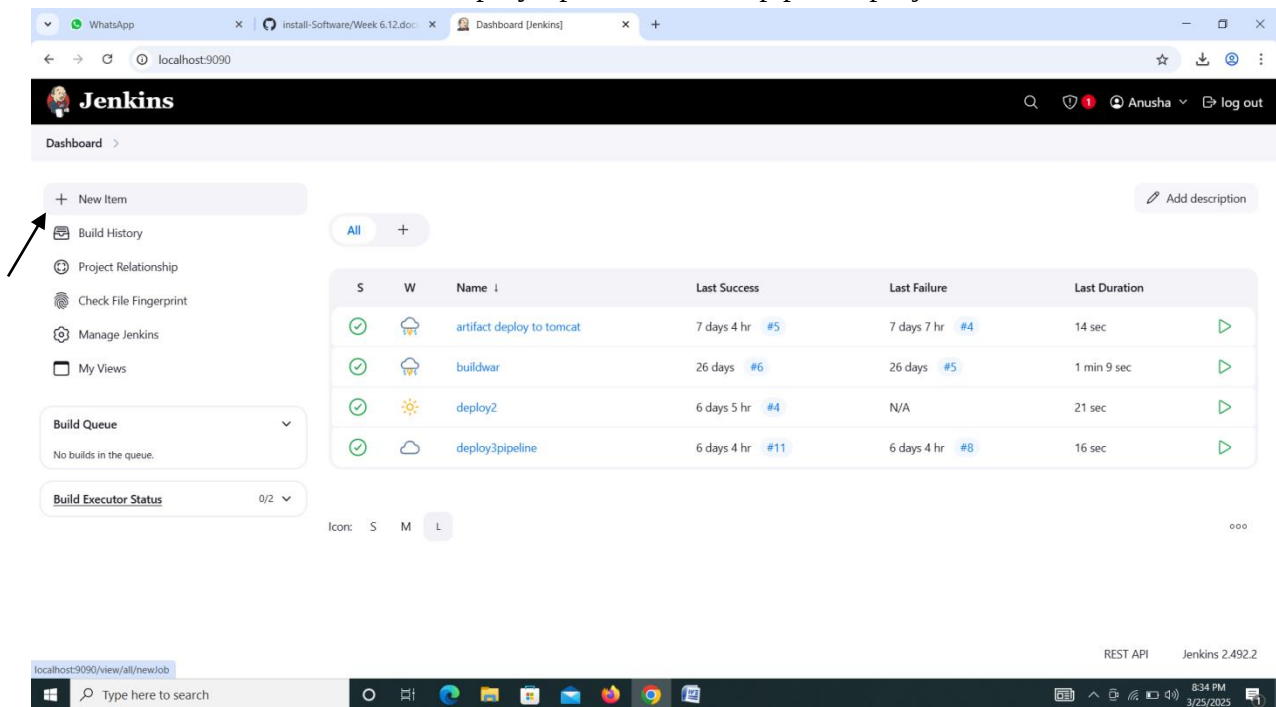
Add Gradle

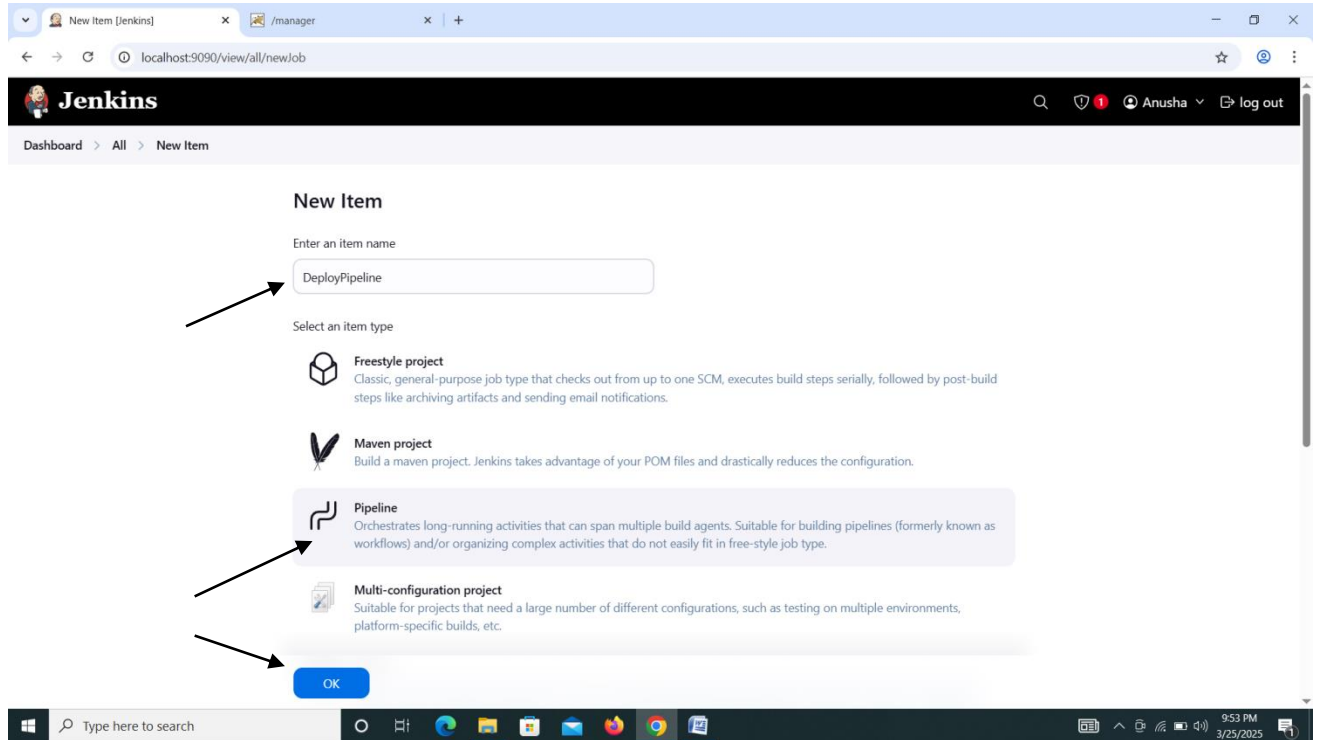
Save | Apply



Step 5:-Creating New Job for deploy artifact in tomcat test server

6. Create New Item → Enter name DeployPipeline → select pipeline project → click ok



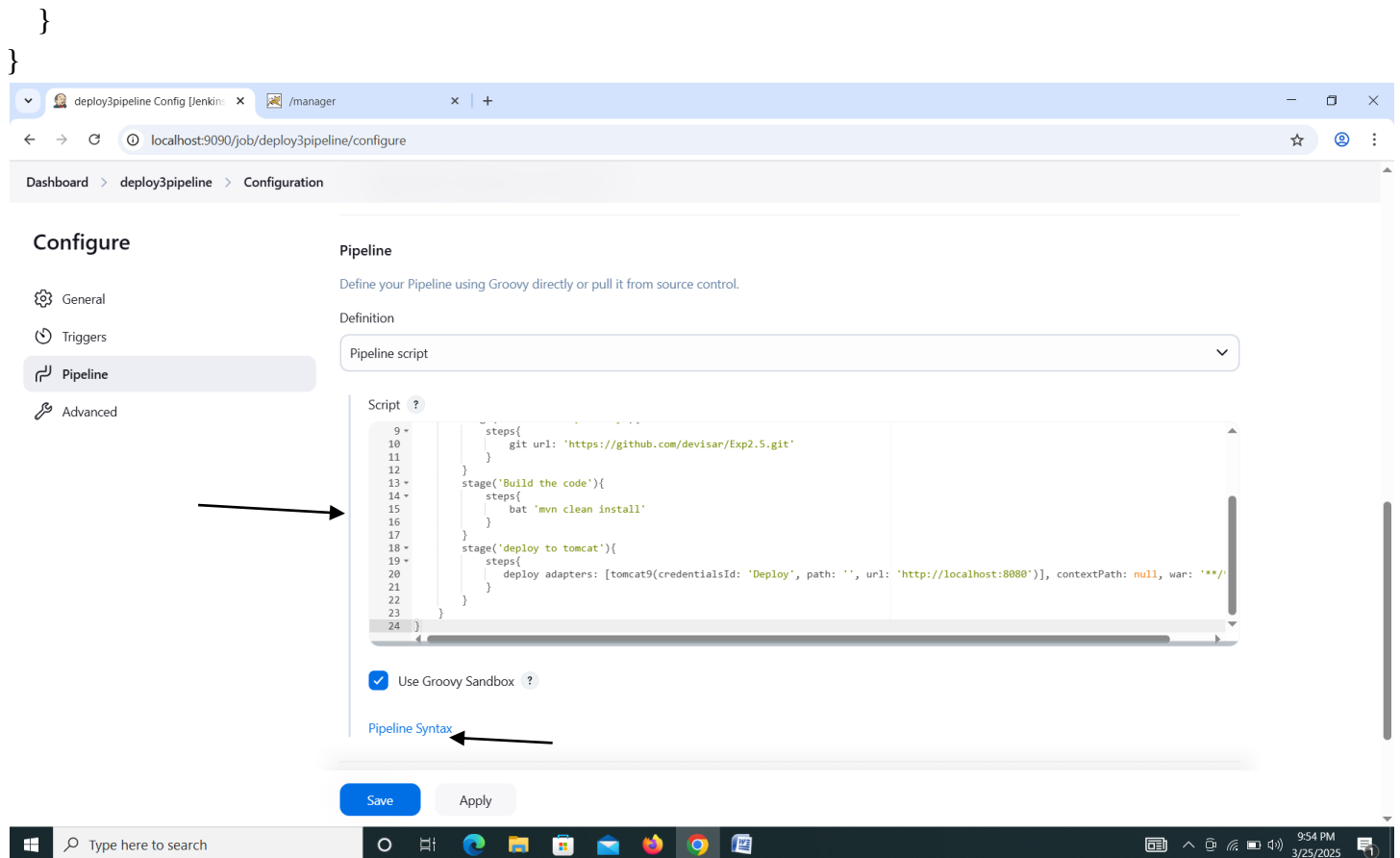


Step 2:-Copy paste below commands pipeline script

```

pipeline{
  agent any
  tools{
    maven 'Maven'
  }
  stages{
    stage('clone the repository'){
      steps{
        git url: 'https://github.com/devisar/Exp2.5.git'
      }
    }
    stage('Build the code'){
      steps{
        bat 'mvn clean install'
      }
    }
    stage('deploy to tomcat'){
      steps{
        deploy adapters: [tomcat9(credentialsId: 'Deploy', path: '', url: 'http://localhost:8080')],
        contextPath: null, war: '**/*.war'
      }
    }
  }
}

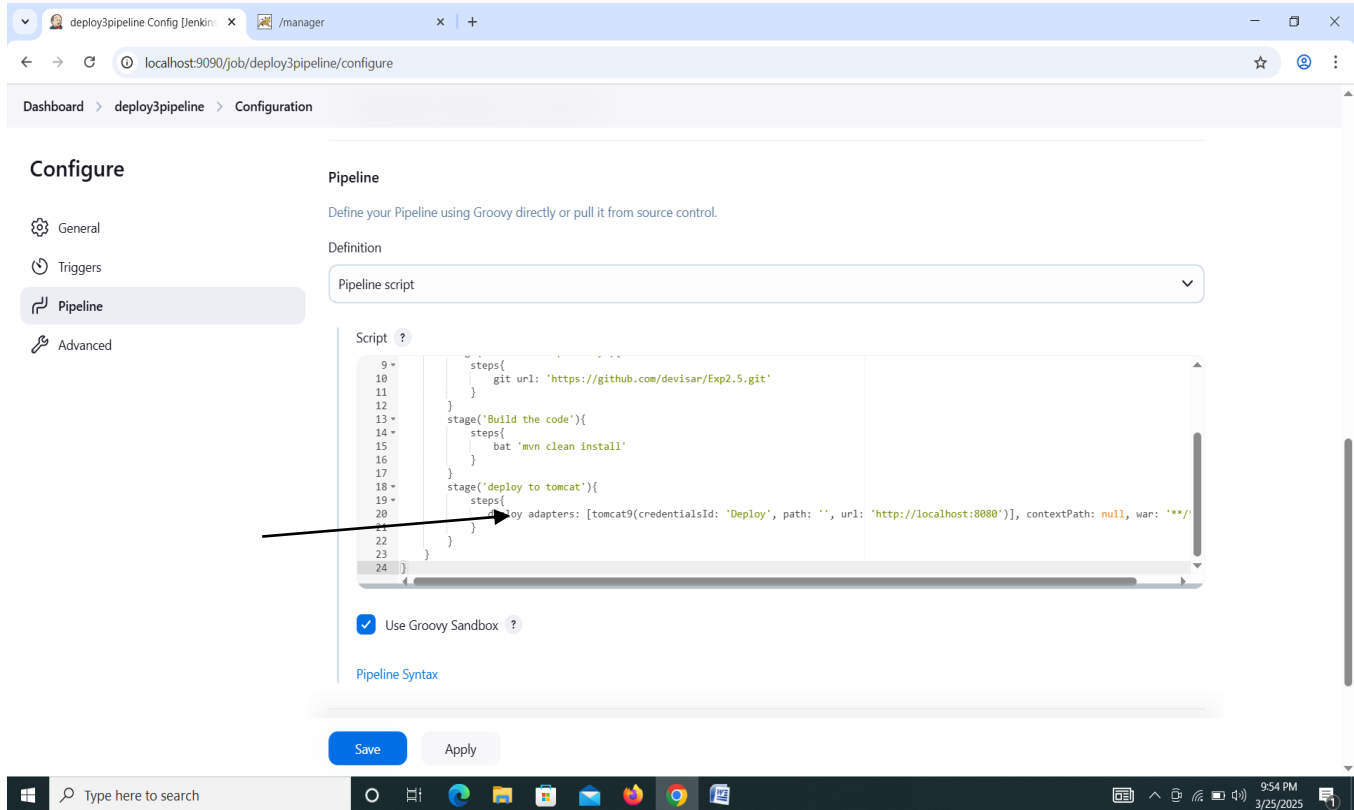
```



Step 6:-deploy to tomcat you want to create pipeline script go to →pipeline syntax→Select in sample step deploy: Deploy war/ear to a container →enter war/ear files:**/*.*.war→Select Containers tomcat 9→select credentials tomcat user name and password. →Enter Tomcat Url like http://localhost:9090. →Click Generate Pipeline Script copy and paste it original pipeline deployment stage.

The screenshot displays the 'Pipeline Syntax' configuration page in Selenium Grid. The page is titled 'localhost:8080/job/DeployPipelineweek7/pipeline-syntax/'. The left sidebar contains links: 'Dashboard', 'DeployPipelineweek7', and 'Pipeline Syntax'. The main content area shows the 'Steps' section with a 'Sample Step' dropdown set to 'deploy: Deploy war/ear to a container'. Below this, the 'WAR/EAR files' field is set to '**/*.war'. The 'Context path' field is empty. The 'Containers' section shows a 'Tomcat 9.x Remote' container with 'Credentials' set to 'deployer/***** (Deployer12345)' and 'Tomcat URL' set to 'http://localhost:9090'. The 'Advanced' dropdown is visible. Below the container configuration, there is a 'Generate Pipeline Script' button. The generated script is shown below the button:

```
deploy adapters: [tomcat9(credentialsId: 'Deployer12345', path: '', url: 'http://localhost:9090'), contextPath: null, war: '**/*.war']
```



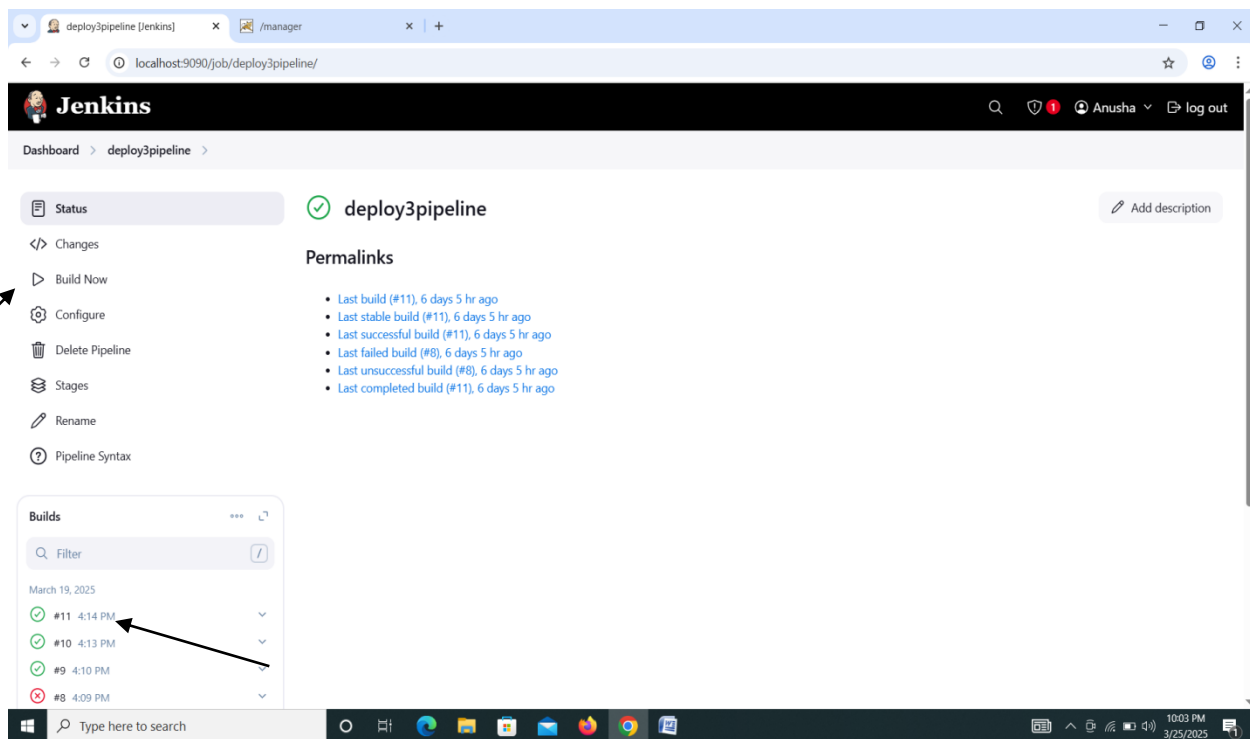
The screenshot shows the Jenkins 'Configure' page for a pipeline named 'deploy3pipeline'. The 'Pipeline' tab is selected in the left sidebar. The 'Definition' section shows a 'Pipeline script' definition. The script is as follows:

```
9  steps{
10      git url: 'https://github.com/devisar/Exp2.5.git'
11  }
12  stage('Build the code'){
13      steps{
14          bat 'mvn clean install'
15      }
16  }
17  stage('deploy to tomcat'){
18      steps{
19          deploy adapters: [tomcat9(credentialsId: 'Deploy', path: '', url: 'http://localhost:8080')], contextPath: null, war: '**/'
20      }
21  }
22  }
23  }
24  }
```

An arrow points to the 'deploy adapters' line in the script. Below the script, the 'Use Groovy Sandbox' checkbox is checked. At the bottom, there are 'Save' and 'Apply' buttons.

Step 7:- Build war File

2. Click Build now → Go to



The screenshot shows the Jenkins 'View' page for the 'deploy3pipeline'. The 'Build Now' button is highlighted with an arrow. The 'Builds' section shows a list of builds:

Build Number	Status	Time
#11	Success	4:14 PM
#10	Success	4:13 PM
#9	Success	4:10 PM
#8	Failure	4:09 PM

An arrow points to build #10. The 'Permalinks' section lists various build links. The 'Build' button is visible in the bottom right corner.

select console output now you got Successfully Finish.

The screenshot shows a web browser window with the address bar displaying `localhost:9090/job/deploy3pipeline/11/console`. The browser tabs include `deploy3pipeline #11 Console` and `/manager`. The page content shows the Jenkins console output for a pipeline named `deploy3pipeline`, job `#11`. The output text is as follows:

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (deploy to tomcat)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] deploy
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war to container Tomcat 9.x Remote
with context null
  Redeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
  Undeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
  Deploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

An arrow points to the `Finished: SUCCESS` line. In the bottom right corner of the console area, it says `REST API Jenkins 2.492.2`. The Windows taskbar is visible at the bottom with the search bar and various application icons.

Output:-

The screenshot displays the Jenkins web interface in a browser window. The top navigation bar shows the Jenkins logo, a search icon, a shield icon with a red exclamation mark, the user 'Anusha', and a 'log out' button. The breadcrumb trail indicates the current location: 'Dashboard > deploy3pipeline > #11'.

The main content area is divided into two sections. On the left, a sidebar contains a 'Status' tab (active), 'Changes', 'Build Now', 'Configure', 'Delete Pipeline', 'Stages', 'Rename', and 'Pipeline Syntax'. The 'Builds' section on the left lists recent builds: #11 (4:14 PM), #10 (4:13 PM), #9 (4:10 PM), and #8 (4:09 PM). The main area shows the 'deploy3pipeline' job status as 'Success' (green checkmark). Below the status, there are 'Permalinks' for various build types: Last build (#11), Last stable build (#11), Last successful build (#11), Last failed build (#8), Last unsuccessful build (#8), and Last completed build (#11).

The console output for build #11 is displayed in a large text area. The output shows the pipeline stages and the deployment process:

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (deploy to tomcat)
[Pipeline] tool
[Pipeline] envVarsForTool
[Pipeline] withEnv
[Pipeline] {
[Pipeline] deploy
[DeployPublisher][INFO] Attempting to deploy 1 war file(s)
[DeployPublisher][INFO] Deploying C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war to container Tomcat 9.x Remote
with context null
  Redeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
  Undeploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
  Deploying [C:\ProgramData\Jenkins\.jenkins\workspace\deploy3pipeline\target\Anusha.war]
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

The bottom of the screenshot shows the Windows taskbar with the search bar and various application icons. The system tray on the right indicates the time as 10:03 PM on 3/25/2025. The bottom right corner of the Jenkins interface shows 'REST API' and 'Jenkins 2.492.2'.

deploy3pipeline #11 Console | x | /manager x +

localhost:8080/manager/html



Tomcat Web Application Manager

Message: OK

Manager

[List Applications](#) [HTML Manager Help](#) [Manager Help](#) [Server Status](#)

Applications

Path	Version	Display Name	Running	Sessions	Commands
/	None specified	Welcome to Tomcat	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/Anusha	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/MyDevOpsPipeline	None specified	Archetype Created Web Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/docs	None specified	Tomcat Documentation	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/examples	None specified	Servlet and JSP Examples	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/host-manager	None specified	Tomcat Host Manager Application	true	0	Start Stop Reload Undeploy Expire sessions with idle ≥ 30 minutes
/manager	None specified	Tomcat Manager Application	true	0	Start Stop Reload Undeploy

Type here to search

deploy3pipeline #11 Console | x | localhost:8080/Anusha/ x +

localhost:8080/Anusha/

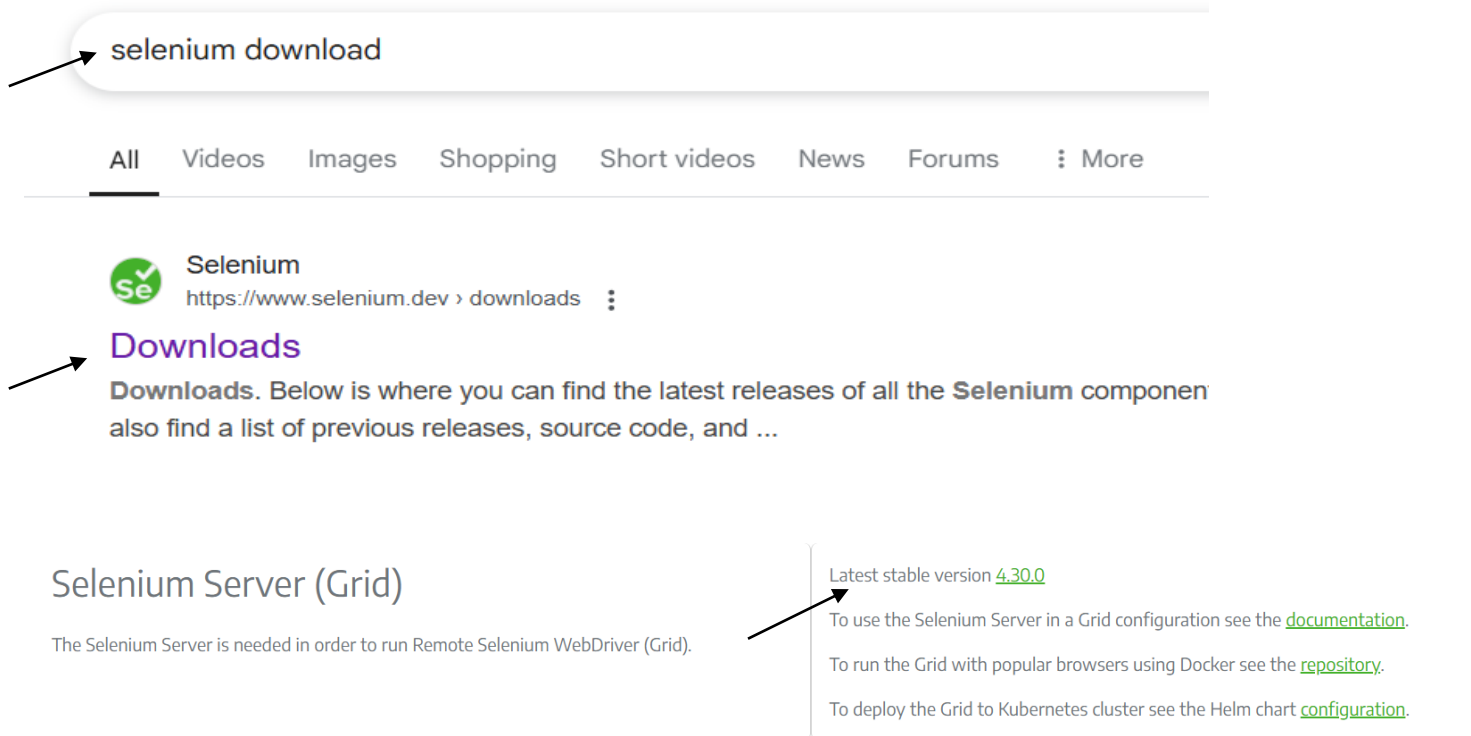
Hello World!

Experiment 8

Week 8

AIM:-Build and deploy a grid for Chrome and Firefox based testing.

Step1:- Download Selenium Grid



The screenshot shows the Selenium website's download page. At the top, there is a search bar with the text "selenium download" and an arrow pointing to it. Below the search bar is a navigation menu with tabs: All, Videos, Images, Shopping, Short videos, News, Forums, and More. The "All" tab is selected. Below the menu, the Selenium logo is displayed, followed by the text "Selenium" and the URL "https://www.selenium.dev > downloads". A purple heading "Downloads" is shown, with an arrow pointing to it. Below this heading, a paragraph states: "Downloads. Below is where you can find the latest releases of all the Selenium components. You can also find a list of previous releases, source code, and ...". Further down, the section "Selenium Server (Grid)" is visible, with a subtext: "The Selenium Server is needed in order to run Remote Selenium WebDriver (Grid)". To the right of this section, a sidebar contains the text "Latest stable version 4.30.0" with an arrow pointing to it, followed by three links: "To use the Selenium Server in a Grid configuration see the documentation.", "To run the Grid with popular browsers using Docker see the repository.", and "To deploy the Grid to Kubernetes cluster see the Helm chart configuration."

selenium download

All Videos Images Shopping Short videos News Forums More

 Selenium
https://www.selenium.dev > downloads

Downloads

Downloads. Below is where you can find the latest releases of all the Selenium components. You can also find a list of previous releases, source code, and ...

Selenium Server (Grid)
The Selenium Server is needed in order to run Remote Selenium WebDriver (Grid).

Latest stable version [4.30.0](#)
To use the Selenium Server in a Grid configuration see the [documentation](#).
To run the Grid with popular browsers using Docker see the [repository](#).
To deploy the Grid to Kubernetes cluster see the Helm chart [configuration](#).

Step2:-Download Chrome Driver

chrome driver download

All

Videos

Images

Short videos

Shopping

News

Forums

More



Chrome for Developers

<https://developer.chrome.com> › Docs › ChromeDriver



Downloads | ChromeDriver - Chrome for Developers

17 Dec 2024 — Earlier Chrome versions · **ChromeDriver** 114.0.5735.90 · **ChromeDriver** 114.0.5735.90 · **ChromeDriver** 113.0.5672.63 · **ChromeDriver** 113.0.5672.24.

Downloads



Warning: If you're using Chrome version 115 or newer, consult the [Chrome for Testing availability dashboard](#). On this dashboard, you'll find [JSON endpoints](#) to download specific ChromeDriver versions.

Earlier Chrome versions

Chrome for Testing availability

This page lists the latest available cross-platform Chrome for Testing versions and assets per Chrome

Consult [our JSON API endpoints](#) if you're looking to build automated scripts based on Chrome for Test

Last updated @ 2025-03-28T07:09:34.087Z

Channel	Version	Revision	Status
<u>Stable</u>	134.0.6998.165	r1415337	☒
<u>Beta</u>	135.0.7049.41	r1427262	☒
<u>Dev</u>	136.0.7091.2	r1437865	☒
<u>Canary</u>	136.0.7093.0	r1438506	☒

Below picture address copy and paste it new tab. Automatically chrome driver downloaded

Version: 134.0.6998.165 (r1415337)

Binary	Platform	URL	HTTP status
chrome	linux64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/linux64/chrome-linux64.zip	200
chrome	mac-arm64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/mac-arm64/chrome-mac-arm64.zip	200
chrome	mac-x64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/mac-x64/chrome-mac-x64.zip	200
chrome	win32	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/win32/chrome-win32.zip	200
chrome	win64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/win64/chrome-win64.zip	200
chromedriver	linux64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/linux64/chromedriver-linux64.zip	200
chromedriver	mac-arm64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/mac-arm64/chromedriver-mac-arm64.zip	200
chromedriver	mac-x64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/mac-x64/chromedriver-mac-x64.zip	200
chromedriver	win32	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/win32/chromedriver-win32.zip	200
chromedriver	win64	https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/win64/chromedriver-win64.zip	200

<https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/win64/chromedriver-win64.zip>

<https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/win64/chromedriver-win64.zip>

<https://storage.googleapis.com/chrome-for-testing-public/134.0.6998.165/win64/chromedriver-win64.zip> - Google Search

Step3:-Download Gecko Driver

firefox driver download

All Images Videos Shopping Short videos News Forums

 [GitHub](https://github.com/mozilla/geckodriver/releases)
<https://github.com/mozilla/geckodriver/releases>

You visit often

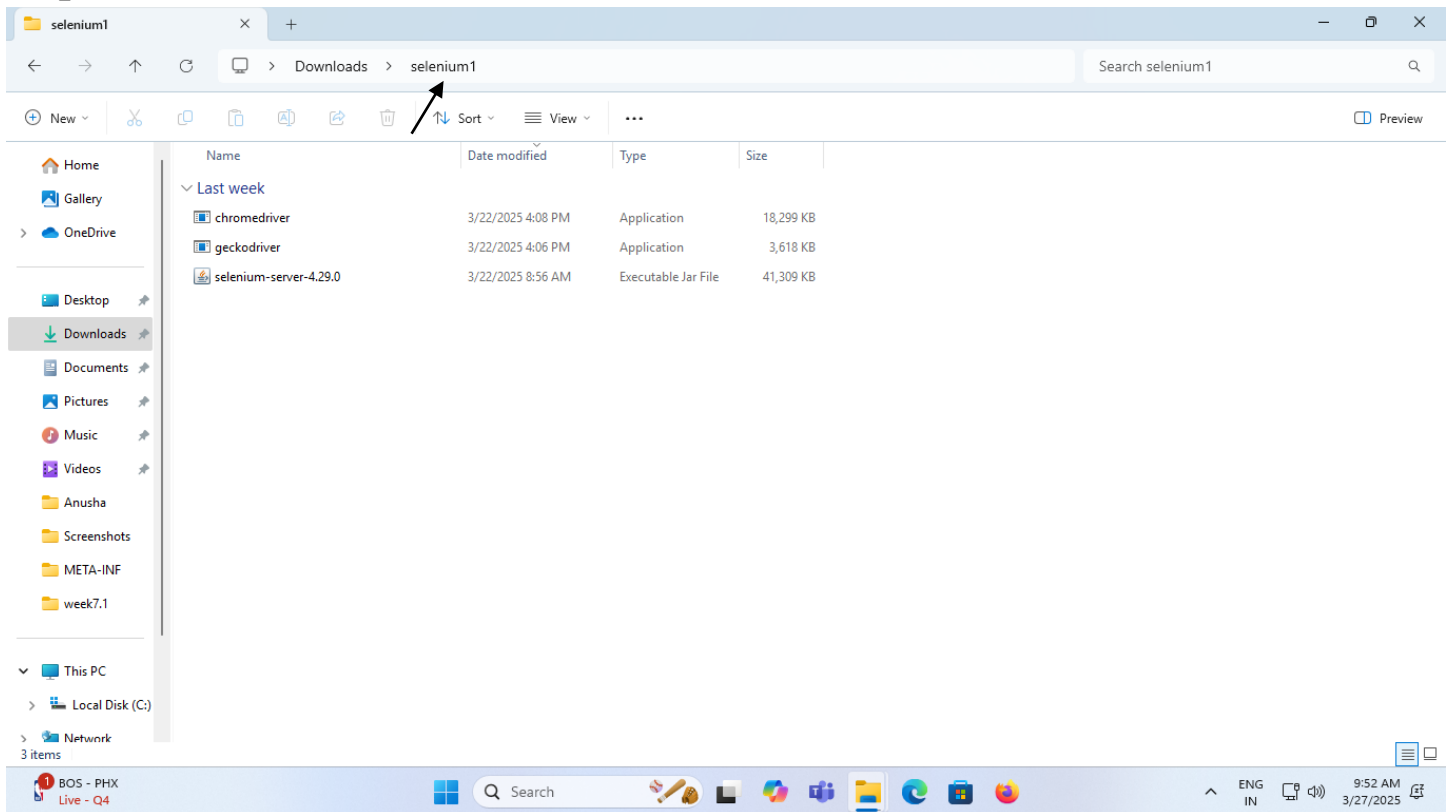
[Releases · mozilla/geckodriver](https://github.com/mozilla/geckodriver/releases)

25 Feb 2025 — **WebDriver for Firefox**. Contribute to mozilla/geckodriver development on GitHub.

▼ Assets

13

geckodriver-v0.36.0-linux-aarch64.tar.gz	2.07 MB	last month
geckodriver-v0.36.0-linux-aarch64.tar.gz.asc	833 Bytes	last month
geckodriver-v0.36.0-linux32.tar.gz	2.16 MB	last month
geckodriver-v0.36.0-linux32.tar.gz.asc	833 Bytes	last month
geckodriver-v0.36.0-linux64.tar.gz	2.2 MB	last month
geckodriver-v0.36.0-linux64.tar.gz.asc	833 Bytes	last month
geckodriver-v0.36.0-macos-aarch64.tar.gz	1.97 MB	last month
geckodriver-v0.36.0-macos.tar.gz	2.19 MB	last month
geckodriver-v0.36.0-win-aarch64.zip	1.62 MB	last month
geckodriver-v0.36.0-win32.zip	1.7 MB	last month
Source code (zip)		last month
Source code (tar.gz)		last month
Show all 13 assets		

Step 4:-Put all drivers one folder

Step 5:-Run hub use following command and node also separate cmd. In cmd change directory where available all drivers

1. `java -jar selenium-server-4.29.0.jar hub`
2. `java -jar selenium-server-4.29.0.jar node --detect-drivers true`

Also run chrome using ip address path it will generate after running hub and verify

The screenshot shows a Windows File Explorer window with a folder named 'selenium1' in the Downloads directory. A Command Prompt window is open, displaying the execution of the Selenium Hub command. The output shows the Hub starting on port 4444. A white arrow points from the URL 'http://172.23.144.1:4444' in the Command Prompt output to the browser's address bar. The browser window shows the Selenium Grid interface at '172.23.144.1:4444/ui/'. The page displays 'The Grid has no registered Nodes yet.' and 'Queue size: 0'. A white arrow points from the browser's address bar to the Selenium Grid URL.

```
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>cd downloads/selenium1

C:\Users\student\Downloads\selenium1>java -jar C:\Users\student\Downloads\selenium1\selenium-server-4.29.0.jar hub
09:52:54.944 INFO [LoggingOptions.configureLogEncoding] - Using the system default encoding
09:52:54.960 INFO [OpenTelemetryTracer.createTracer] - Using OpenTelemetry for tracing
09:52:55.167 INFO [BoundZmqEventBus.<init>] - XPUB binding to [binding to tcp://*:4442, advertising as tcp://172.23.144.1:4442], XSUB binding to [binding to tcp://*:4443, advertising as tcp://172.23.144.1:4443]
09:52:55.220 INFO [UnboundZmqEventBus.<init>] - Connecting to tcp://172.23.144.1:4442 and tcp://172.23.144.1:4443
09:52:55.245 INFO [UnboundZmqEventBus.<init>] - Sockets created
09:52:56.248 INFO [UnboundZmqEventBus.<init>] - Event bus ready
09:52:56.838 INFO [Hub.execute] - Started Selenium Hub 4.29.0 (revision 18ae989): http://172.23.144.1:4444
```

Selenium Grid
4.29.0 (revision 18ae989)

Overview
Sessions
Help

The Grid has no registered Nodes yet.
More information about Selenium Grid can be found at the [Help](#) section.

Queue size: 0

Help - All rights reserved - Software Freedom Conservancy 2025.

```
Command Prompt - java -jar x + v
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

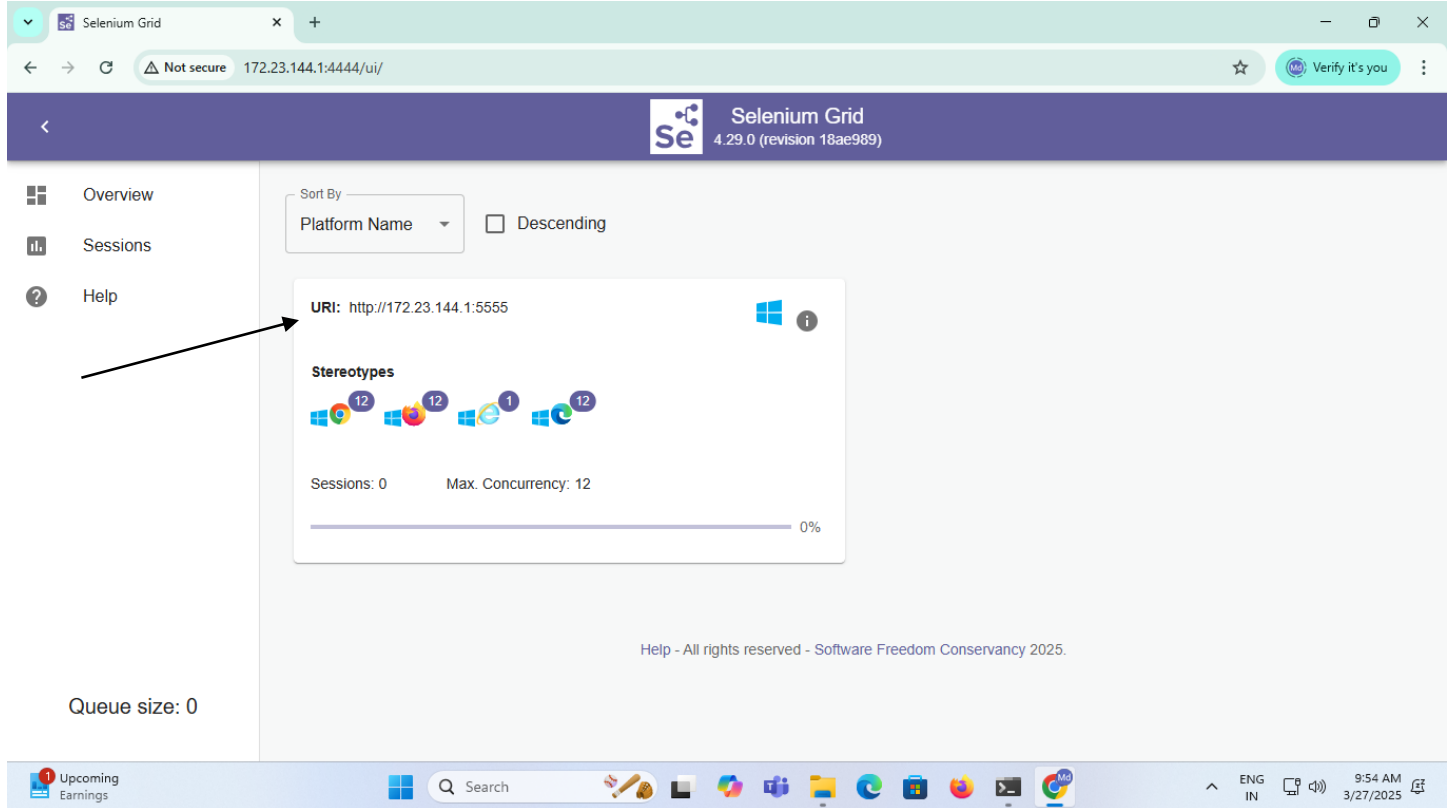
C:\Users\student>cd downloads\selenium1

C:\Users\student\Downloads\selenium1>java -jar C:\Users\student\Downloads\selenium1\selenium-server-4.29.0.jar node --detect-drivers true
09:54:10.855 INFO [LoggingOptions.configureLogEncoding] - Using the system default encoding
09:54:10.859 INFO [OpenTelemetryTracer.createTracer] - Using OpenTelemetry for tracing
09:54:11.052 INFO [UnboundZmqEventBus.<init>] - Connecting to tcp://0.0.0.0:4442 and tcp://0.0.0.0:4443
09:54:11.110 INFO [UnboundZmqEventBus.<init>] - Sockets created
09:54:12.123 INFO [UnboundZmqEventBus.<init>] - Event bus ready
09:54:12.248 INFO [NodeServer.createHandlers] - Reporting self as: http://172.23.144.1:5555
09:54:12.291 INFO [NodeOptions.getSessionFactories] - Detected 12 available processors
09:54:12.291 INFO [NodeOptions.discoverDrivers] - Looking for existing drivers on the PATH.
09:54:12.291 INFO [NodeOptions.discoverDrivers] - Add '--selenium-manager true' to the startup command to setup drivers automatically.
09:54:12.908 WARN [SeleniumManager.lambda$runCommand$1] - Unable to discover proper msedgedriver version in offline mode
09:54:13.345 WARN [SeleniumManager.lambda$runCommand$1] - Unable to discover proper IEDriverServer version in offline mode
09:54:13.367 INFO [NodeOptions.report] - Adding Internet Explorer for {"browserName": "internet explorer", "platformName": "Windows 11"} 1 times
09:54:13.367 INFO [NodeOptions.report] - Adding Firefox for {"browserName": "firefox", "platformName": "Windows 11"} 12 times
09:54:13.367 INFO [NodeOptions.report] - Adding Edge for {"browserName": "MicrosoftEdge", "platformName": "Windows 11"} 12 times
09:54:13.367 INFO [NodeOptions.report] - Adding Chrome for {"browserName": "chrome", "platformName": "Windows 11"} 12 times
09:54:13.414 INFO [Node.<init>] - Binding additional locator mechanisms: relative
09:54:13.576 INFO [NodeServer$2.start] - Starting registration process for Node http://172.23.144.1:5555
09:54:13.581 INFO [NodeServer.execute] - Started Selenium node 4.29.0 (revision 18ae989): http://172.23.144.1:5555
09:54:13.587 INFO [NodeServer$2.lambda$start$2] - Sending registration event...
09:54:13.909 INFO [NodeServer.lambda$createHandlers$2] - Node has been added
```

```
Command Prompt - java -jar x + v
Microsoft Windows [Version 10.0.26100.3476]
(c) Microsoft Corporation. All rights reserved.

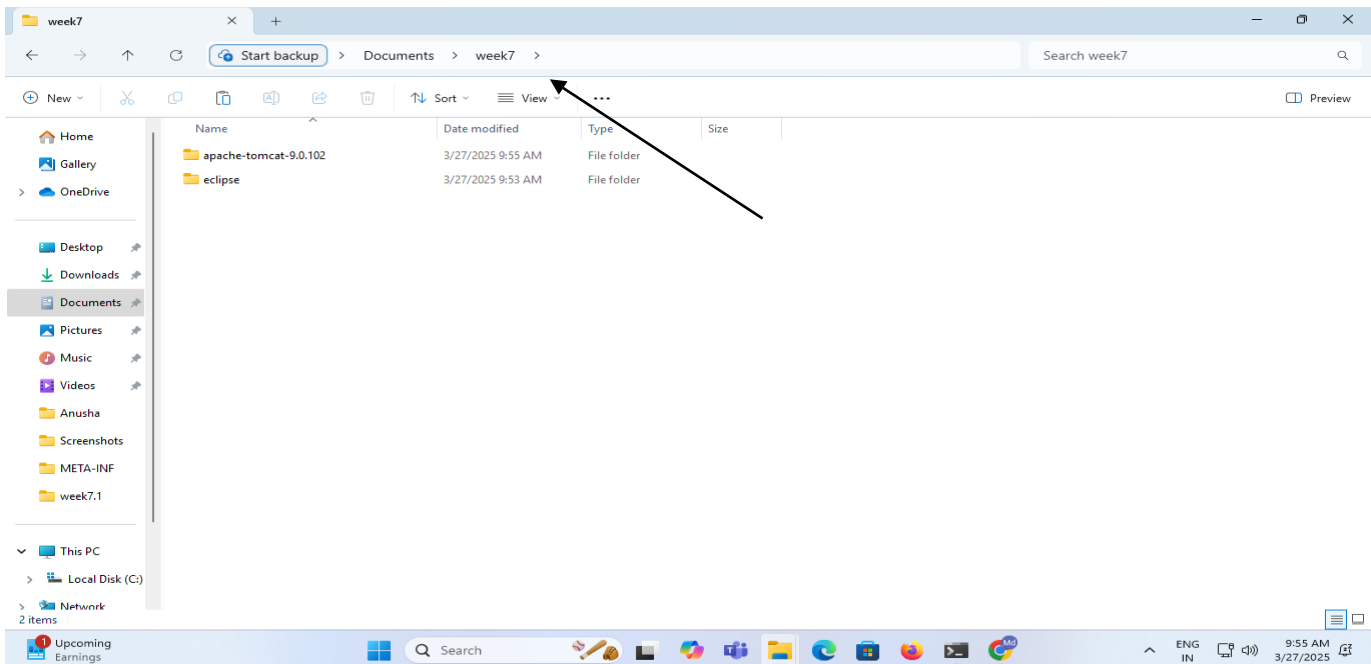
C:\Users\student>cd downloads\selenium1

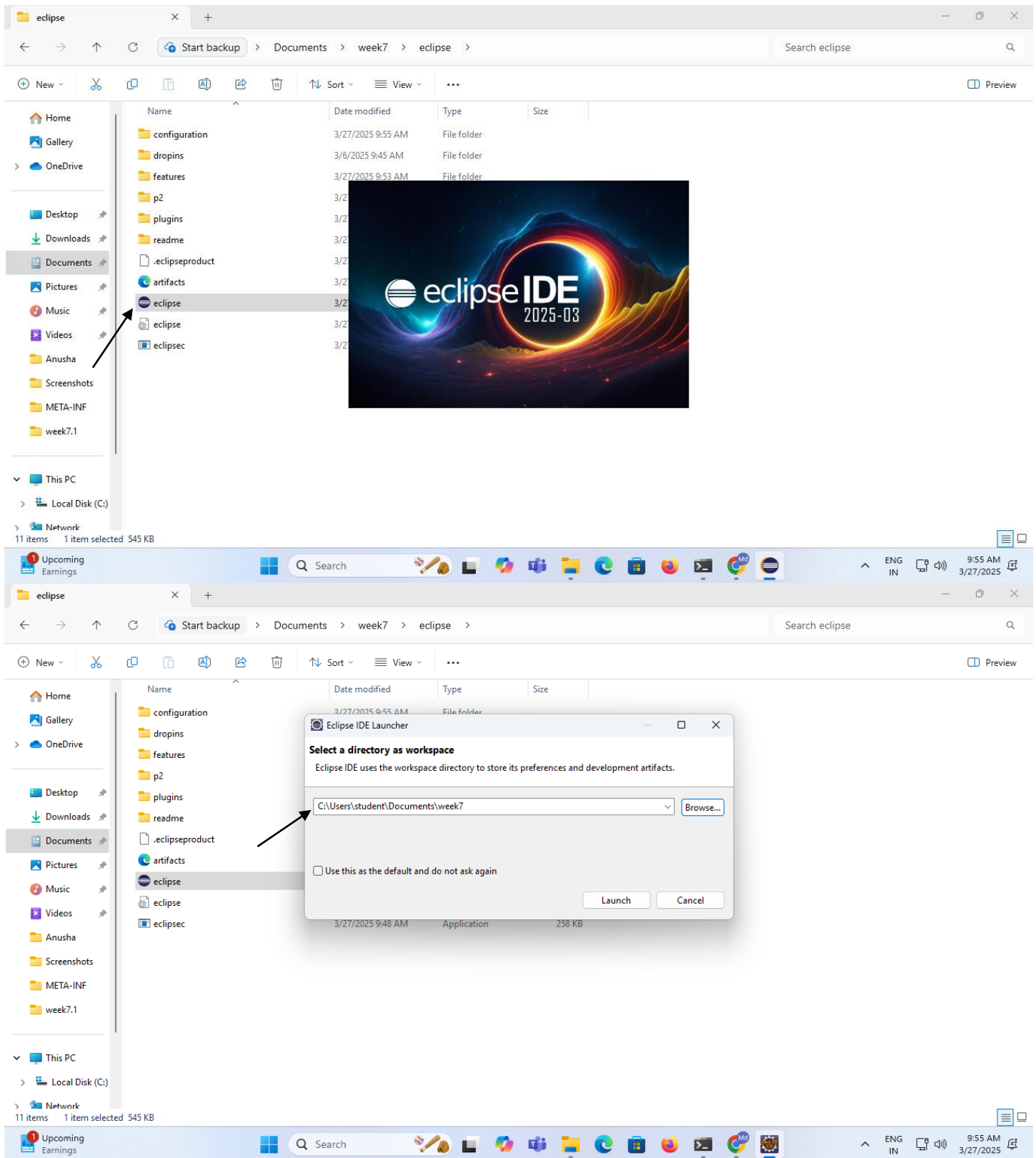
C:\Users\student\Downloads\selenium1>java -jar C:\Users\student\Downloads\selenium1\selenium-server-4.29.0.jar hub
09:52:54.944 INFO [LoggingOptions.configureLogEncoding] - Using the system default encoding
09:52:54.960 INFO [OpenTelemetryTracer.createTracer] - Using OpenTelemetry for tracing
09:52:55.167 INFO [BoundZmqEventBus.<init>] - XPUB binding to [binding to tcp://*:4442, advertising as tcp://172.23.144.1:4442], XSUB binding to [binding to tcp://*:4443, advertising as tcp://172.23.144.1:4443]
09:52:55.220 INFO [UnboundZmqEventBus.<init>] - Connecting to tcp://172.23.144.1:4442 and tcp://172.23.144.1:4443
09:52:55.245 INFO [UnboundZmqEventBus.<init>] - Sockets created
09:52:56.248 INFO [UnboundZmqEventBus.<init>] - Event bus ready
09:52:56.838 INFO [Hub.execute] - Started Selenium Hub 4.29.0 (revision 18ae989): http://172.23.144.1:4444
09:54:13.625 INFO [Node.<init>] - Binding additional locator mechanisms: relative
09:54:13.904 INFO [GridModel.setAvailability] - Switching Node 0026ba90-db0e-46f6-8f58-329ecab899af (uri: http://172.23.144.1:5555) from DOWN to UP
09:54:13.904 INFO [LocalDistributor.add] - Added node 0026ba90-db0e-46f6-8f58-329ecab899af at http://172.23.144.1:5555. Health check every 120s
```



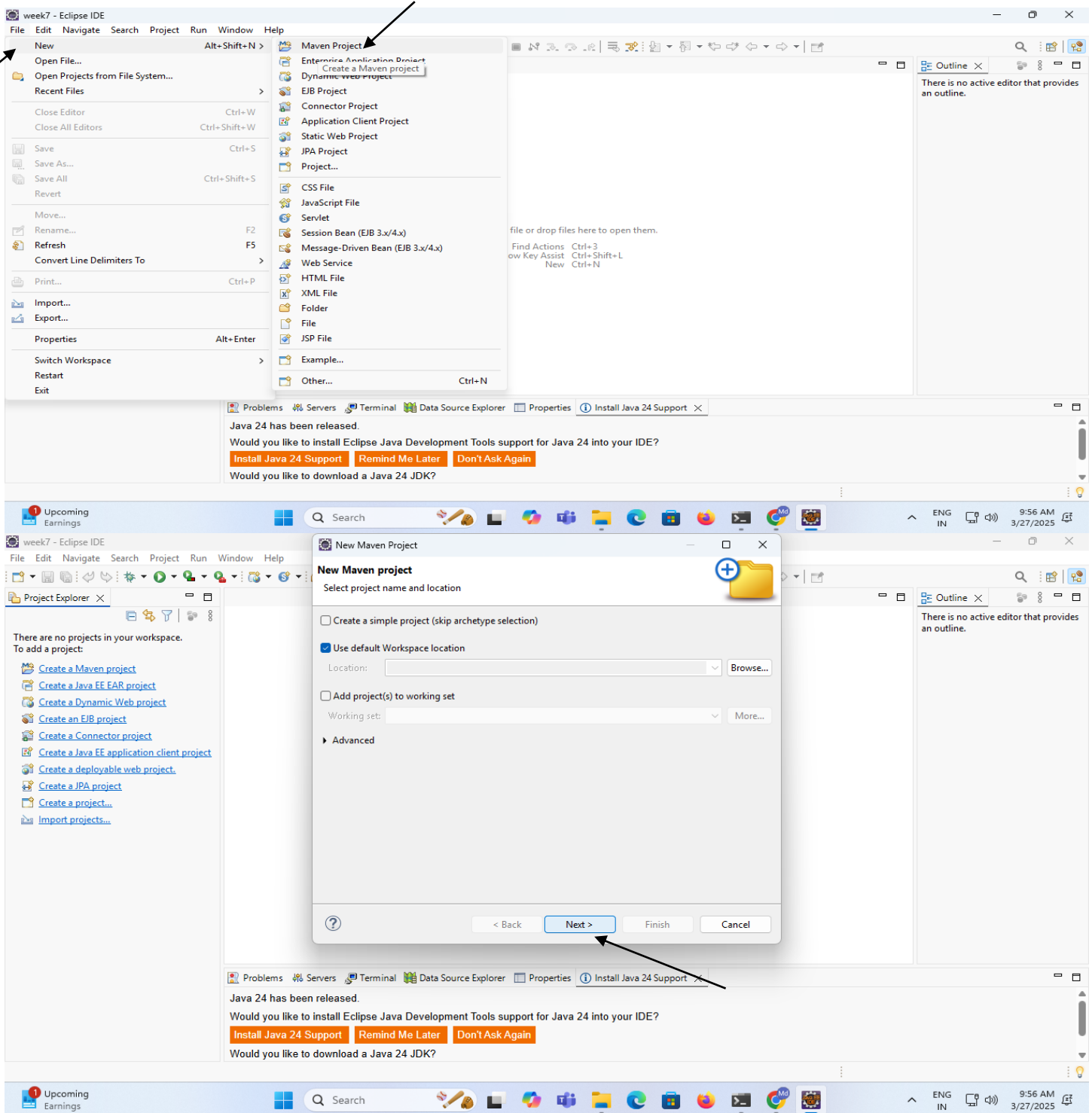
Step 6:-Open Eclipse Attach Apache Tomcat and TestNG

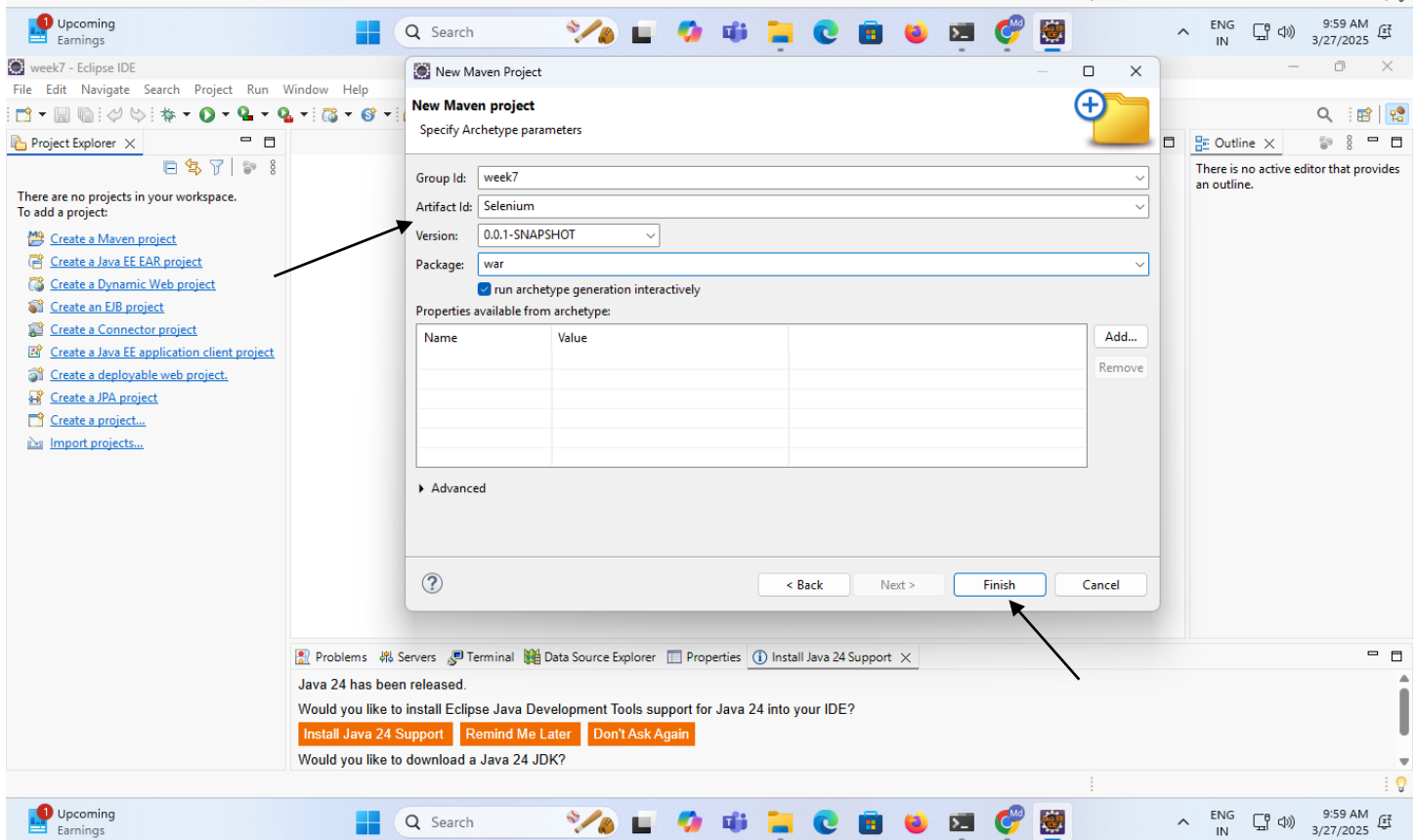
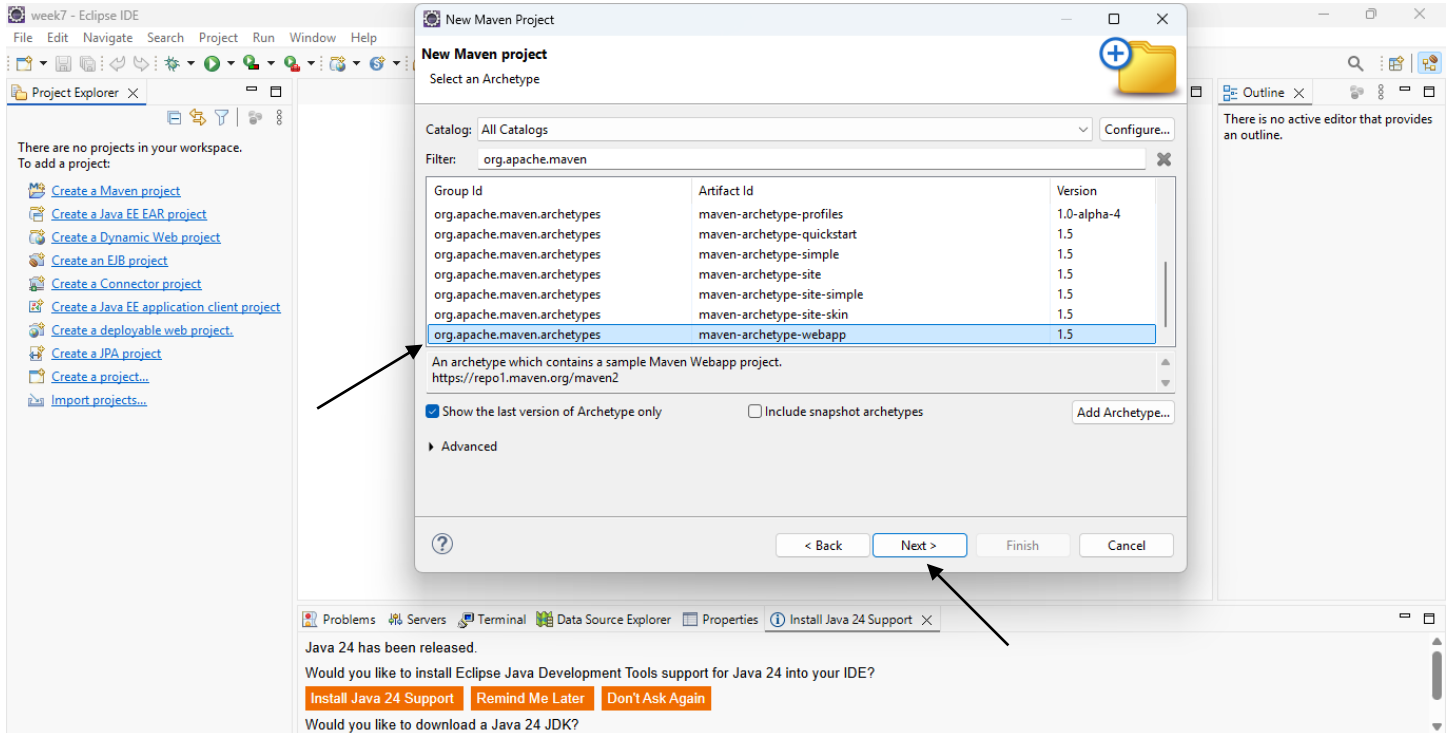
1. Store Apache and eclipse folders only single folder and lunch Eclipse same folder

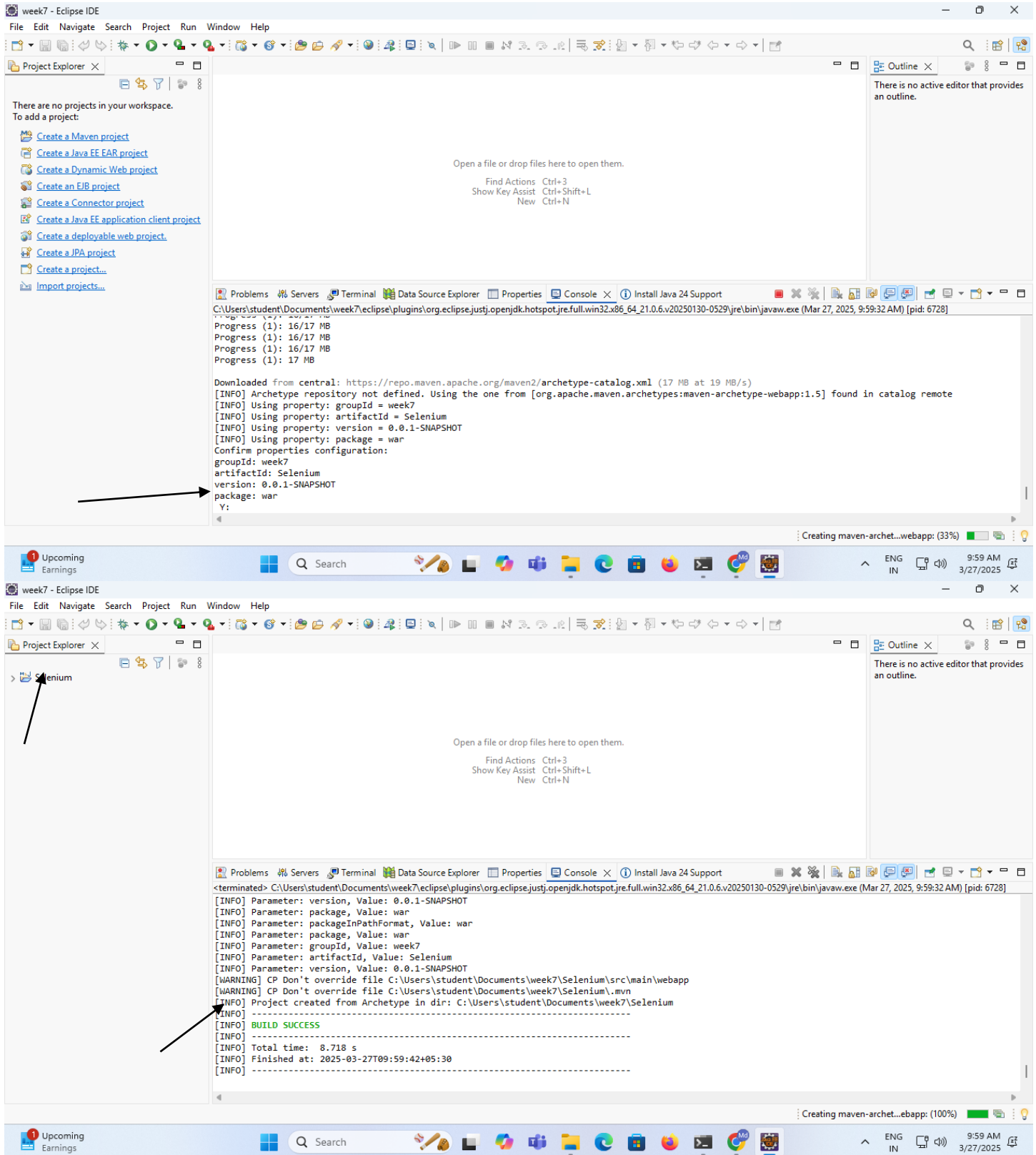




3. Create maven project goto File→New→Maven Project Select→Next→select org.apache.maven.archetypes select webapp→Enter Group Id:week7 ArtifactId:selenium Package:war→Finish. Enter Y you get Build Success in console.







**3. Open Pom.xml copy paste below code this all I added dependencies and plugins. Next Right click Project→Maven→Update project→ok.
(Note:If you not update every changes it will not save.)**

```

<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>DevOps</groupId>
  <artifactId>MyDevPipeline</artifactId>
  <packaging>war</packaging>
  <version>0.0.1-SNAPSHOT</version>
  <name>MyDevPipeline Maven Webapp</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  <!--

```

```

https://mvnrepository.com/artifact/javax.servlet/javax.servlet-api -->
  <dependency>
    <groupId>javax.servlet</groupId>
    <artifactId>javax.servlet-api</artifactId>
    <version>3.1.0</version>
    <scope>provided</scope>
  </dependency>
  <!--

```

```

https://mvnrepository.com/artifact/org.seleniumhq.selenium/seleniu
m-api -->
  <dependency>

```



```

        <groupId>org.seleniumhq.selenium</groupId>
        <artifactId>selenium-server</artifactId>
        <version>3.141.59</version>
    </dependency>
    <!-- https://mvnrepository.com/artifact/org.testng/testng -->
    <dependency>
        <groupId>org.testng</groupId>
        <artifactId>testng</artifactId>
        <version>6.14.3</version>    <!-- Use the latest version
available(7.11.0) -->
        <scope>test</scope>
    </dependency>

    <dependency>
        <groupId>io.github.bonigarcia</groupId>
        <artifactId>webdrivermanager</artifactId>
        <version>5.3.0</version>
    </dependency>

</dependencies>
<build>
    <finalName>MyDevPipeline</finalName>
    <pluginManagement>
        <plugins>
            <plugin>

                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-clean-plugin</artifactId>
                <version>3.4.0</version>
            </plugin>
            <plugin>

```

```
<groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-resources-
plugin</artifactId>
    <version>3.3.1</version>
</plugin>
<plugin>
```

```
<groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-compiler-
plugin</artifactId>
    <version>3.13.0</version>
    <configuration>
        <source>17.0.12</source>
        <target>17.0.12</target>
    </configuration>
</plugin>
<plugin>
```

```
<groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-surefire-
plugin</artifactId>
    <version>3.3.0</version>
</plugin>
<plugin>
```

```
<groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-war-plugin</artifactId>
    <version>3.4.0</version>
</plugin>
<plugin>
```

```

<groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-install-plugin</artifactId>
  <version>3.1.2</version>
</plugin>
<plugin>

```

```

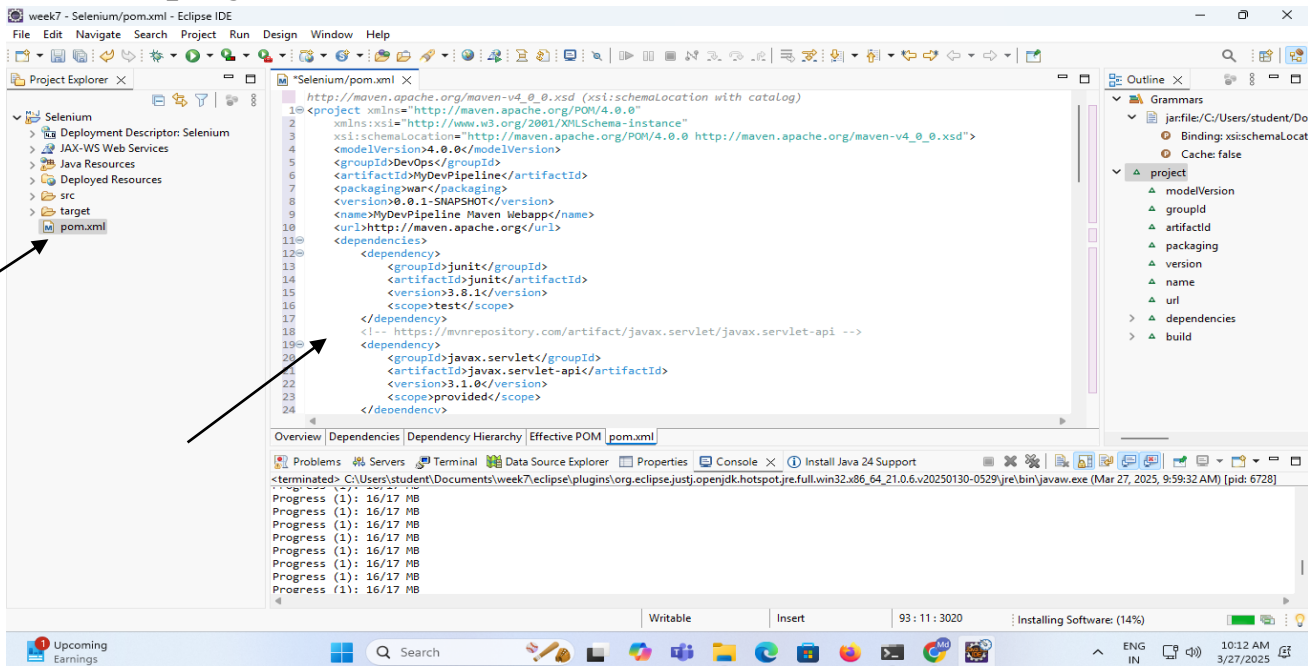
<groupId>org.apache.maven.plugins</groupId>
  <artifactId>maven-deploy-plugin</artifactId>
  <version>3.1.2</version>
</plugin>
</plugins>
</pluginManagement>

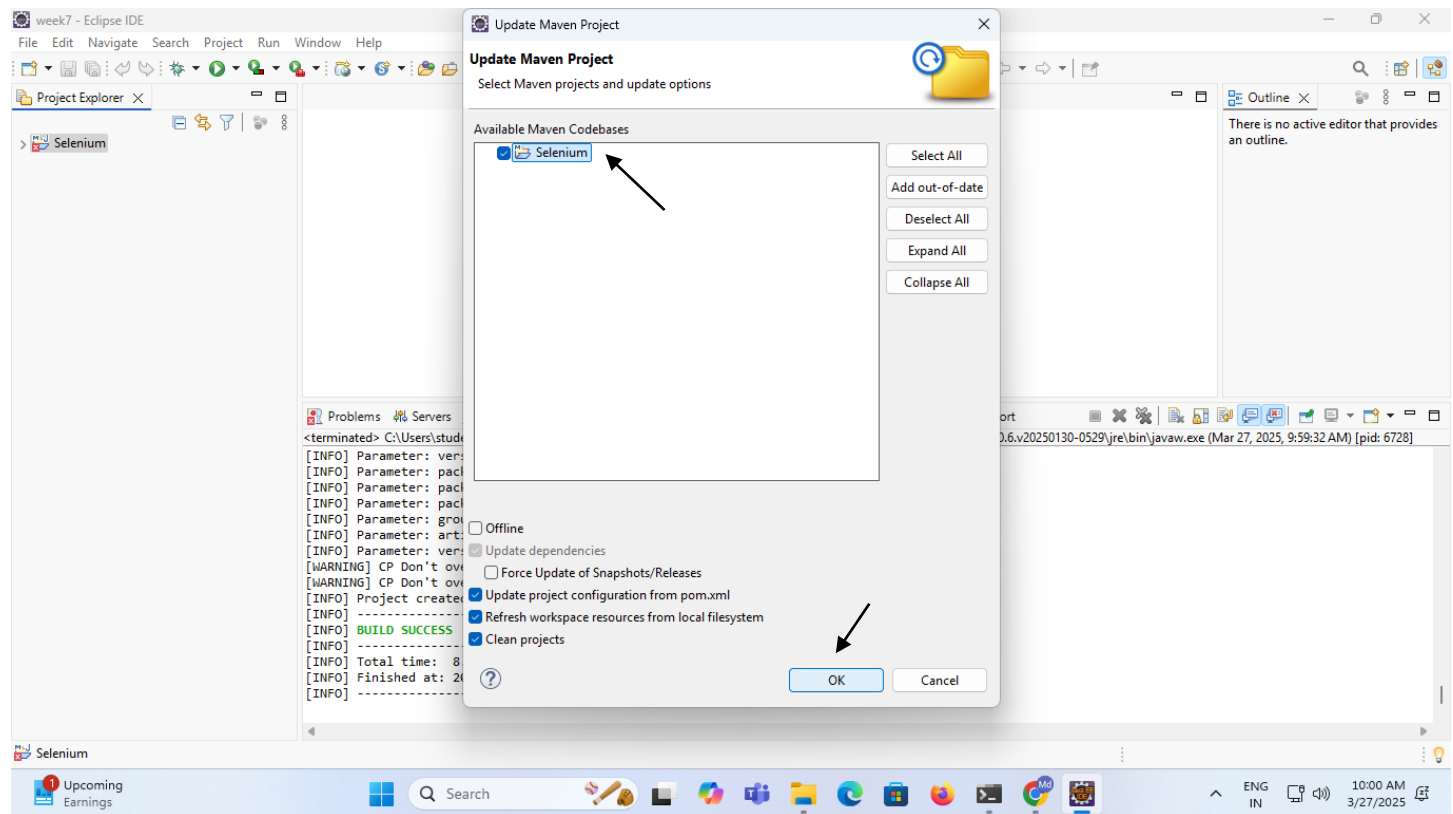
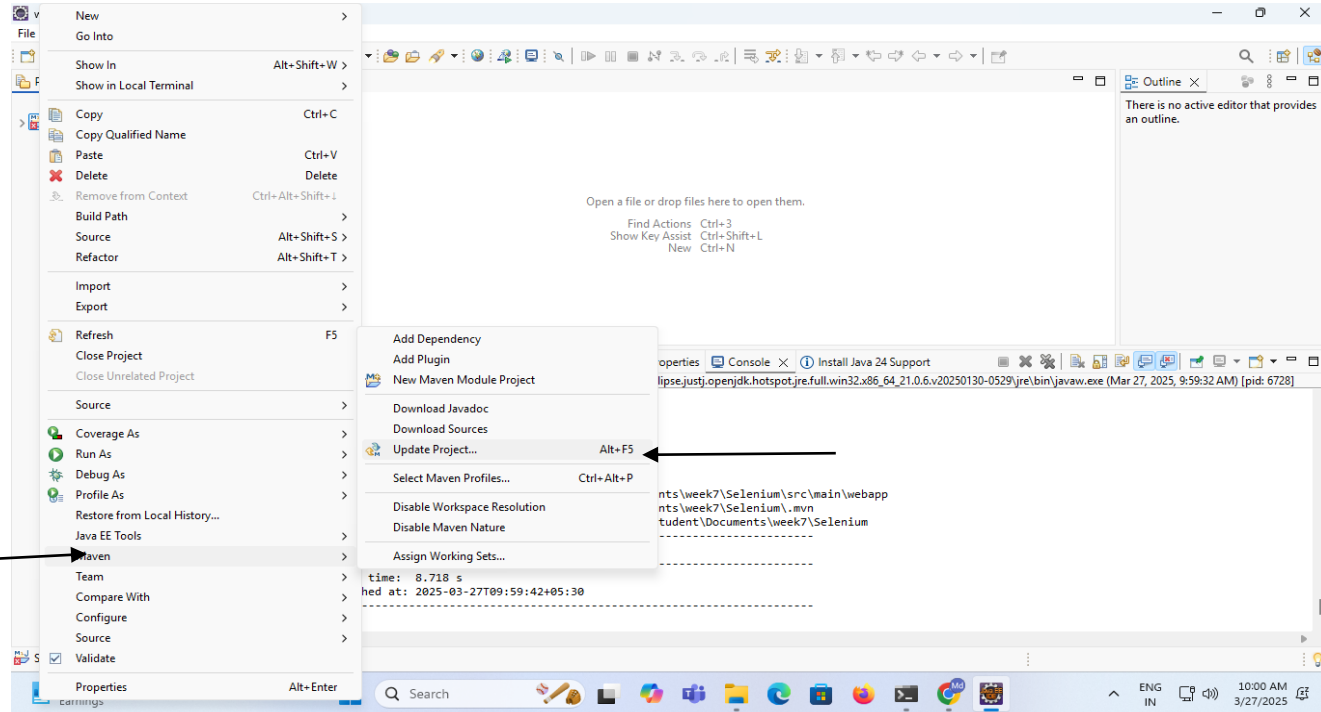
```

```

</build>
</project>

```





4.Adding Apache tomcat server

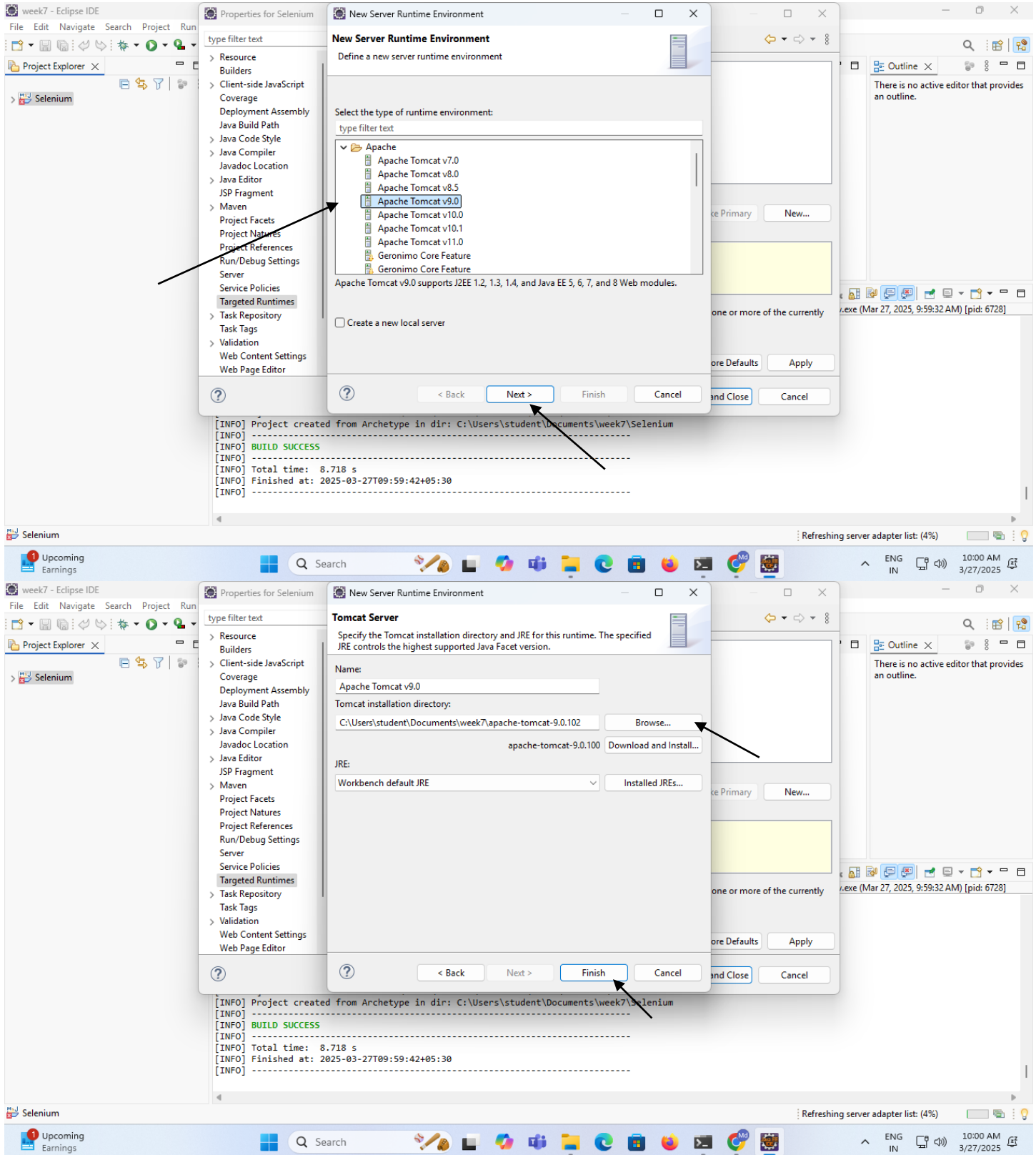
Go to Project→Properties→Maven→Targeted Runtimes→new→select apache tomcat 9→next→browse apache folder address→finish→click check box apache 9 and apply→apply and close.

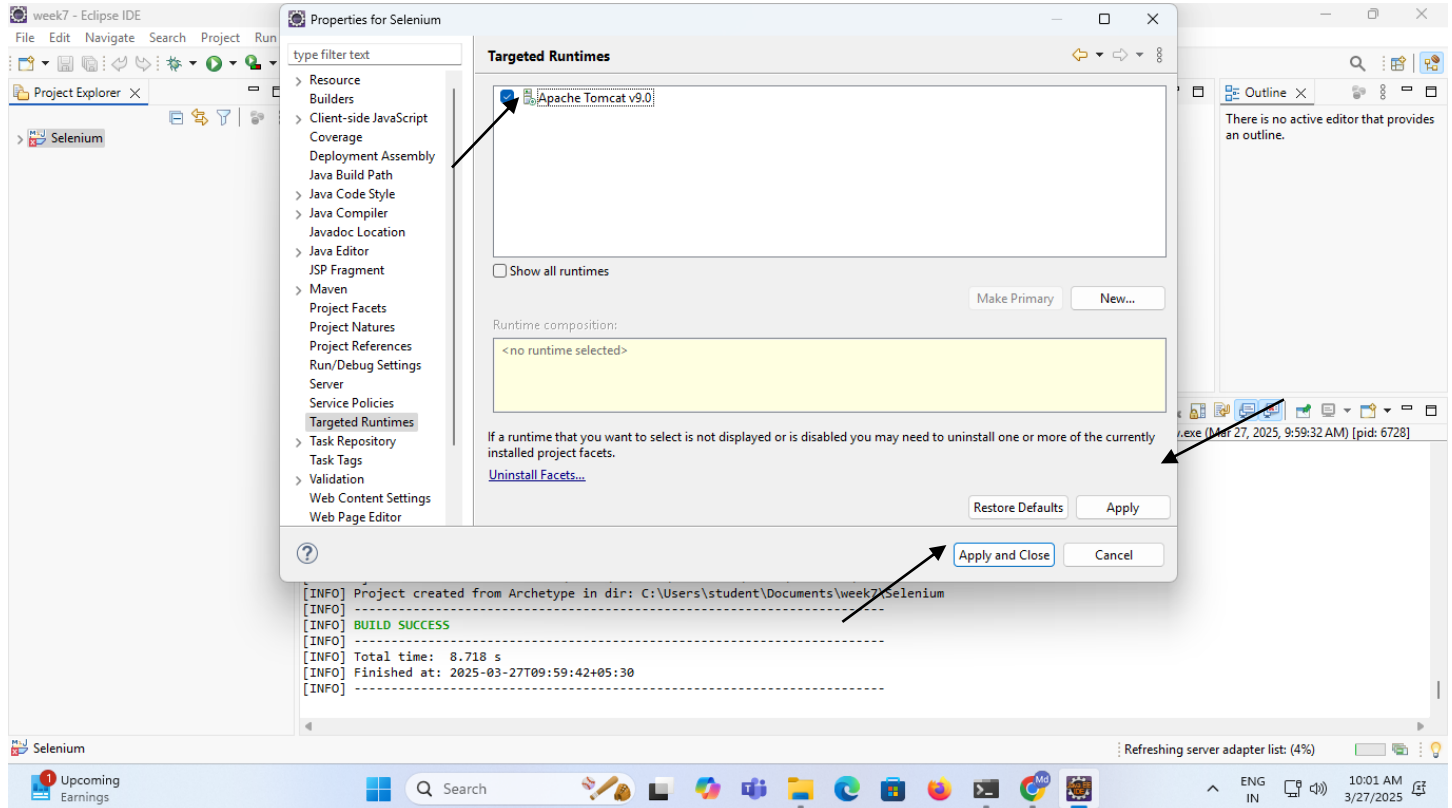
(Note:-because running src→index.js file running purpose if you not adding it will show errors.)

The screenshot displays the Eclipse IDE interface with the Selenium project selected in the Project Explorer. The 'Project' menu is open, and the 'Properties' option is highlighted. The console output shows the build process for the Selenium project, including parameters like version (0.0.1-SNAPSHOT), package (war), and groupId (week7). The build is successful, with a total time of 8.718 s and a finish time of 2025-03-27T09:59:42+05:30.

The 'Properties for Selenium' dialog is open, showing the 'Targeted Runtimes' tab. The 'Runtime composition' section indicates that no runtime is selected. The 'New...' button is highlighted, indicating the next step in the configuration process.

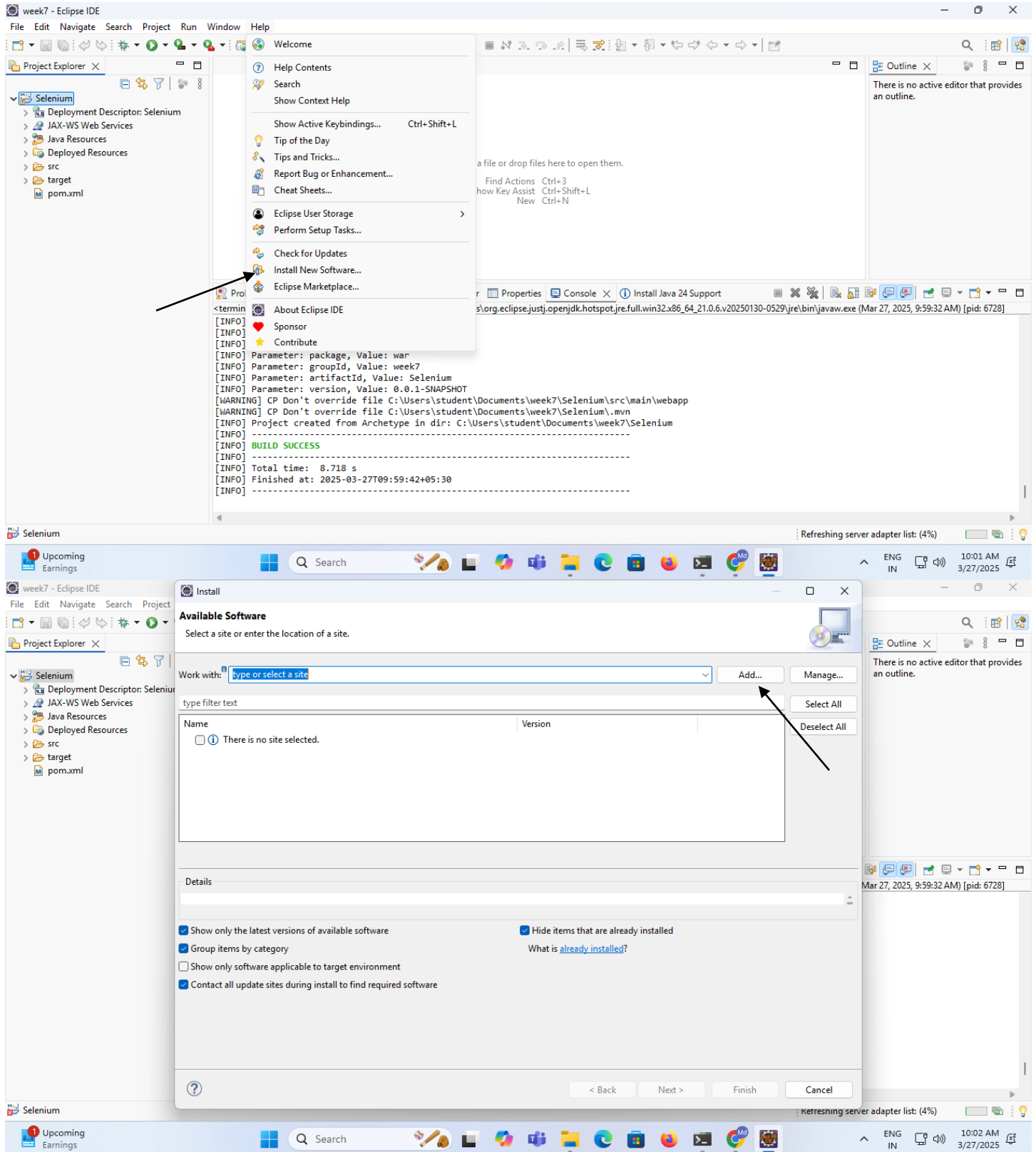
```
<terminated> C:\Users\student\Documents\week7\eclipse\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.6.v20250130-0529\jre\bin\javaw.exe (Mar 27, 2025, 9:59:32 AM) [pid: 6728]
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[INFO] Parameter: package, Value: war
[INFO] Parameter: packageInPathFormat, Value: war
[INFO] Parameter: package, Value: war
[INFO] Parameter: groupId, Value: week7
[INFO] Parameter: artifactId, Value: Selenium
[INFO] Parameter: version, Value: 0.0.1-SNAPSHOT
[WARNING] CP Don't override file C:\Users\student\Documents\week7\Selenium\src\main\webapp
[WARNING] CP Don't override file C:\Users\student\Documents\week7\Selenium\mvn
[INFO] Project created from Archetype in dir: C:\Users\student\Documents\week7\Selenium
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 8.718 s
[INFO] Finished at: 2025-03-27T09:59:42+05:30
[INFO] -----
```

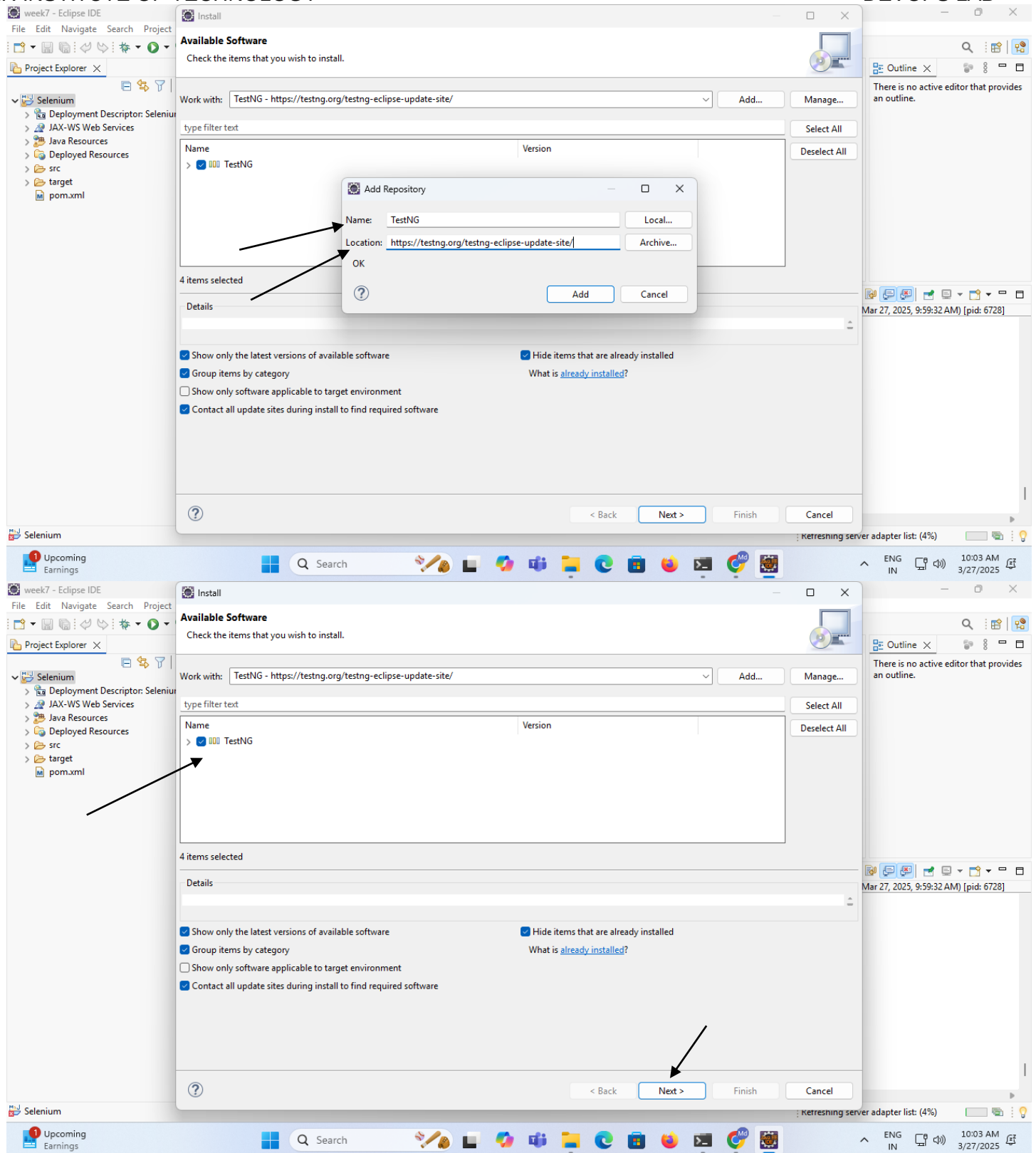




5. Adding TestNG in Eclipse.

Go to help→Install New Software→Add→Enter Name:TestNG,
 Location:<https://testng.org/testing-eclipse-update-site/>→Add.
 Select TestNG check box→Next→Next→I accept→Finish
 Trust Selected→Trust Selected.





The screenshot displays the Eclipse IDE interface with the 'Install' dialog box open. The 'Available Software' tab is selected, showing a list of software items. The 'TestNG' item is selected, and the 'Details' tab is also visible, showing a list of software items including Guava, TestNG, and TestNG M2E Integration. A progress bar indicates 'Calculating requirements and dependencies.' An arrow points to the 'Next >' button.

Available Software

Check the items that you wish to install.

Work with: TestNG - <https://testng.org/testng-eclipse-update-site/>

Name	Version
TestNG	

4 items selected

Details

☒ Show only the latest versions of available software ☒ Hide items that are already installed

☒ Group items by category ☐ Show only software applicable to target environment

☒ Contact all update sites during install to find required software

What is already installed?

Calculating requirements and dependencies.

< Back Next > Finish Cancel

Install Details

Your original request has been modified. See the details.

Name	Version	Id
Guava: Google Core Libraries for Java	32.1.2-jre	com.google.guava
TestNG	7.11.0.202503091801	org.testng.eclipse.feature.group
TestNG M2E (Maven) Integration (Optional)	7.11.0.202503091801	org.testng.eclipse.maven.feature.f...
TestNG P2 Feature	7.11.0.202503091654	org.testng.p2.feature.feature.group

Size: Unknown

Details

Your original request has been modified.
"Guava: Google Core Libraries for Java" will be ignored because a newer version is already installed.

< Back Next > Finish Cancel

The screenshot shows the Eclipse IDE interface with the 'Review Licenses' dialog box open. The 'Apache License' is selected in the 'Licenses' list. The 'License text' pane displays the Apache License Version 2.0, January 2004. The 'I accept the terms of the license agreement' radio button is selected. The 'Finish' button is highlighted with a red arrow.

Review Licenses
Licenses must be reviewed and accepted before the software can be installed.

Licenses:

- Apache License

License text:

Apache License
Version 2.0, January 2004
<http://www.apache.org/licenses/>
TERMS AND CONDITIONS FOR USE, REPRODUCTION, AND DISTRIBUTION

1. Definitions.

"License" shall mean the terms and conditions for use, reproduction, and distribution as defined by Sections 1 through 9 of this document.

"Licensor" shall mean the copyright owner or entity authorized by the copyright owner that is granting the License.

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< Back Next > Finish Cancel

Trust Authorities
Do you trust content originating from the following authorities? Installing content involves performing actions that alter the installation's configuration and may potentially be used for malicious purposes.

Authority / Update Site	Units	Secured
☒ https://download.oracle.com	1	☒
☒ https://testng.org	16	☒

☒ Remember selected authorities ☐ Always trust all authorities Select All Deselect All Expand All Collapse All

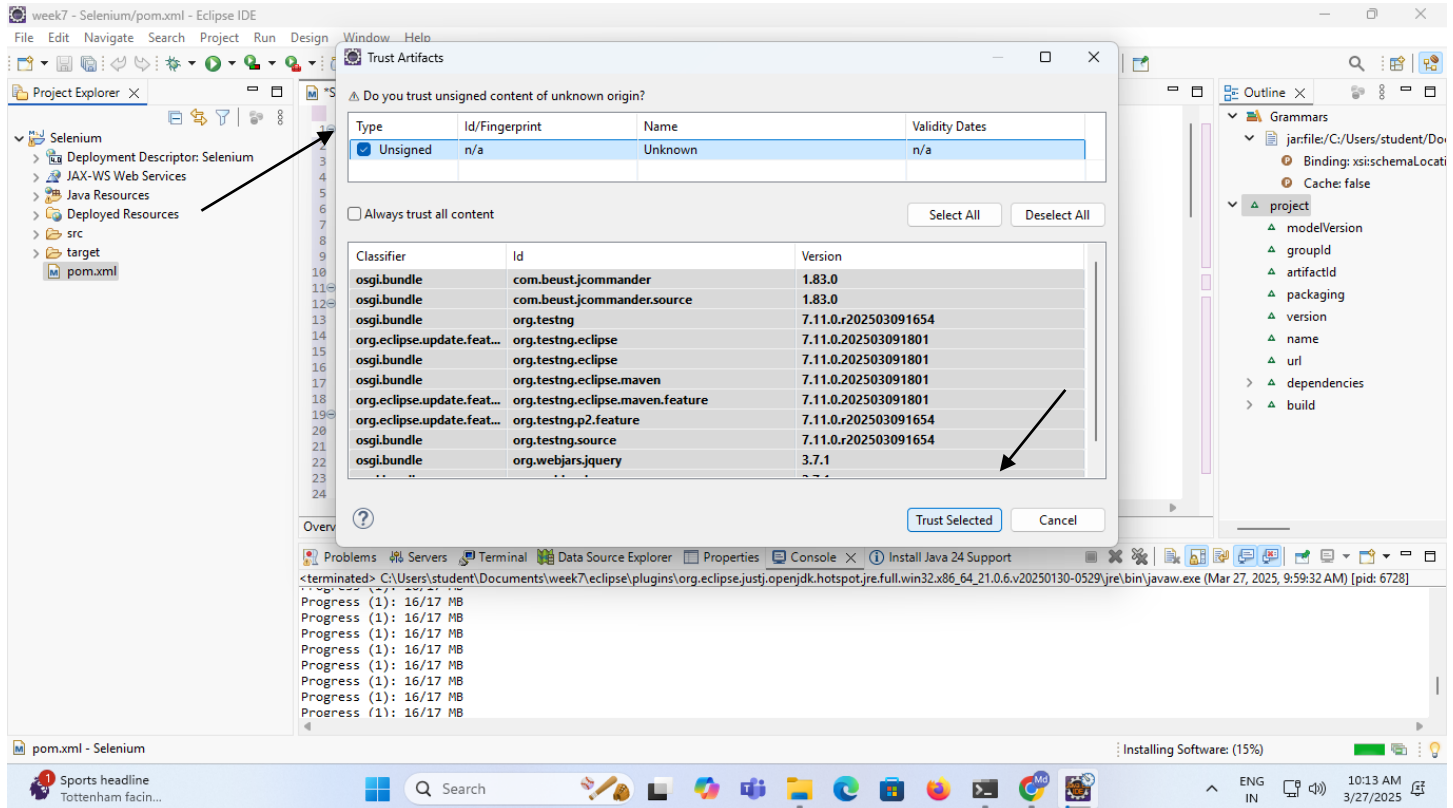
download.oracle.com; Oracle Corporation
DigiCert TLS RSA SHA256 2020 CA1; DigiCert Inc

Details... Export...

Id	Version
javaewah	1.1.6.v20160919-1400
com.beust.jcommander	1.83.0
com.beust.jcommander.source	1.83.0
org.testng	7.11.0.202503091654
org.testng.eclipse	7.11.0.202503091801
org.testng.eclipse.feature.group	7.11.0.202503091801
org.testng.eclipse.feature.jar	7.11.0.202503091801
org.testng.eclipse.maven	7.11.0.202503091801
org.testng.eclipse.maven.feature.feature.g...	7.11.0.202503091801
org.testng.eclipse.maven.feature.feature.jar	7.11.0.202503091801

Details...

Trust Selected Cancel



6.Create Folder inside src →test inside java.java folder select build path →current folder.Right click that folder→new→others→class→next→give package name and class name→ok

Java file created copy paste below code.change hub address must and should.

Same page right click→run→TestNG →Finally its run test passes.

SeleniumGridDemo.java

package com.testautomation.grid;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.chrome.ChromeDriver;

import org.openqa.selenium.firefox.FirefoxDriver;

import org.openqa.selenium.remote.DesiredCapabilities;

import org.openqa.selenium.remote.RemoteWebDriver;

import org.testng.annotations.Test;

import java.net.URL;

public class SeleniumGridDemo {

@Test

public void runTestOnChrome() throws Exception {

DesiredCapabilities capabilities = DesiredCapabilities.chrome();

WebDriver driver = new RemoteWebDriver(new URL("http://172.16.226.24:4444/wd/hub"), capabilities);

driver.get("https://www.google.com");

System.out.println("Page title on Chrome: " + driver.getTitle());

driver.quit();

}

@Test

public void runTestOnFirefox() throws Exception {

DesiredCapabilities capabilities = DesiredCapabilities.firefox();

WebDriver driver = new RemoteWebDriver(new URL("http://172.16.226.24:4444/wd/hub"), capabilities);

driver.get("https://www.google.com");

System.out.println("Page title on Firefox: " + driver.getTitle());

driver.quit();

}

}



The screenshot shows the Eclipse IDE interface. A 'Select a wizard' dialog box is open, displaying a list of wizards. The 'Class' wizard is selected. The 'Next >' button is highlighted. The background shows the 'Selenium/pom.xml' file and the 'SeleniumTesting.java' file.

Wizards:

- Class
- Interface
- Java Project
- Java Project from Existing Ant Buildfile
- Plug-in Project
- General
- Connection Profiles
- Eclipse Modeling Framework
- EJB
- Git
- Gradle
- Java

Selenium/pom.xml

```
<?xml version="1.0" encoding="UTF-8" ?>
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>DevOps</groupId>
<artifactId>MyDevOps</artifactId>
<url>http://www.example.com</url>
<dependencies>
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.0.0</version>
<scope>test</scope>
</dependency>
</dependencies>
</project>
```

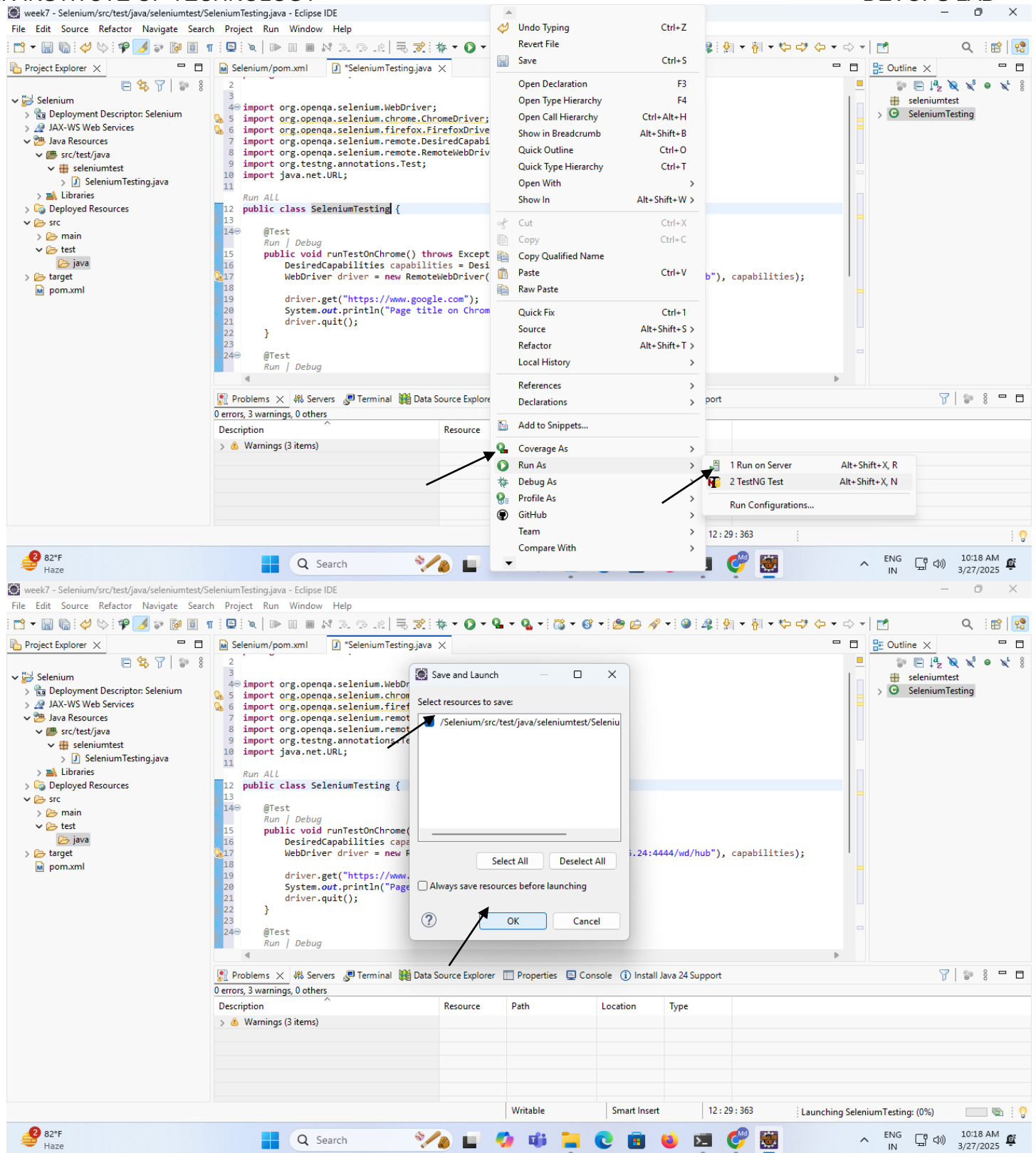
SeleniumTesting.java

```
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.remote.DesiredCapabilities;
import org.openqa.selenium.remote.RemoteWebDriver;
import org.testng.annotations.Test;
import java.net.URL;

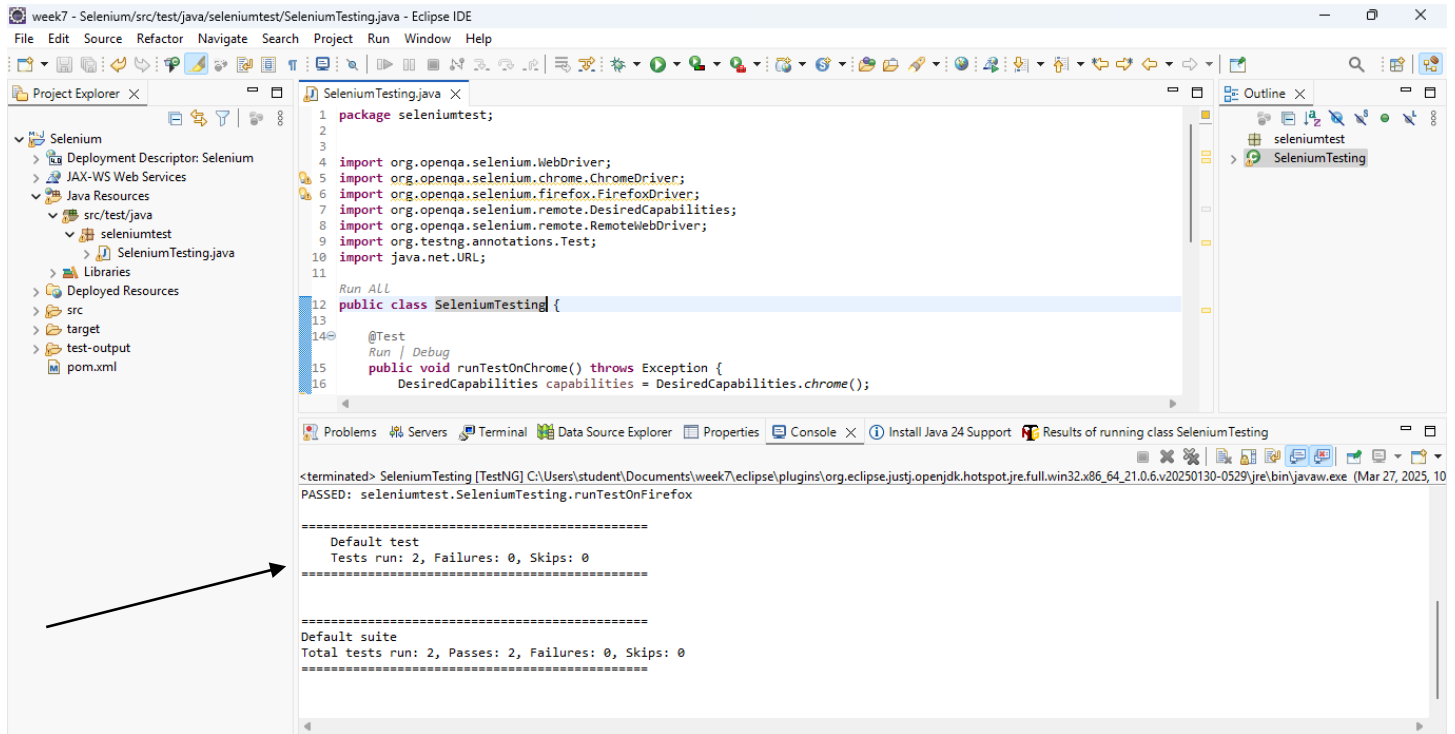
public class SeleniumTesting {

    @Test
    public void runTestOnChrome() throws Exception {
        DesiredCapabilities capabilities = DesiredCapabilities.chrome();
        WebDriver driver = new RemoteWebDriver(new URL("http://172.16.226.24:4444/wd/hub"), capabilities);

        driver.get("https://www.google.com");
        System.out.println("Page title on Chrome: " + driver.getTitle());
        driver.quit();
    }
}
```



Output:-



The screenshot shows the Eclipse IDE with the SeleniumTesting.java file open. The code defines a package, imports Selenium WebDriver and TestNG classes, and contains a test class SeleniumTesting with a test method runTestOnChrome. The console output shows the test results for SeleniumTesting.

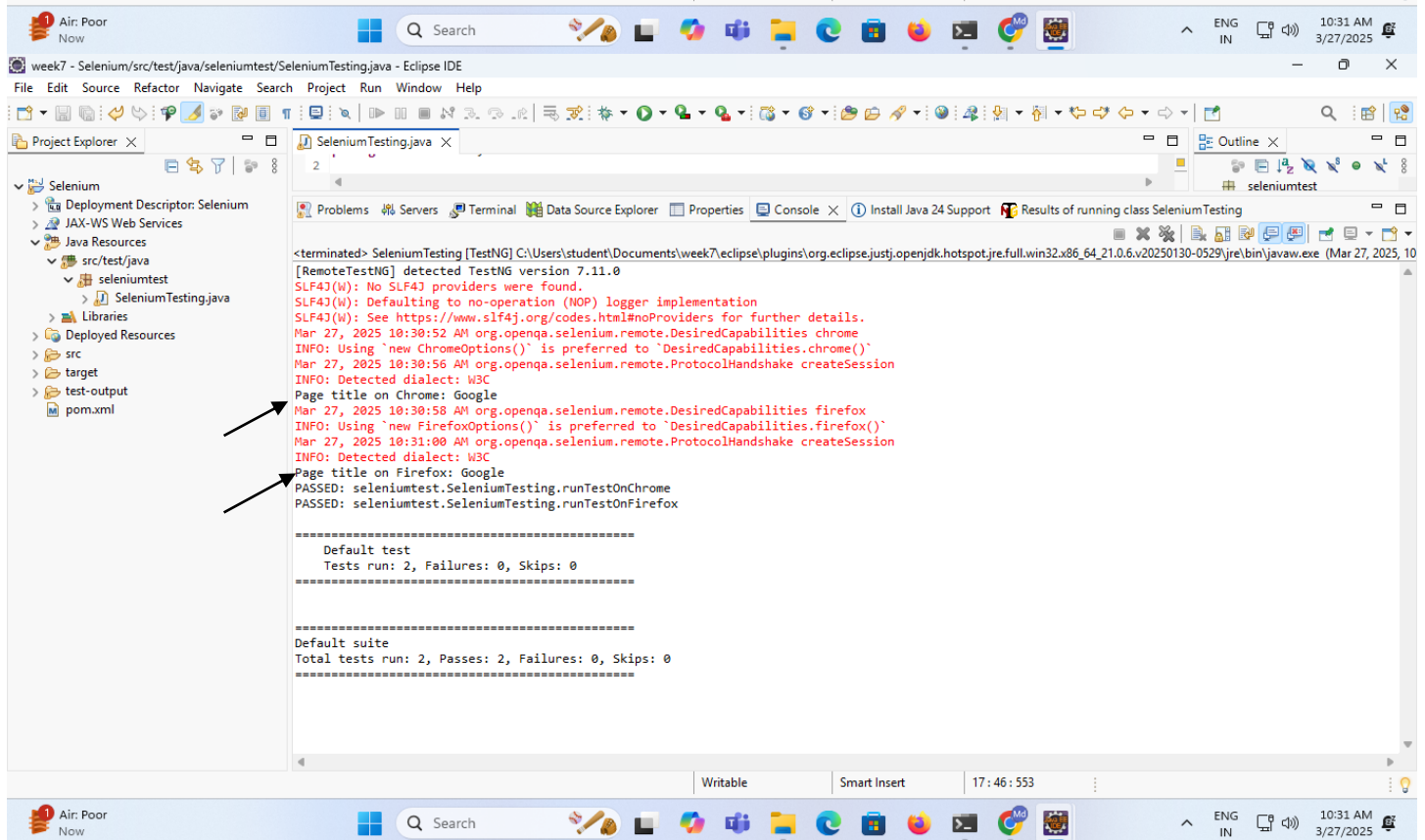
```
1 package seleniumtest;
2
3
4 import org.openqa.selenium.WebDriver;
5 import org.openqa.selenium.chrome.ChromeDriver;
6 import org.openqa.selenium.firefox.FirefoxDriver;
7 import org.openqa.selenium.remote.DesiredCapabilities;
8 import org.openqa.selenium.remote.RemoteWebDriver;
9 import org.testng.annotations.Test;
10 import java.net.URL;
11
12 public class SeleniumTesting {
13
14     @Test
15     public void runTestOnChrome() throws Exception {
16         DesiredCapabilities capabilities = DesiredCapabilities.chrome();
17     }
18 }
```

Console Output:

```
<terminated> SeleniumTesting [TestNG] C:\Users\student\Documents\week7\eclipse\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.6.v20250130-0529\jre\bin\javaw.exe (Mar 27, 2025, 10:31:30 AM)
PASSED: seleniumtest.SeleniumTesting.runTestOnFirefox

=====
Default test
Tests run: 2, Failures: 0, Skips: 0
=====

Default suite
Total tests run: 2, Passes: 2, Failures: 0, Skips: 0
=====
```



The screenshot shows the Eclipse IDE with the SeleniumTesting.java file open. The code is the same as in the previous screenshot. The console output shows detailed logs for the test execution, including TestNG version, SLF4J warnings, and the test results for SeleniumTesting.

```
<terminated> SeleniumTesting [TestNG] C:\Users\student\Documents\week7\eclipse\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64_21.0.6.v20250130-0529\jre\bin\javaw.exe (Mar 27, 2025, 10:31:30 AM)
[RemoteTestNG] detected TestNG version 7.11.0
SLF4J(W): No SLF4J providers were found.
SLF4J(W): Defaulting to no-operation (NOP) logger implementation
SLF4J(W): See https://www.slf4j.org/codes.html#noProviders for further details.
Mar 27, 2025 10:30:52 AM org.openqa.selenium.remote.DesiredCapabilities chrome
INFO: Using 'new ChromeOptions()' is preferred to 'DesiredCapabilities.chrome()'
Mar 27, 2025 10:30:56 AM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
Page title on Chrome: Google
Mar 27, 2025 10:30:58 AM org.openqa.selenium.remote.DesiredCapabilities firefox
INFO: Using 'new FirefoxOptions()' is preferred to 'DesiredCapabilities.firefox()'
Mar 27, 2025 10:31:00 AM org.openqa.selenium.remote.ProtocolHandshake createSession
INFO: Detected dialect: W3C
Page title on Firefox: Google
PASSED: seleniumtest.SeleniumTesting.runTestOnChrome
PASSED: seleniumtest.SeleniumTesting.runTestOnFirefox

=====
Default test
Tests run: 2, Failures: 0, Skips: 0
=====

Default suite
Total tests run: 2, Passes: 2, Failures: 0, Skips: 0
=====
```

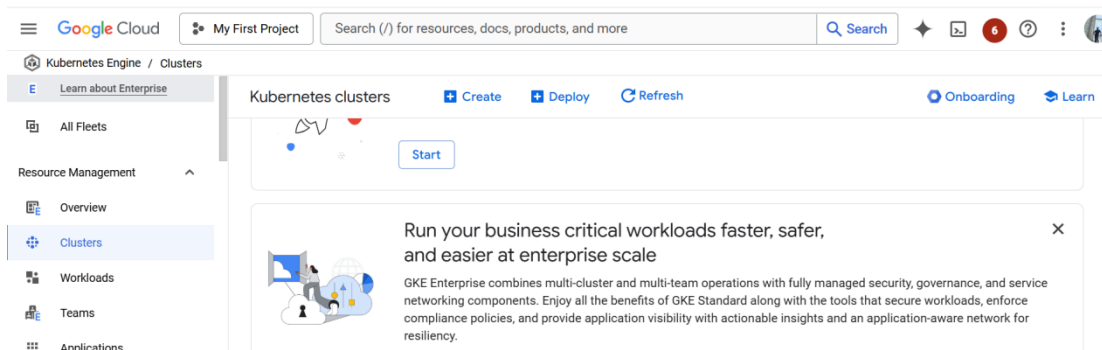
Experiment-9

Aim:-Create deployment resource using Kubernetes.

- Create the **google cloud console** free account
- It is a two step process
- It is deducting the 2 rupees from your account and it will give the 330\$ free credit points.

NOTE: Don't active the full account

- Once the account is created u can login to google cloud console



- NOW CREATE THE KUBERNETES CLUSTER

Open the cloud shell



To see the cluster list run the below command

Gcloud container clusters list (no clusters are there)

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ gcloud container clusters list
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

You create the cluster with below command

gcloud container clusters create my-cluster --zone us-central1-a

Cluster creation is taking 5 to 10 mints time

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ gcloud container clusters create my-cluster --zone us-central1-a
Note: The Kubelet readonly port (10255) is now deprecated. Please update your workloads to use the recommended alternatives. See https://cloud.google.com/kube
rnetes-engine/docs/how-to/disable-kubelet-readonly-port for ways to check usage and for migration instructions.
Note: Your Pod address range ('--cluster-ipv4-cidr') can accommodate at most 1008 node(s).
Creating cluster my-cluster in us-central1-a...Working.
```

Once the cluster is created u can see the below message automatically

```
To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/workload/_gcloud/us-central1-a/my-cluster?project=hidden-solstice-454006-n0
kubeconfig entry generated for my-cluster.
NAME: my-cluster
LOCATION: us-central1-a
MASTER VERSION: 1.31.6-gke.1020000
MASTER IP: 34.66.171.95
MACHINE TYPE: e2-medium
NODE VERSION: 1.31.6-gke.1020000
NUM NODES: 3
STATUS: RUNNING
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

Now u go and check kubernetes engine--->cluster , you can see the my-cluster is running

Kubernetes clusters							Onboarding	Learn
Filter Enter property name or value								
☐ Status	Name ↑	Location	Tier ⓘ	Number of nodes	Total vCPUs	Total memory		
☒	my-cluster	us-central1-a	Standard	3	6	12 GB		

Run the below command

gcloud container clusters get-credentials my-cluster --zone us-central1-a

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ gcloud container clusters get-credentials my-cluster --zone us-central1-a
Fetching cluster endpoint and auth data.
kubeconfig entry generated for my-cluster.
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

To see the list of nodes

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ kubectl get nodes
NAME                                STATUS    ROLES    AGE    VERSION
gke-my-cluster-default-pool-020e8447-03bv    Ready    <none>    8m43s    v1.31.6-gke.1020000
gke-my-cluster-default-pool-020e8447-1510    Ready    <none>    8m43s    v1.31.6-gke.1020000
gke-my-cluster-default-pool-020e8447-rv2r    Ready    <none>    8m43s    v1.31.6-gke.1020000
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

Create the pods

Kubectl run --image tomcat webserver

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ kubectl run --image tomcat webserver
pod/webserver created
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

To see the pods list

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
webserver     1/1     Running   0           20m
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

To get the list of pods along with ip address and which node the pod is running

Kubectl get pods -o wide

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ kubectl get pods -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP        NODE                                NOMINATED NODE   READINESS GATES
webserver     1/1     Running   0           22m   10.40.0.4  gke-my-cluster-default-pool-020e8447-1510    <none>           <none>
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

Actually u can create the pod using definition file

Create pd-df1.yaml

Vim pd-df1.yaml

```
apiVersion: v1
kind: Pod
metadata:
  name: jenkins-pod
spec:
  containers:
  - name: myjenkins
    image: jenkins/jenkins
  ports:
  - containerPort: 8080
    hostPort: 8080
```

for accessing the application u need to open the port

How to open the port

gcloud compute firewall-rules create rule2 --allow tcp:8080

```
sarvesh_ambala@cloudshell:~/sample (hidden-solstice-454006-n0)$ gcloud compute firewall-rules create rule2 --allow tcp:8080
Creating firewall...working..Created [https://www.googleapis.com/compute/v1/projects/hidden-solstice-454006-n0/global/firewalls/rule2].
Creating firewall...done.
NAME: rule2
NETWORK: default
DIRECTION: INGRESS
PRIORITY: 1000
ALLOW: tcp:8080
DENY:
DISABLED: False
```

Kubectl create -f pd-df1.yaml

Kubectl get pods -o wide

```
sarvesh_ambala@cloudshell:~/sample (hidden-solstice-454006-n0)$ kubectl get pods -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
jenkins-pod	0/1	ImagePullBackOff	0	3h25m	10.40.0.5	gke-my-cluster-default-pool-020e8447-1510	<none>	<none>
jenkins-pod1	0/1	ImagePullBackOff	0	3h8m	10.40.0.6	gke-my-cluster-default-pool-020e8447-1510	<none>	<none>
jk-pod	1/1	Running	0	4m33s	10.40.1.5	gke-my-cluster-default-pool-020e8447-rv2r	<none>	<none>
webserver	1/1	Running	0	4h5m	10.40.0.4	gke-my-cluster-default-pool-020e8447-1510	<none>	<none>

Kubectl get nodes -o wide

```
sarvesh_ambala@cloudshell:~/sample1 (hidden-solstice-454006-n0)$ kubectl get nodes -o wide
NAME                                STATUS    ROLES    AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE
KERNEL-VERSION   CONTAINER-RUNTIME
gke-my-cluster-default-pool-020e8447-03bv   Ready    <none>   4h19m   v1.31.6-gke.1020000   10.128.0.15   35.225.52.28   Container-Optimized OS from Google
6.6.72+          containerd://1.7.24
gke-my-cluster-default-pool-020e8447-1510   Ready    <none>   4h19m   v1.31.6-gke.1020000   10.128.0.13   34.60.92.233   Container-Optimized OS from Google
6.6.72+          containerd://1.7.24
gke-my-cluster-default-pool-020e8447-rv2r   Ready    <none>   4h19m   v1.31.6-gke.1020000   10.128.0.14   34.56.2.233    Container-Optimized OS from Google
6.6.72+          containerd://1.7.24
sarvesh_ambala@cloudshell:~/sample1 (hidden-solstice-454006-n0)$
```

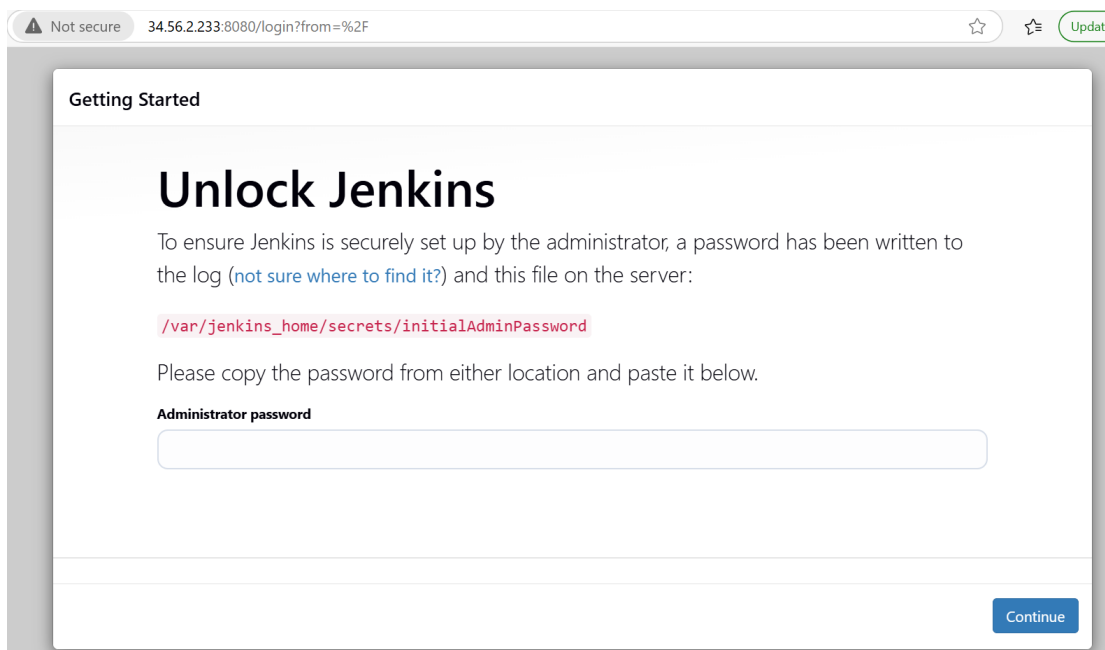
How can we access the pod

Take the external ip add the port no 8080

Open the browser paste ipaddress:8080

Now u can able to see the Jenkins

Output:-



Experiment 10

AIM:-Create a docker image for any application using docker file and push it to Docker Hub.

Step 1:-[Connecting AWS Instance Ubuntu using Mobaxterm](#)

1. Login AWS(Amazon Web Services) Account
2. Lunch Instance name Docker
3. Connect to Ubuntu or mobaxterm

(note:-Follow this url for docker file and application—<https://github.com/devisar/devopslab>)

Step 2:-[Create Docker Hub Account and create repository in Docker Hub](#)

Step 3:-Install Docker and Check Status and Start Docker in mobaxterm

1. sudo apt update -y
2. sudo apt install docker.io -y
3. sudo systemctl status docker(come outside use command ctrl+z)

```
ubuntu@ip-172-31-90-47:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-02-28 03:50:40 UTC; 2min 5s ago
     TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
      Main PID: 2110 (dockerd)
         Tasks: 8
        Memory: 35.8M (peak: 38.1M)
          CPU: 253ms
       CGroup: /system.slice/docker.service
              └─2110 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Feb 28 03:50:38 ip-172-31-90-47 systemd[1]: Starting docker.service - Docker Application Container Engine...
Feb 28 03:50:38 ip-172-31-90-47 dockerd[2110]: time="2025-02-28T03:50:38.974271527Z" level=info msg="Starting up"
Feb 28 03:50:38 ip-172-31-90-47 dockerd[2110]: time="2025-02-28T03:50:38.975783980Z" level=info msg="detected 127.0.0.53 name"
Feb 28 03:50:39 ip-172-31-90-47 dockerd[2110]: time="2025-02-28T03:50:39.094061737Z" level=info msg="Loading containers: star"
Feb 28 03:50:39 ip-172-31-90-47 dockerd[2110]: time="2025-02-28T03:50:39.560107166Z" level=info msg="Loading containers: done"
Feb 28 03:50:40 ip-172-31-90-47 dockerd[2110]: time="2025-02-28T03:50:40.287913914Z" level=info msg="Docker daemon" commit="2"
Feb 28 03:50:40 ip-172-31-90-47 dockerd[2110]: time="2025-02-28T03:50:40.288020817Z" level=info msg="Daemon has completed ini"
Feb 28 03:50:40 ip-172-31-90-47 dockerd[2110]: time="2025-02-28T03:50:40.338547546Z" level=info msg="API listen on /run/docke"
Feb 28 03:50:40 ip-172-31-90-47 systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-21/21 (END)
```

Above status command is docker running means no problem if not run use command below to run

4. sudo systemctl start docker

Step 4:- Grant Access

Why we give grant access means

A easy way to verify your Docker installation is by running the below command

docker run hello-world

If the output says:

```
ubuntu@ip-172-31-90-47:~$ docker run hello-world
docker: permission denied while trying to connect to the Docker daemon socket at unix:///var/run/docker.sock: Head "http://%2Fvar%2Frun%2Fdocker.sock/_ping": dial unix /var/run/docker.sock: connect: permission denied.
See 'docker run --help'.
ubuntu@ip-172-31-90-47:~$
```

This can mean two things,

1. Docker daemon is not running.(start docker using “sudo systemctl start docker”)
2. Your user does not have access to run docker commands.

Grant Access to your user to run docker commands

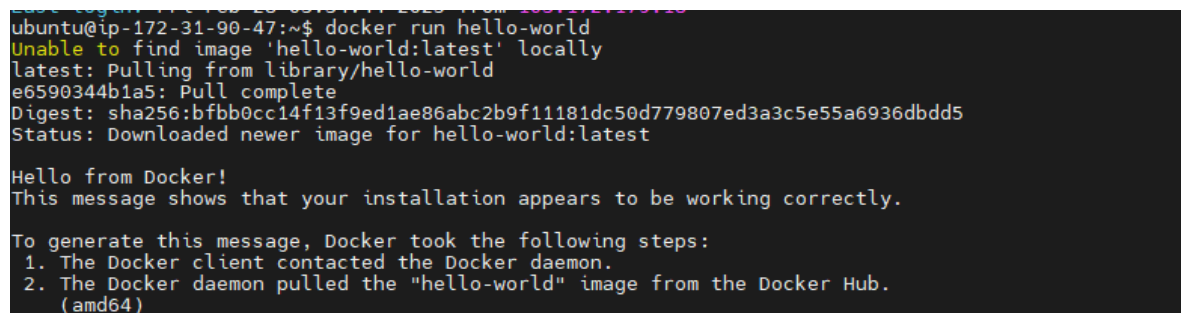
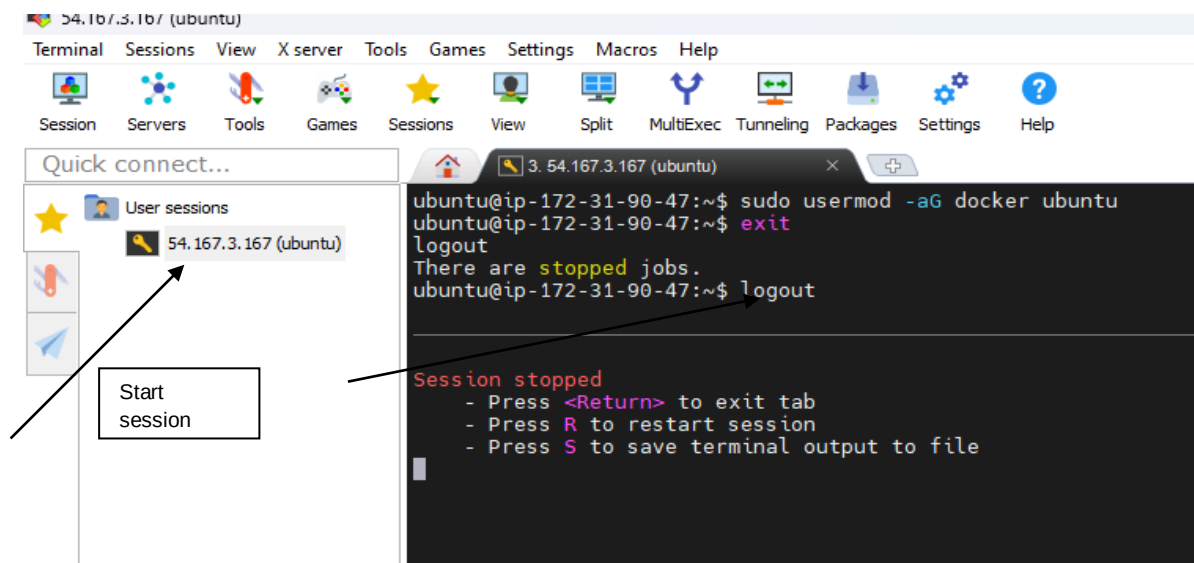
1. `sudo usermod -aG docker ubuntu`

In the above command ubuntu is the name of the user, you can change the username appropriately.

NOTE: : You need to logout and login back for the changes to be reflected.

2. Logout purpose use commands exit or logout

Again run command “docker run hello-world”

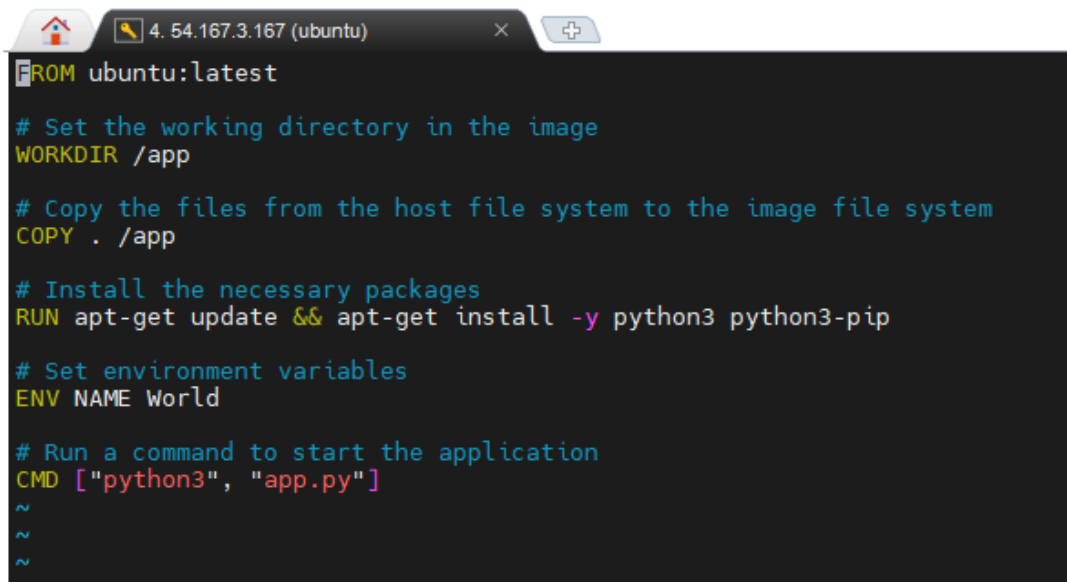


Step 5:-Creating application and Docker file

1. `Mkdir docker1`
2. `cd docker1`
3. `vim app.py`

`print("hello world")`
4. `cat app.py`

5. vim Dockerfile (below pic commands write lab record) typing command vim before click “i” for insert data after completion Docker file commands save before click esc use :wq!



```
FROM ubuntu:latest

# Set the working directory in the image
WORKDIR /app

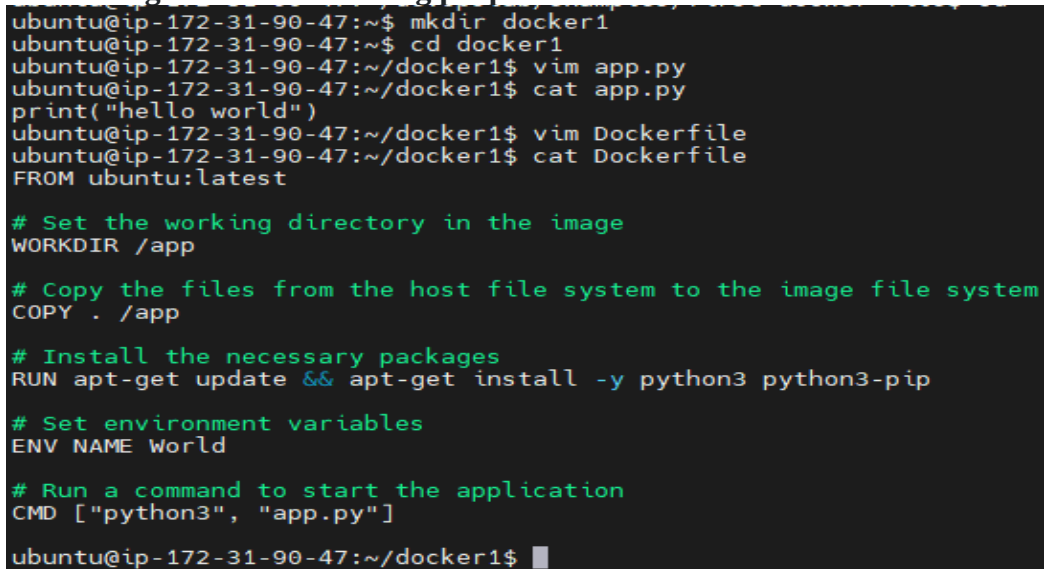
# Copy the files from the host file system to the image file system
COPY . /app

# Install the necessary packages
RUN apt-get update && apt-get install -y python3 python3-pip

# Set environment variables
ENV NAME World

# Run a command to start the application
CMD ["python3", "app.py"]
~
~
~
```

Below image for understanding purpose



```
ubuntu@ip-172-31-90-47:~$ mkdir docker1
ubuntu@ip-172-31-90-47:~$ cd docker1
ubuntu@ip-172-31-90-47:~/docker1$ vim app.py
ubuntu@ip-172-31-90-47:~/docker1$ cat app.py
print("hello world")
ubuntu@ip-172-31-90-47:~/docker1$ vim Dockerfile
ubuntu@ip-172-31-90-47:~/docker1$ cat Dockerfile
FROM ubuntu:latest

# Set the working directory in the image
WORKDIR /app

# Copy the files from the host file system to the image file system
COPY . /app

# Install the necessary packages
RUN apt-get update && apt-get install -y python3 python3-pip

# Set environment variables
ENV NAME World

# Run a command to start the application
CMD ["python3", "app.py"]
ubuntu@ip-172-31-90-47:~/docker1$
```

se

Step 6:- Build and Check Docker image

Syntax:- docker build -t dockerhub_username/repositoryname:tag .

1. docker build -t anu1308/dockerimage:latest .
2. docker images

Below images for understanding purpose

```

Building an image from a Dockerfile
ubuntu@ip-172-31-90-47:~/docker1$ docker build -t anu1308/dockerimage:latest .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
             Install the buildx component to build images with BuildKit:
             https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 3.072kB
Step 1/6 : FROM ubuntu:latest
latest: Pulling from library/ubuntu
5a7813e071bf: Pull complete
Digest: sha256:72297848456d5d37d1262630108ab308d3e9ec7ed1c3286a32fe09856619a782
Status: Downloaded newer image for ubuntu:latest
--> a04dc4851cbc
Step 2/6 : WORKDIR /app
--> Running in 490762c2b766
--> Removed intermediate container 490762c2b766
--> f19d0296889a
Step 3/6 : COPY . /app
--> 05e2e2564f3a
Step 4/6 : RUN apt-get update && apt-get install -y python3 python3-pip
--> Running in 0cad8878ec6b
Get:1 http://archive.ubuntu.com/ubuntu noble InRelease [256 kB]
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [842 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:6 http://archive.ubuntu.com/ubuntu noble/restricted amd64 Packages [117 kB]
Get:7 http://archive.ubuntu.com/ubuntu noble/universe amd64 Packages [19.3 MB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [24.2 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [807 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1053 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [331 kB]
Get:12 http://archive.ubuntu.com/ubuntu noble/main amd64 Packages [1808 kB]
Get:13 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1131 kB]
Get:14 http://archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [28.8 kB]
Get:15 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [881 kB]
Get:16 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1336 kB]
Get:17 http://archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [16.0 kB]
Fetched 28.3 MB in 3s (8996 kB/s)
Successfully tagged anu1308/dockerimage:latest
ubuntu@ip-172-31-90-47:~/docker1$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED           SIZE
anu1308/dockerimage latest       cdbf156a1eda     56 seconds ago   574MB
ubuntu              latest       a04dc4851cbc     4 weeks ago      78.1MB
hello-world         latest       74cc54e27dc4     5 weeks ago      10.1kB
ubuntu@ip-172-31-90-47:~/docker1$

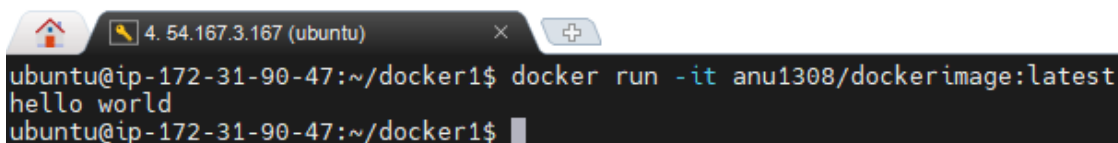
```

Step 7:- Run your First Docker Container

1. `docker run -it anu1308/dockerimage:latest`

Output

Hello World



```

ubuntu@ip-172-31-90-47:~/docker1$ docker run -it anu1308/dockerimage:latest
hello world
ubuntu@ip-172-31-90-47:~/docker1$

```

Step 8:-Docker Login

1. docker login

```
ubuntu@ip-172-31-90-47:~/docker1$ docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, you can create one by visiting https://docs.docker.com/engine/reference/commandline/login/#credentials-store.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security. See https://docs.docker.com/engine/reference/commandline/login/#credentials-store for more.

Username: anu1308
Password:
WARNING! Your password will be stored unencrypted in /home/ubuntu/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credentials-store

Login Succeeded
ubuntu@ip-172-31-90-47:~/docker1$
```

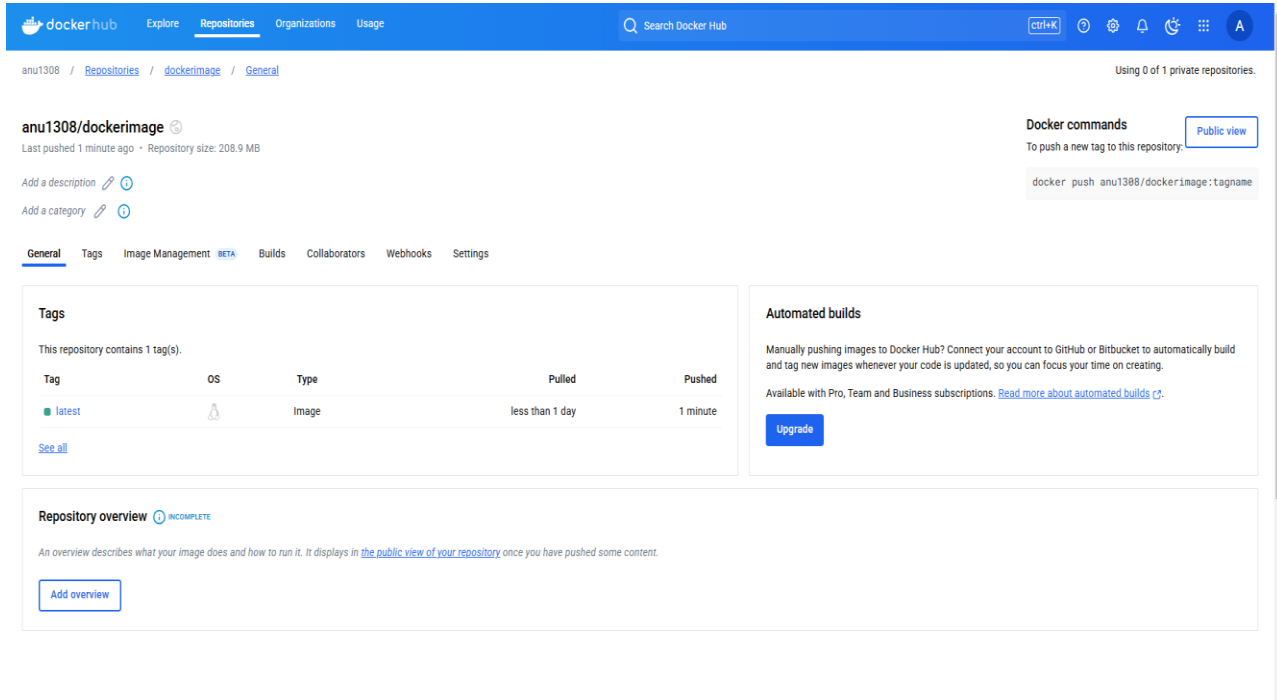
Step 9:- Push the Image to DockerHub and share it with the world

1. docker push anu1308/dockerimage:latest

```
ubuntu@ip-172-31-90-47:~/docker1$ docker push anu1308/dockerimage:latest
The push refers to repository [docker.io/anu1308/dockerimage]
2d65eeb8aca1: Pushed
9863a1399ed4: Pushed
8870c5a70606: Pushed
4b7c01ed0534: Mounted from library/ubuntu
latest: digest: sha256:ec420e83236afe2170d2feac0ba4ff0af40d1eb580bb3c83a5e50a374c5d1746 size: 1155
ubuntu@ip-172-31-90-47:~/docker1$
```

Output:-

```
ubuntu@ip-172-31-90-47:~/docker1$ docker images
REPOSITORY          TAG          IMAGE ID          CREATED          SIZE
anu1308/dockerimage latest       cdbf156a1eda     56 seconds ago  574MB
ubuntu               latest       a04dc4851cbc     4 weeks ago     78.1MB
hello-world          latest       74cc54e27dc4     5 weeks ago     10.1kB
ubuntu@ip-172-31-90-47:~/docker1$
```



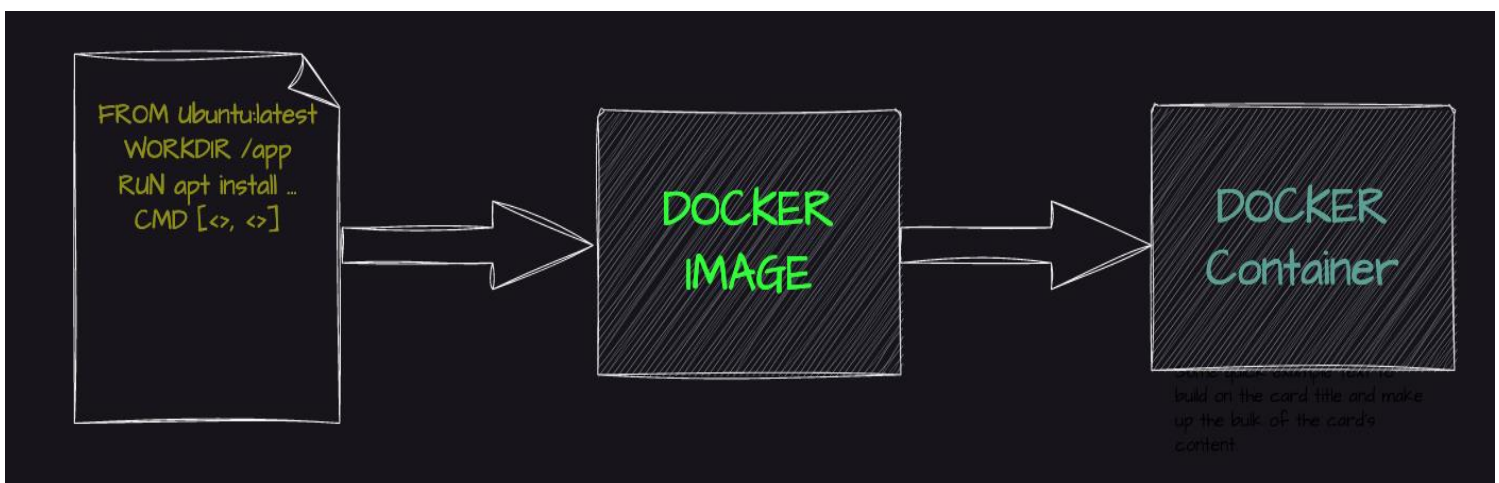
Definitions:-

Docker LifeCycle

We can use the above Image as reference to understand the lifecycle of Docker.

There are three important things,

1. docker build -> builds docker images from Dockerfile
2. docker run -> runs container from docker images
3. docker push -> push the container image to public/private registries(docker hub) to share the docker images.



What is a container ?

A container is a standard unit of software that packages up code and all its dependencies so the application runs quickly and reliably from one computing environment to another. A Docker container image is a lightweight, standalone, executable package of software that includes everything needed to run an application: code, runtime, system tools, system libraries and settings.

Why are containers light weight ?

Containers are lightweight because they use a technology called containerization, which allows them to share the host operating system's kernel and libraries, while still providing isolation for the application and its dependencies. This results in a smaller footprint compared to traditional virtual machines, as the containers do not need to include a full operating system. Additionally, Docker containers are designed to be minimal, only including what is necessary for the application to run, further reducing their size.

What is Docker ?

Docker is a containerization platform that provides easy way to containerize your applications, which means, using Docker you can build container images, run the images to create containers and also push these containers to container registries such as DockerHub

Docker daemon

The Docker daemon (dockerd) listens for Docker API requests and manages Docker objects such as images, containers, networks, and volumes. A daemon can also communicate with other daemons to manage Docker services.

Docker client

The Docker client (docker) is the primary way that many Docker users interact with Docker. When you use commands such as docker run, the client sends these commands to dockerd, which carries them out. The docker command uses the Docker API. The Docker client can communicate with more than one daemon.

Docker Desktop

Docker Desktop is an easy-to-install application for your Mac, Windows or Linux environment that enables you to build and share containerized applications and microservices. Docker Desktop includes the Docker daemon (dockerd), the Docker client (docker), Docker Compose, Docker Content Trust, Kubernetes, and Credential Helper. For more information, see Docker Desktop.

Docker registries

A Docker registry stores Docker images. Docker Hub is a public registry that anyone can use, and Docker is configured to look for images on Docker Hub by default. You can even run your own private registry.

When you use the docker pull or docker run commands, the required images are pulled from your configured registry. When you use the docker push command, your image is pushed to your configured registry. Docker objects

When you use Docker, you are creating and using images, containers, networks, volumes, plugins, and other objects. This section is a brief overview of some of those objects.

Dockerfile

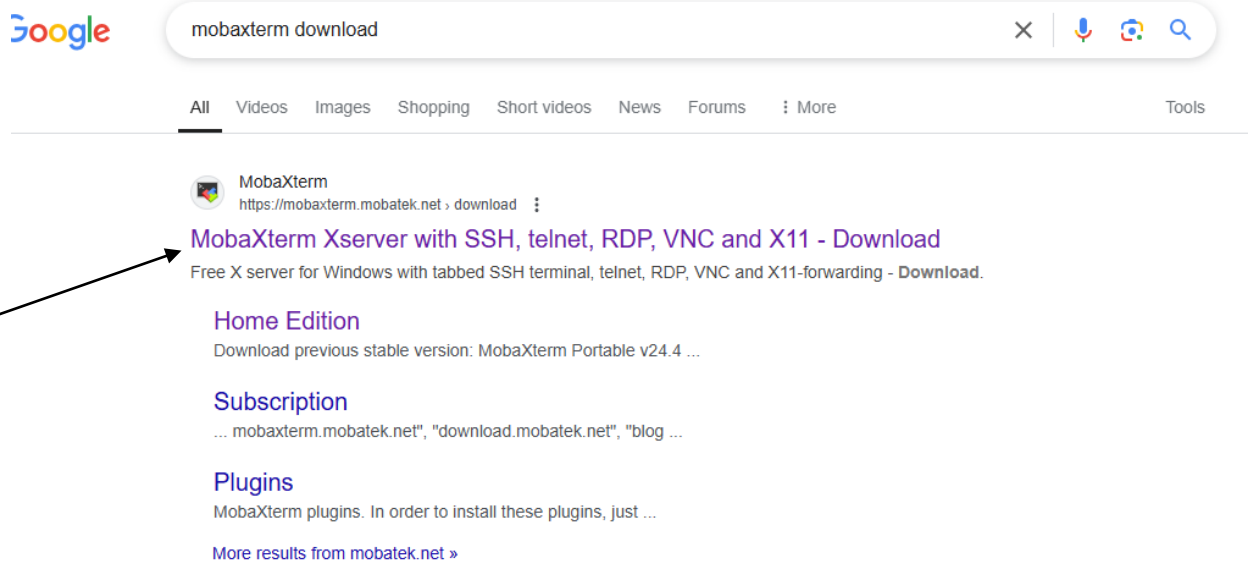
Dockerfile is a file where you provide the steps to build your Docker Image.

Images

An image is a read-only template with instructions for creating a Docker container. Often, an image is based on another image, with some additional customization. For example, you may build an image which is based on the ubuntu image, but installs the Apache web server and your application, as well as the configuration details needed to make your application run.

Below Images for Understanding purpose

Connecting AWS Instance to Mobaxterm



The screenshot displays the MobaXterm website's pricing page. The navigation bar at the top includes links for Home, Demo, Features, Download, Plugins, Help, and Contact, along with social media icons and buttons for 'Customer area' and 'Buy'. The main content is divided into two columns: 'Home Edition' (Free) and 'Professional Edition' (\$69 / 49€ per user*). The Home Edition lists features such as Full X server and SSH support, Remote desktop (RDP, VNC, Xdmcp), Remote terminal (SSH, telnet, rlogin, Mosh), X11-Forwarding, Automatic SFTP browser, Master password protection, Plugins support, Portable and installer versions, Full documentation, Max. 12 sessions, Max. 2 SSH tunnels, Max. 4 macros, and Max. 360 seconds for Tftp, Nfs and Cron. A 'Download now' button is located at the bottom of the Home Edition column, with an arrow pointing to it from the bottom left. The Professional Edition lists features including Every feature from Home Edition +, Customize your startup message and logo, Modify your profile script, Remove unwanted games, screensaver or tools, Unlimited number of sessions, Unlimited number of tunnels and macros, Unlimited run time for network daemons, Enhanced security settings, 12-months updates included, Deployment inside company, and Lifetime right to use. A 'Subscribe online / Get a quote' button is located at the bottom of the Professional Edition column.

Home Edition

Free

- Full **X server** and **SSH** support
- Remote desktop (RDP, VNC, Xdmcp)
- Remote terminal (SSH, telnet, rlogin, Mosh)
- X11-Forwarding
- Automatic SFTP browser
- Master password protection
- Plugins support
- Portable and installer versions
- Full documentation
- Max. **12** sessions
- Max. **2** SSH tunnels
- Max. **4** macros
- Max. **360** seconds for Tftp, Nfs and Cron

[Download now](#)

Professional Edition

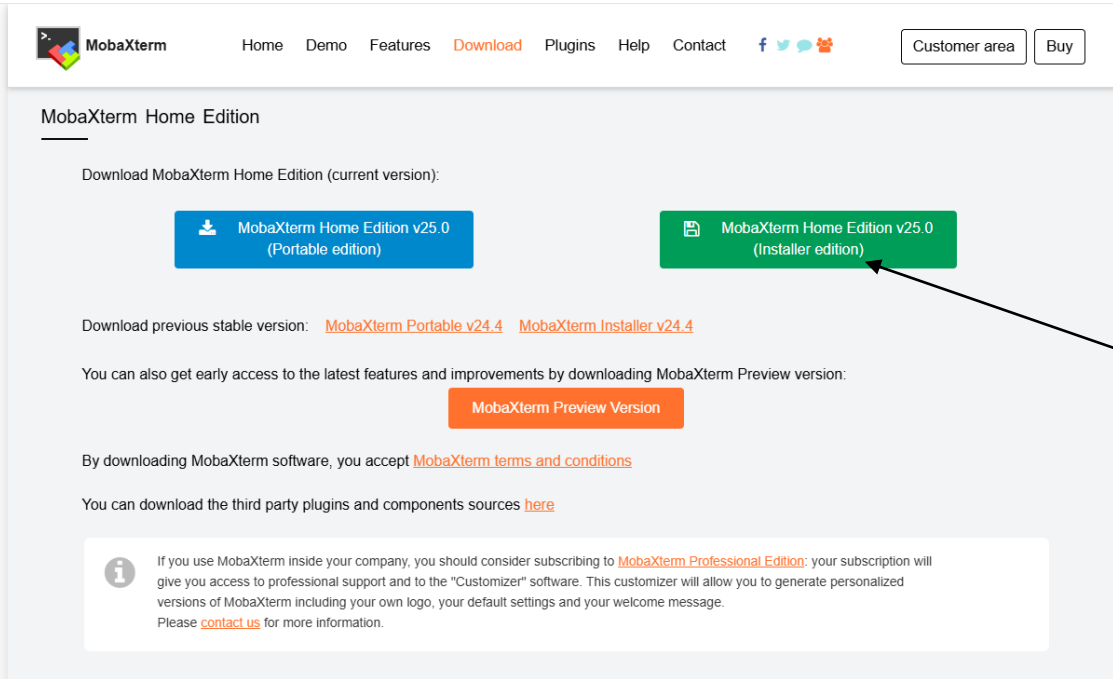
\$69 / 49€ per user*

* Excluding tax. Volume discounts [available](#)

Every feature from Home Edition +

- Customize your startup message and logo
- Modify your profile script
- Remove unwanted games, screensaver or tools
- Unlimited number of sessions
- Unlimited number of tunnels and macros
- Unlimited run time for network daemons
- Enhanced security settings
- 12-months updates included
- Deployment inside company
- Lifetime right to use

[Subscribe online / Get a quote](#)



MobaXterm Home Edition

Download MobaXterm Home Edition (current version):

[MobaXterm Home Edition v25.0 \(Portable edition\)](#) [MobaXterm Home Edition v25.0 \(Installer edition\)](#)

Download previous stable version: [MobaXterm Portable v24.4](#) [MobaXterm Installer v24.4](#)

You can also get early access to the latest features and improvements by downloading MobaXterm Preview version: [MobaXterm Preview Version](#)

By downloading MobaXterm software, you accept [MobaXterm terms and conditions](#)

You can download the third party plugins and components sources [here](#)

If you use MobaXterm inside your company, you should consider subscribing to [MobaXterm Professional Edition](#): your subscription will give you access to professional support and to the "Customizer" software. This customizer will allow you to generate personalized versions of MobaXterm including your own logo, your default settings and your welcome message. Please [contact us](#) for more information.

 MobaXterm_Installer_v25.0 Date modified

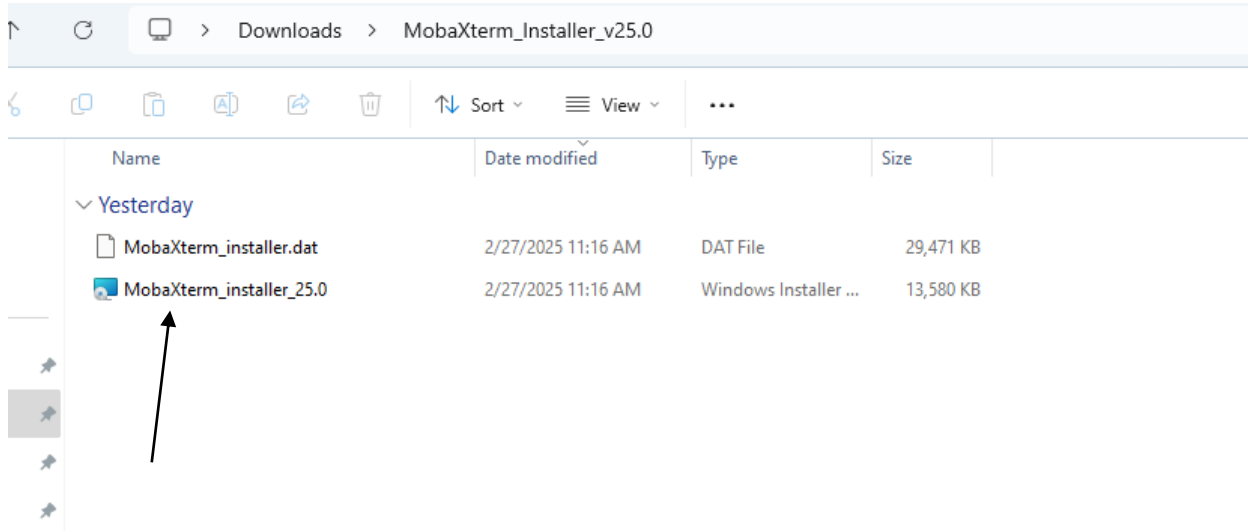
 p5.pem Date modified
Type: PEM File Size: 1.0 KB

 Docker-Zero-to-Hero-main Date modified

 MobaXterm_Installer_v25.0 Date modified

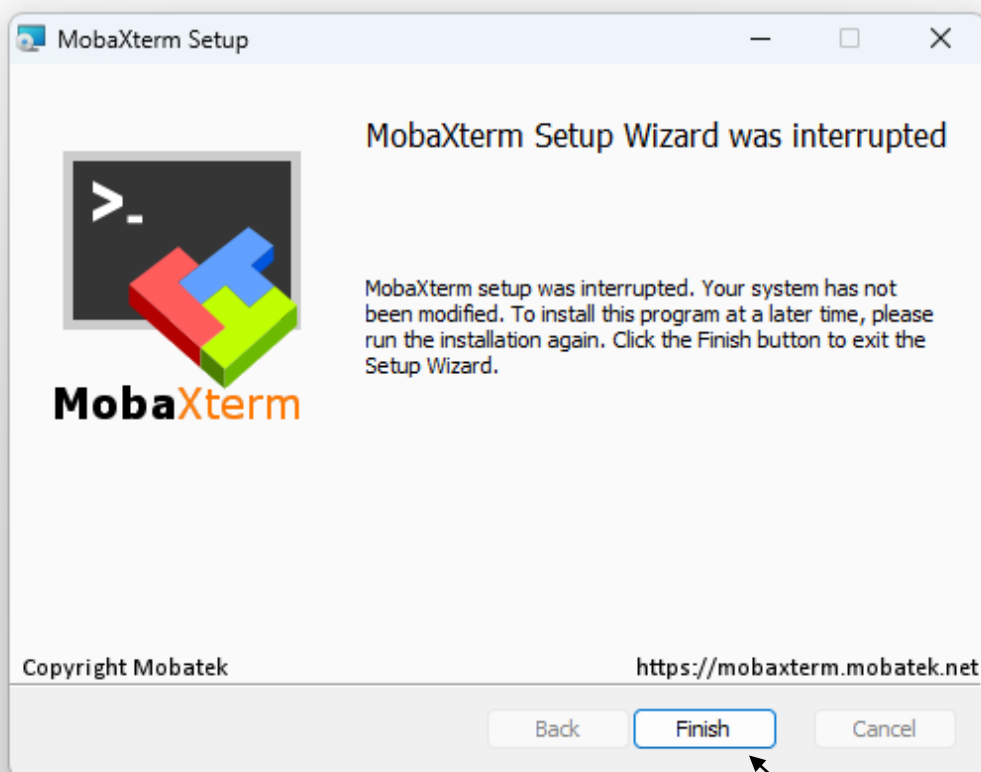
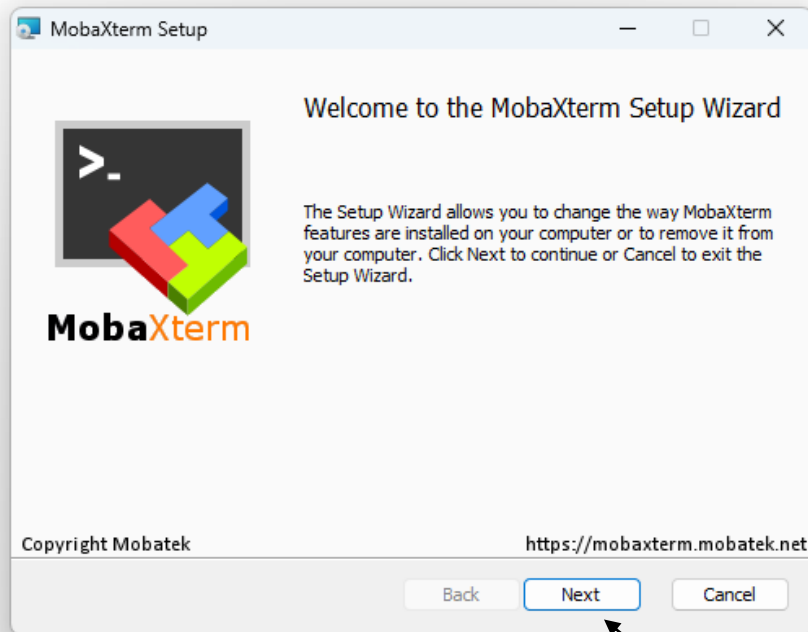
Earlier this week

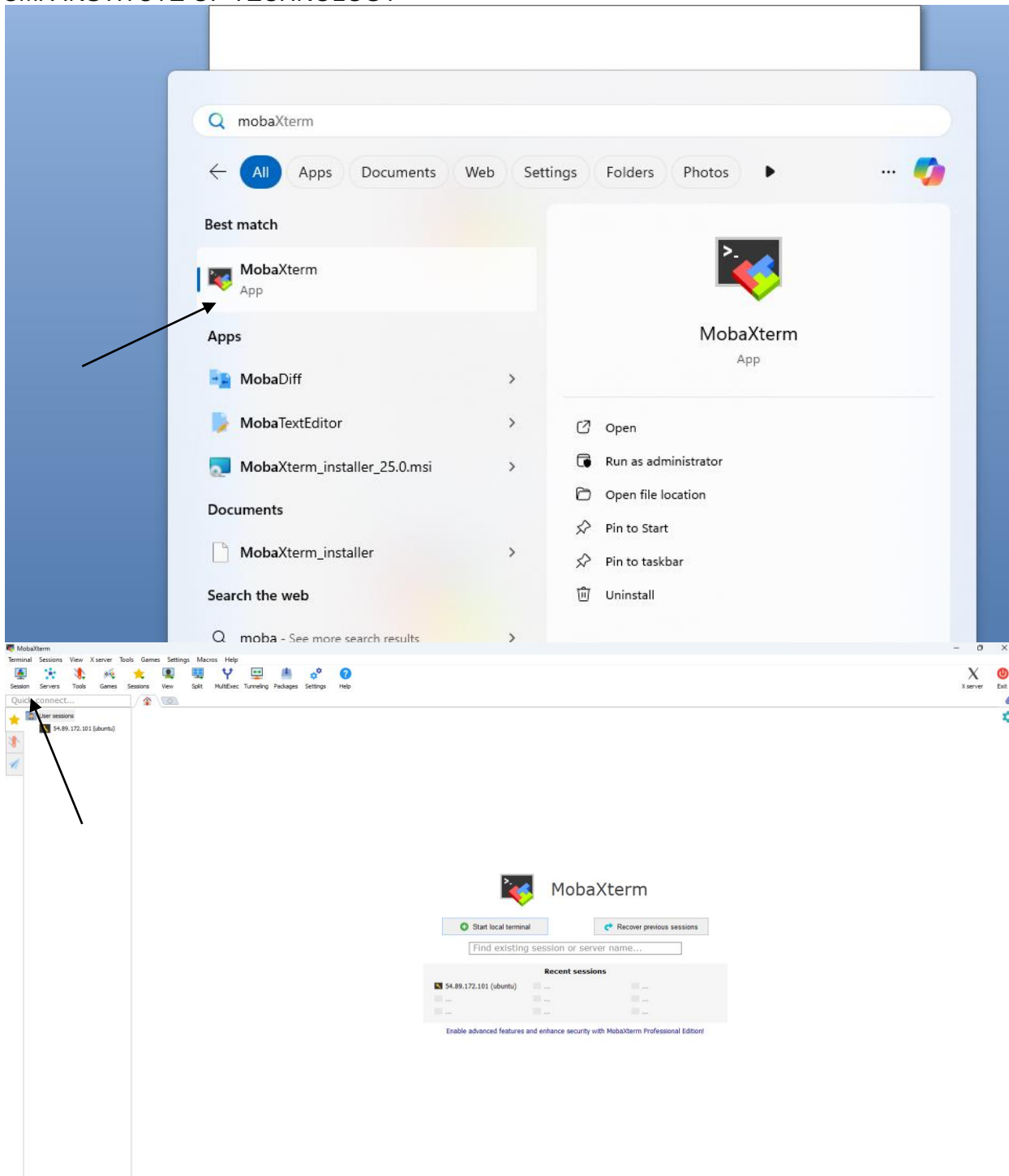
 exp4-main (1) Date modified

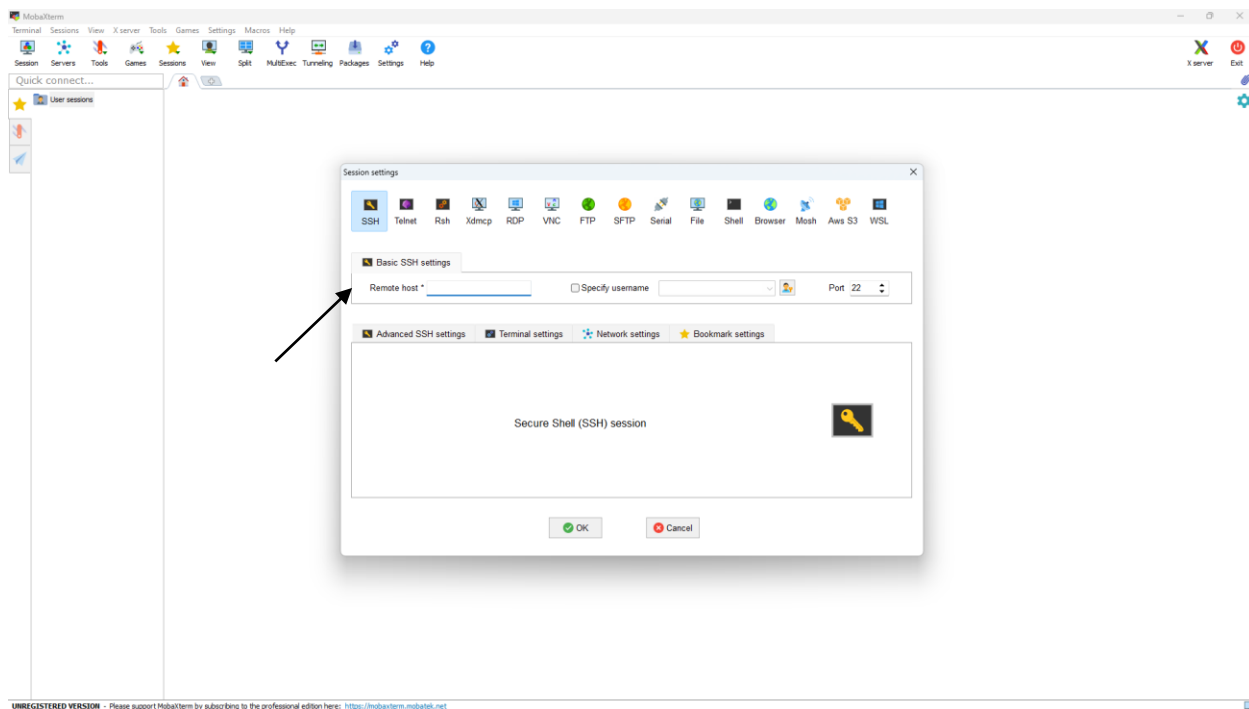
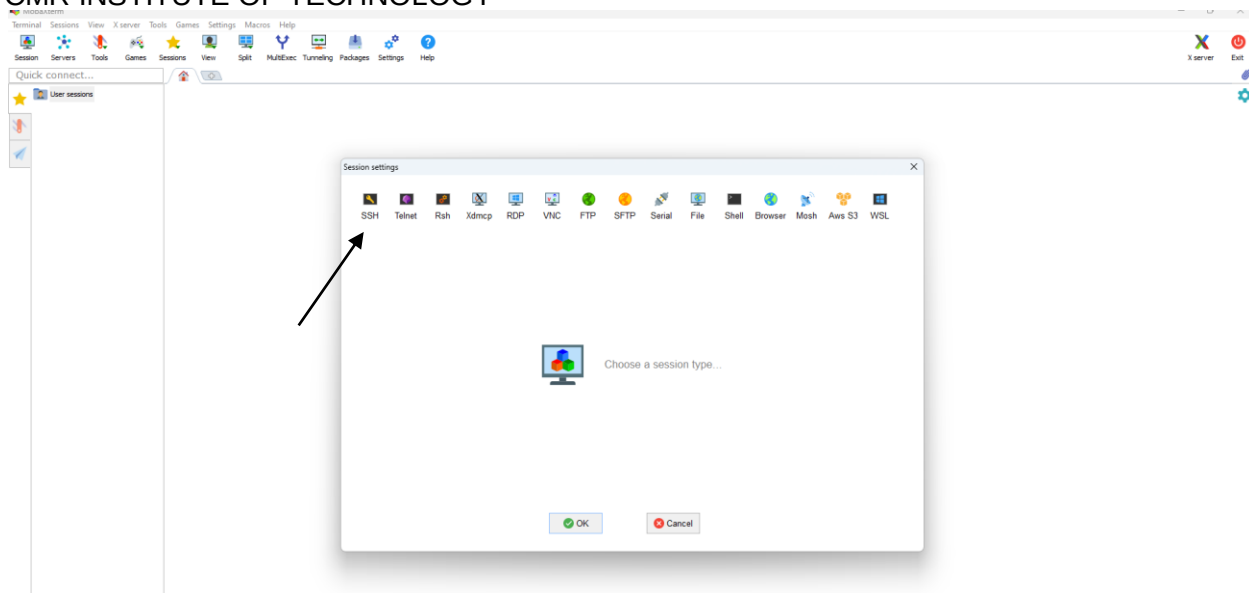


File Explorer path: Downloads > MobaXterm_Installer_v25.0

Name	Date modified	Type	Size
Yesterday			
MobaXterm_installer.dat	2/27/2025 11:16 AM	DAT File	29,471 KB
MobaXterm_installer_25.0	2/27/2025 11:16 AM	Windows Installer ...	13,580 KB







Remote host we want public key and .pem file also so login to AWS Account

Try the new sign in UI
See our new improved Amazon Web Services sign in experience before we officially launch.

Enable new sign in



Sign in

☒ **Root user**
Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**
User within an account that performs daily tasks. [Learn more](#)

Root user email address

harithachennagari@gmail.com

penugondaanusha12345@gm...

By continuing, you agree to the [AWS Terms of Service](#), [AWS Privacy Notice](#), and [AWS Customer Agreement](#).

New to AWS?

Create a new AWS account



AWS re:Invent
DEC. 2 - DEC. 6, 2024 | LAS VEGAS, NV

Amazon Nova, new foundation models that deliver frontier intelligence and industry leading price performance

See what Amazon Nova can do

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Multi-session disabled English

EC2 > Instances

Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Images

AMIs

AMI Catalog

Elastic Block Store

Volumes

Snapshots

Lifecycle Manager

Network & Security

Security Groups

Elastic IPs

Placement Groups

Key Pairs

Network Interfaces

Instances

Find Instance by attribute or tag (case-sensitive)

All states

Last updated less than a minute ago

Connect

Instance state

Actions

Launch instances

Name

Instance ID

Instance state

Instance type

Status check

Alarm status

Availability Zone

Public IPv4 DNS

Public IPv4 ...

Elastic IP

No instances

You do not have any instances in this region

Launch instances

Select an instance

135

The screenshot shows the AWS Management Console 'Launch an instance' page. The 'Name and tags' section has 'Docker' entered in the name field. The 'Application and OS Images (Amazon Machine Image)' section shows the 'Ubuntu Server 24.04 LTS (HVM)' AMI selected. The 'Summary' section on the right shows the instance configuration: 1 instance, Canonical Ubuntu 24.04 AMI, t2.micro instance type, new security group, and 1 8 GiB volume. A 'Free tier' notification is visible. A 'Create key pair' modal is open in the foreground, with 'p5' entered in the 'Key pair name' field. The 'Key pair type' is set to 'RSA' and the 'Private key file format' is set to '.pem'. A warning message at the bottom of the modal states: 'When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. Learn more'. The 'Create key pair' button is highlighted.

Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices. Do not show me this message again Take a walkthrough

Launch an instance info

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags info

Name
Docker Add additional tags

Application and OS Images (Amazon Machine Image) info

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or browse for AMIs if you don't see what you are looking for below.

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

aws ubuntu Microsoft Red Hat SUSE debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
ami-04b4f1a9cf54c11d0 (64-bit x86) / ami-0a7a4d879338439934 (64-bit ARM)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Summary

Number of instances 1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more
ami-04b4f1a9cf54c11d0

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet.

Cancel Launch instance Preview code

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.
p5
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA
RSA encrypted private and public key pair

☐ ED25519
ED25519 encrypted private and public key pair

Private key file format

☒ .pem
For use with OpenSSH

☐ .ppk
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

Cancel Create key pair

Family: t2 1 vCPU 1 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0154 USD per Hour
On-Demand SUSE base pricing: 0.0118 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour
On-Demand Linux base pricing: 0.0116 USD per Hour

Additional costs apply for AMIs with pre-installed software

Key pair (login) info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

p5

Create new key pair

Network settings info

Network info

vpc-0a06788c976618b9d

Subnet info

No preference (Default subnet in any availability zone)

Auto-assign public IP info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ Allow SSH traffic from Anywhere 0.0.0.0/0

☒ Allow HTTPS traffic from the internet To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet To set up an endpoint, for example when creating a web server

Summary

Number of instances info

1

Software Image (AMI)

Canonical, Ubuntu, 24.04, amd64...read more
ami-04b4f1e9f54c11d0

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet.

Cancel Launch instance Preview code

Here you can Directly connect Ubuntu also but better use mobaxterm

Instances (1/1) info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4	Elastic IP
Docker	i-09d3e11cc114228c5	Running	t2.micro	Initializing		us-east-1c	ec2-54-167-3-167.com...	54.167.3.167	-

i-09d3e11cc114228c5 (Docker)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID: i-09d3e11cc114228c5

IPv6 address: -

Hostname type: IP name: ip-172-31-90-47.ec2.internal

Answer private resource DNS name: IPv4 (A)

Public IPv4 address: 54.167.3.167 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-90-47.ec2.internal

Instance type: t2.micro

Private IPv4 addresses: 172.31.90.47

Public IPv4 DNS: ec2-54-167-3-167.compute-1.amazonaws.com | open address

Elastic IP addresses: -

Connect to instance info

Connect to your instance i-09d3e11cc114228c5 (Docker) using any of these options

EC2 Instance Connect Session Manager SSH client EC2 serial console

Instance ID: i-09d3e11cc114228c5 (Docker)

Connection Type

☒ Connect using EC2 Instance Connect
Connect using the EC2 Instance Connect browser-based client, with a public IPv4 or IPv6 address.

☐ Connect using EC2 Instance Connect Endpoint
Connect using the EC2 Instance Connect browser-based client, with a private IPv4 address and a VPC endpoint.

Public IPv4 address

54.167.3.167

IPv6 address

-

Username

Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.

ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel Connect

Find instance by instance ID or tag (i-09d3e11cc)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
Docker	i-09d3e11cc114228c5	Running	t2.micro	Initializing	View alarms +	us-east-1c	ec2-54-167-3-167.com...	54.167.3.167	-

i-09d3e11cc114228c5 (Docker)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID: i-09d3e11cc114228c5

IPv6 address: -

Hostname type: -

Public IPv4 address: 54.167.3.167 | open address

Instance state: Running

Private IP DNS name (IPv4 only): -

Private IPv4 addresses: 172.31.90.47

Public IPv4 DNS: ec2-54-167-3-167.compute-1.amazonaws.com | open address

Session settings

SSH | Telnet | Rsh | Xdmcp | RDP | VNC | FTP | SFTP | Serial | File | Shell | Browser | Mosh | Aws S3 | WSL

Basic SSH settings

Remote host: 54.167.3.167 | Specify username: ubuntu | Port: 22

Advanced SSH settings | Terminal settings | Network settings | Bookmark settings

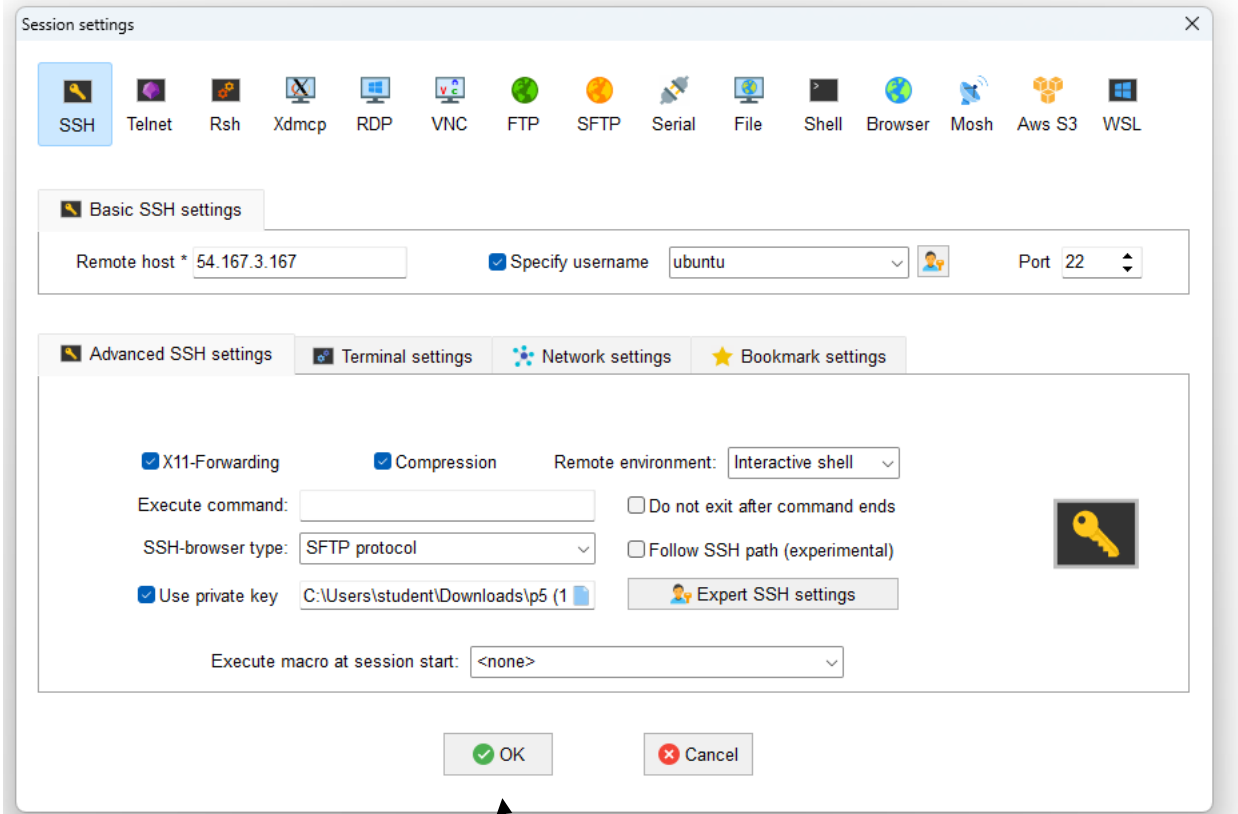
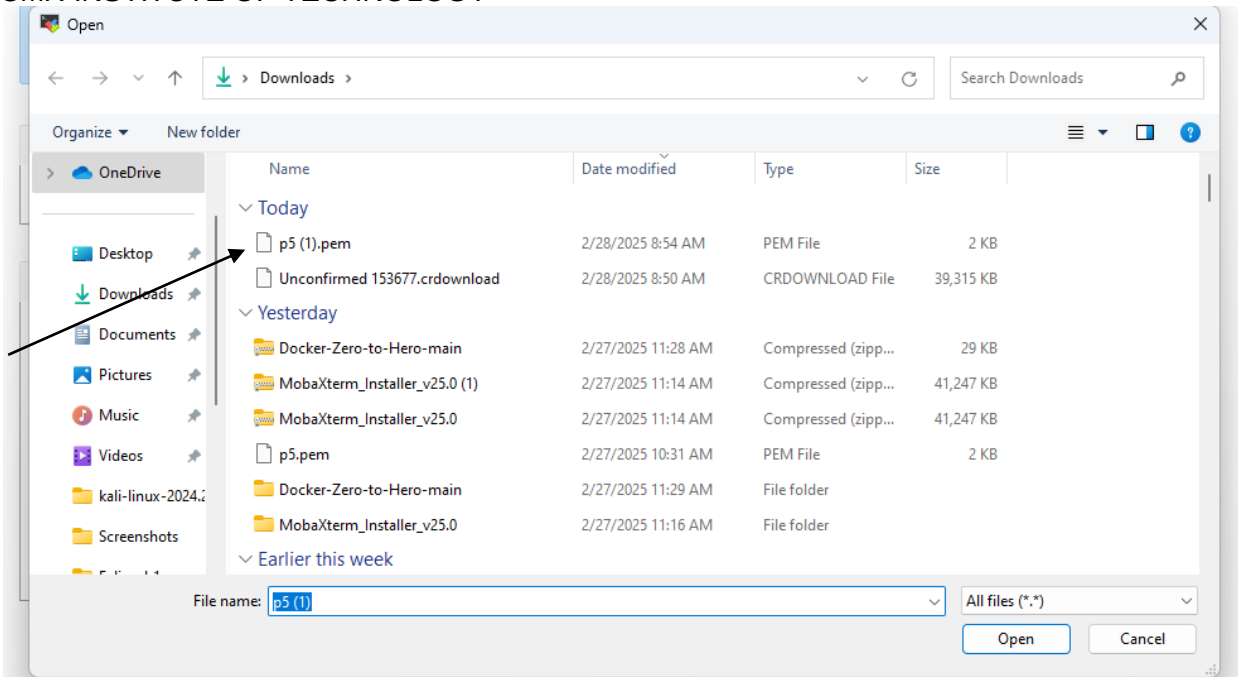
☒ X11-Forwarding | ☒ Compression | Remote environment: Interactive shell

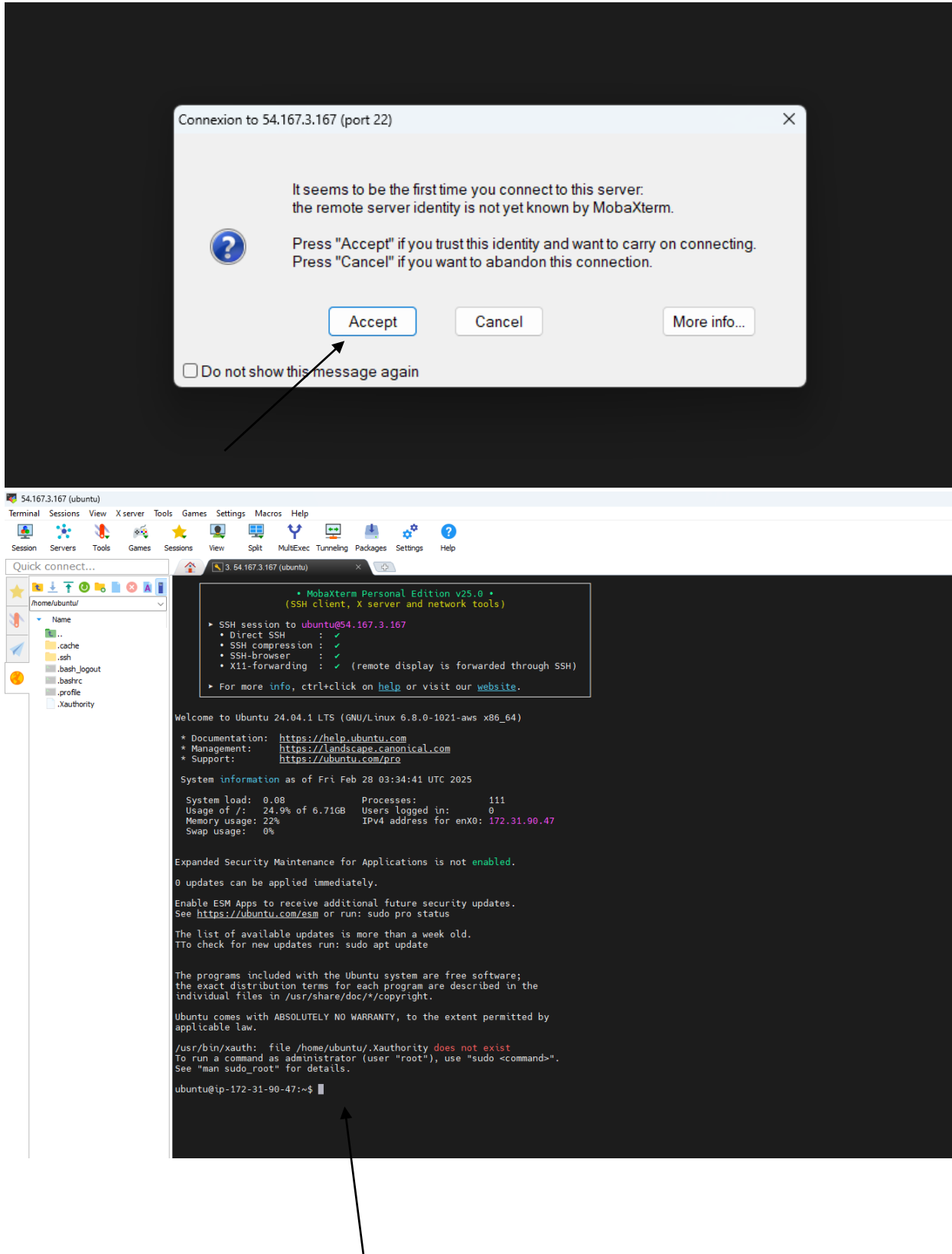
Execute command: | ☐ Do not exit after command ends

SSH-browser type: SFTP protocol | ☐ Follow SSH path (experimental)

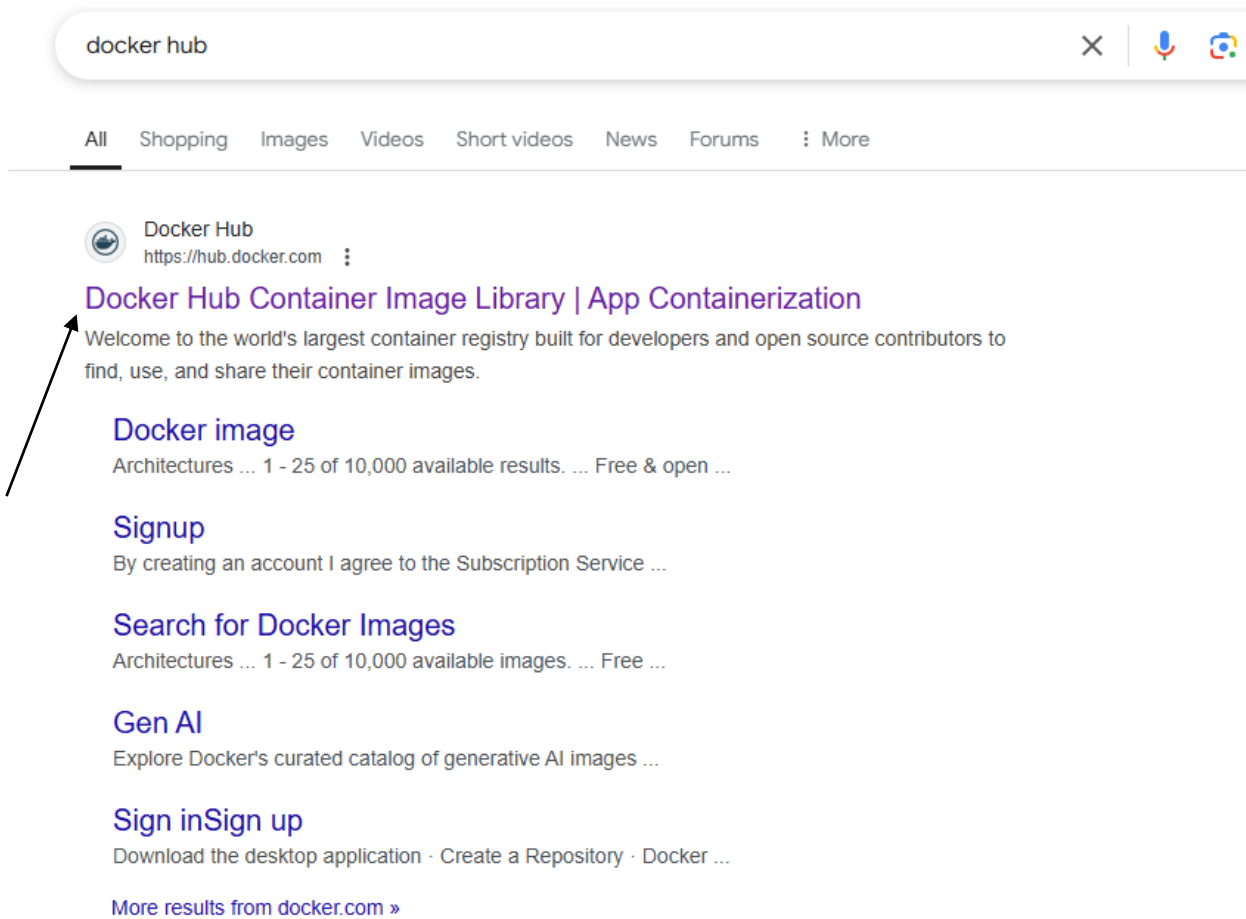
☒ Use private key | |

Execute macro at session start: <none>





Create Docker Hub Account and create repository in Docker Hub



A Google search for "docker hub" is shown. The search bar at the top contains "docker hub" with a close button (X) and voice search icons. Below the search bar, navigation tabs include "All", "Shopping", "Images", "Videos", "Short videos", "News", "Forums", and "More". The "All" tab is selected. The first search result is for "Docker Hub" with the URL "https://hub.docker.com". The title of the result is "Docker Hub Container Image Library | App Containerization". A black arrow points from the left margin to this title. The description below the title reads: "Welcome to the world's largest container registry built for developers and open source contributors to find, use, and share their container images." Below the description are several links: "Docker image" (with subtext "Architectures ... 1 - 25 of 10,000 available results. ... Free & open ..."), "Signup" (with subtext "By creating an account I agree to the Subscription Service ..."), "Search for Docker Images" (with subtext "Architectures ... 1 - 25 of 10,000 available images. ... Free ..."), "Gen AI" (with subtext "Explore Docker's curated catalog of generative AI images ..."), and "Sign inSign up" (with subtext "Download the desktop application · Create a Repository · Docker ..."). At the bottom of the result is a link "More results from docker.com »".

docker hub

All Shopping Images Videos Short videos News Forums More

Docker Hub
https://hub.docker.com

Docker Hub Container Image Library | App Containerization

Welcome to the world's largest container registry built for developers and open source contributors to find, use, and share their container images.

Docker image
Architectures ... 1 - 25 of 10,000 available results. ... Free & open ...

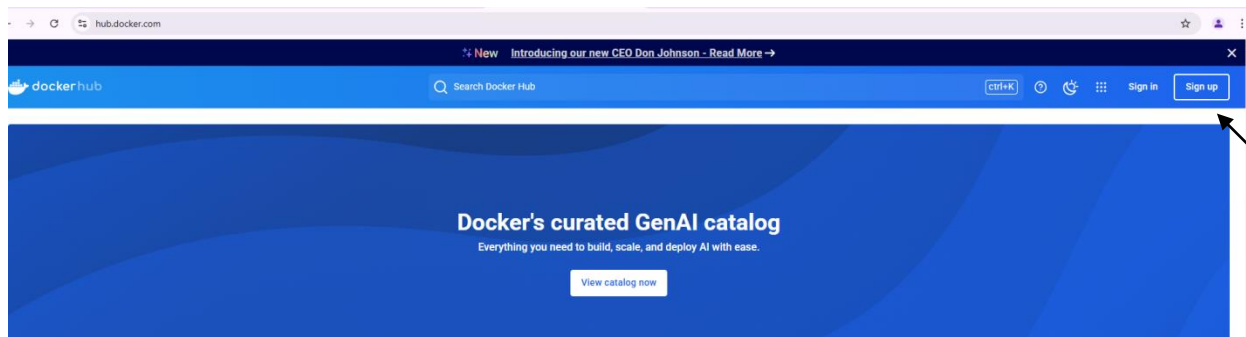
Signup
By creating an account I agree to the Subscription Service ...

Search for Docker Images
Architectures ... 1 - 25 of 10,000 available images. ... Free ...

Gen AI
Explore Docker's curated catalog of generative AI images ...

Sign inSign up
Download the desktop application · Create a Repository · Docker ...

[More results from docker.com »](#)



A screenshot of the Docker Hub website homepage. The browser address bar shows "hub.docker.com". The website has a dark blue header with the Docker Hub logo, a search bar, and links for "Sign in" and "Sign up". A black arrow points from the right margin to the "Sign up" button. Below the header is a large blue banner with the text "Docker's curated GenAI catalog" and "Everything you need to build, scale, and deploy AI with ease." A button labeled "View catalog now" is centered at the bottom of the banner.

hub.docker.com

New Introducing our new CEO Don Johnson - Read More →

dockerhub Search Docker Hub C++ Sign in Sign up

Docker's curated GenAI catalog
Everything you need to build, scale, and deploy AI with ease.

[View catalog now](#)



Create your account

We suggest signing up with your work email address.



Send me occasional product updates and announcements.

Sign up

OR



Continue with Google



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docker.

Sign in

Using Docker for work? We recommend signing in with your work email address.

Username or email address*

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dockerhub Explore Repositories Organizations Usage

Create a repository

Name	Last Pushed ↑	Contains	Visibility	Scout
anuj308/devopsipeline_demo	about 22 hours ago		Public	Inactive

1-1 of 1

dockerhub

Explore

Repositories

Organizations

Usage

Search Docker Hub

CS14K

A

Repositories / Create

Using 0 of 1 private repositories.

Create repository

Repository Name *

dockerimage

Short description

Visibility

Using 0 of 1 private repositories. [Get more](#)

Public

Appears in Docker Hub search results

Private

Only visible to you

Cancel

Create

Pushing images

You can push a new image to this repository using the CLI:

docker tag local-image:tagname new-repo:tagname

docker push new-repo:tagname

Make sure to replace tagname with your desired image repository tag.

anul308

Search by repository name

All content

Create a repository

Name	Last Pushed	Contains	Visibility	Scout
anul308/dockerimage	1 minute ago		Public	Inactive

1-1 of 1

anul308

Search by repository name

All content

Create a repository

Name	Last Pushed	Contains	Visibility	Scout
anul308/dockerimage	1 minute ago		Public	Inactive

1-1 of 1

Experiment 11

Aim:-Setup Grafana for Devops.

ACCESS GRAFANA :

kubectl get secret prometheus-grafana -n monitoring -o jsonpath="{.data.admin-user}" | base64 --decode ; echo

If you run the above command u can see the username for grafana (**admin**)

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0) $ kubectl get secret prometheus-grafana -n monitoring -o jsonpath="{.data.admin-user}" | base64 --decode ; echo
admin
```

kubectl get secret prometheus-grafana -n monitoring -o jsonpath="{.data.admin-password}" | base64 --decode ; echo

If you run the above command u can see the password for grafana (**prom-operator**)

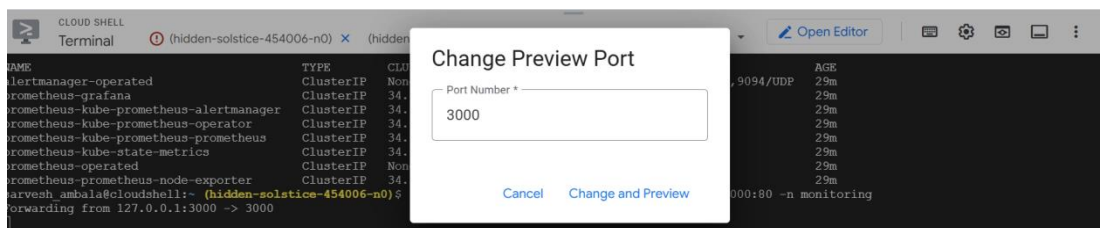
```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0) $ kubectl get secret prometheus-grafana -n monitoring -o jsonpath="{.data.admin-password}" | base64 --decode ; echo
prom-operator
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0) $
```

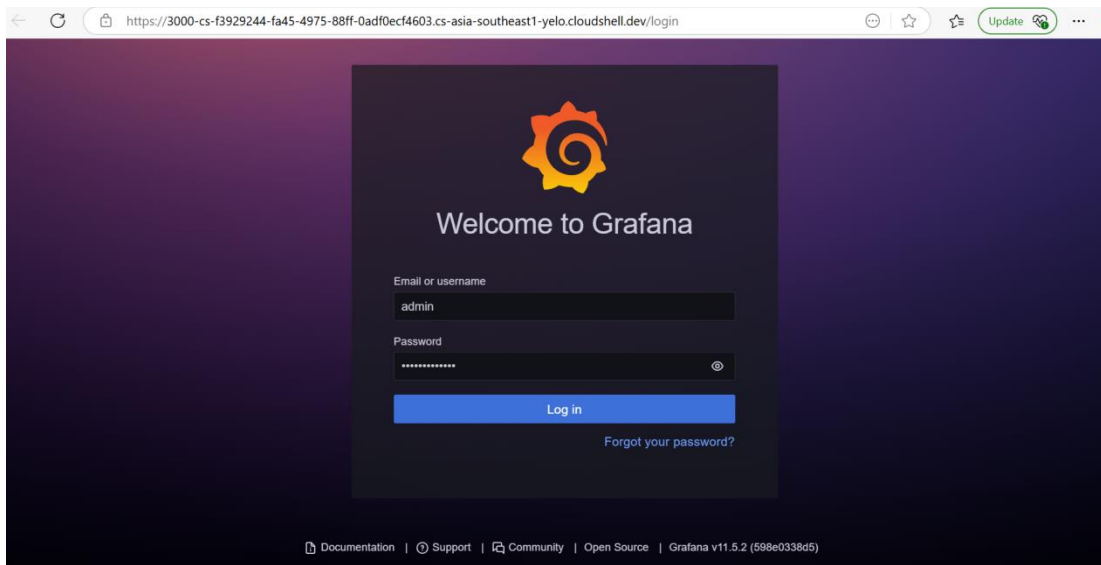
PORT FORWARDING

kubectl port-forward svc/prometheus-grafana 3000:80 -n monitoring

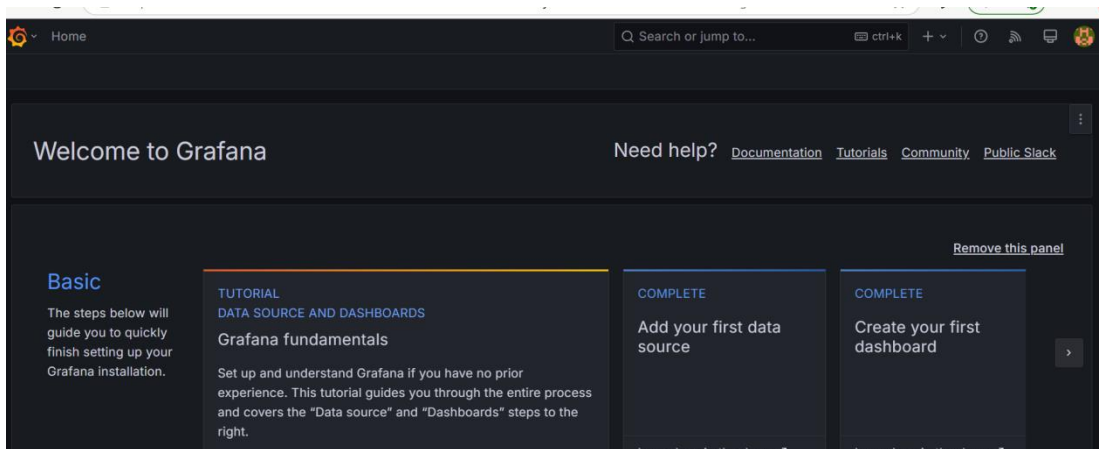
```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0) $ kubectl port-forward svc/prometheus-grafana 3000:80 -n monitoring
Forwarding from 127.0.0.1:3000 -> 3000
```

Click on the web preview give the port no 3000 and click on change and preview u can see the grafana



Output:-

You can login with admin and prom-operator



Experiment 12

Aim:-Setup Prometheus for Devops.

PROMETHEUS SETUP:

helm repo add prometheus <https://prometheus-community.github.io/helm-charts>

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ helm repo add prometheus https://prometheus-community.github.io/helm-charts
"prometheus" has been added to your repositories
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$
```

helm repo update

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "grafana" chart repository
...Successfully got an update from the "prometheus1" chart repository
...Successfully got an update from the "prometheus" chart repository
...Successfully got an update from the "prometheus-community" chart repository
...Successfully got an update from the "pc" chart repository
```

helm install prometheus prometheus-community/kube-prometheus-stack --namespace monitoring --create-namespace

This will install prometheus,alertmanager and grafana

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ helm install prometheus prometheus-community/kube-prometheus-stack --namespace monitoring --create-namespace

```

Check the prometheus pods and services

kubectl get pods -n monitoring

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0)$ kubectl get pods -n monitoring
NAME                                READY   STATUS    RESTARTS   AGE
alertmanager-prometheus-kube-prometheus-alertmanager-0  2/2     Running   0          3m12s
prometheus-grafana-75bb7d6986-rzg4q  3/3     Running   0          3m19s
prometheus-kube-prometheus-operator-65c669f8f9-qcjwt  1/1     Running   0          3m19s
prometheus-kube-state-metrics-645c667b6-6bwwl  1/1     Running   0          3m19s
prometheus-prometheus-kube-prometheus-prometheus-0  2/2     Running   0          3m11s
prometheus-prometheus-node-exporter-f87x4  1/1     Running   0          3m20s
prometheus-prometheus-node-exporter-gz8rs  1/1     Running   0          3m20s
prometheus-prometheus-node-exporter-vnb86  1/1     Running   0          3m20s
```

kubectl get svc -n monitoring


```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0) $ kubectl get svc -n monitoring
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT (S)	AGE
alertmanager-operated	ClusterIP	None	<none>	9093/TCP, 9094/TCP, 9094/UDP	4m27s
prometheus-grafana	ClusterIP	34.118.235.147	<none>	80/TCP	4m36s
prometheus-kube-prometheus-alertmanager	ClusterIP	34.118.225.122	<none>	9093/TCP, 8080/TCP	4m36s
prometheus-kube-prometheus-operator	ClusterIP	34.118.232.165	<none>	443/TCP	4m36s
prometheus-kube-prometheus-prometheus	ClusterIP	34.118.232.239	<none>	9090/TCP, 8080/TCP	4m36s
prometheus-kube-state-metrics	ClusterIP	34.118.228.217	<none>	8080/TCP	4m36s
prometheus-operated	ClusterIP	None	<none>	9090/TCP	4m26s
prometheus-prometheus-node-exporter	ClusterIP	34.118.232.54	<none>	9100/TCP	4m36s

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0) $
```

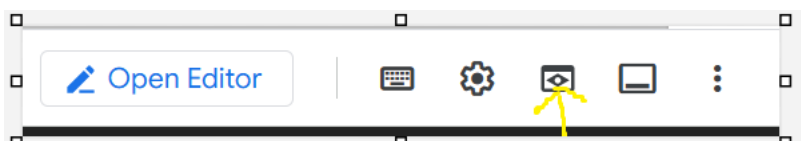
Access prometheus and port forwarding

kubectl port-forward svc/prometheus-kube-prometheus-prometheus 9090:9090 -n monitoring

```
sarvesh_ambala@cloudshell:~ (hidden-solstice-454006-n0) $ kubectl port-forward svc/prometheus-kube-prometheus-prometheus 9090:9090 -n monitoring
```

Forwarding from 127.0.0.1:9090 -> 9090

Click on the webpreview



Change port no to 9090

Status	Name	Location	Tier	Number of nodes	Total vCPUs	Total memory
☒	my-cluster	us-central1-a	Standard	3	6	12 GB

ClusterIP 34.118.235.147 <none> 80/TCP
ClusterIP 34.118.225.122 <none> 9093/TCP
ClusterIP 34.118.232.165 <none> 443/TCP
ClusterIP 34.118.232.239 <none> 9090/TCP
ClusterIP 34.118.228.217 <none> 8080/TCP
ClusterIP None <none> 9090/TCP
ClusterIP 34.118.232.54 <none> 9100/TCP

Preview on port 8080
Change port
About web preview

```
(hidden-solstice-454006-n0) $ kubectl port-forward svc/prometheus-kube-prometheus-prometheus 9090:9090 -n monitoring
```

Click on change and preview

Status	Name	Location	Tier	Number of nodes	Total vCPUs	Total memory
☒	my-cluster	us-central1-a	Standard	3	6	12 GB

ClusterIP 34.118.235.147 <none> 80/TCP
ClusterIP 34.118.225.122 <none> 9093/TCP
ClusterIP 34.118.232.165 <none> 443/TCP
ClusterIP 34.118.232.239 <none> 9090/TCP
ClusterIP 34.118.228.217 <none> 8080/TCP
ClusterIP None <none> 9090/TCP
ClusterIP 34.118.232.54 <none> 9100/TCP

Change Preview Port

Port Number *

9090

Cancel Change and Preview

```
(hidden-solstice-454006-n0) $ kubectl port-forward svc/prometheus-kube-prometheus-prometheus 9090:9090 -n monitoring
```

Now u can able to see prometheus in the browser

Output:-

